Global Unichip | Research update | 27 October 2026

Global Unichip (3443.TW) helps customers implement custom chips and procure their production. Revenue growth is real, but the proportion retained after wafers, packaging, engineers and working capital matters more than the size of the customer chip invoice.

The selected model adds a bounded risk that the existing reference turnkey margin budget overstates retained economics. It does not assume a newly announced fee cut or insert the anonymous source's undefined single-digit margin as a reported GUC gross margin. The reference revenue budgets, four prior event probabilities, 20x method and end-window shares are retained.

Measure	Current selection	Boundary
Q4 2027 model ceiling	2,613.12	Research estimate, TWD per ordinary share
Previous same-quarter ceiling	2,657.03	Prior approved model vintage; same method
Revision	-1.6526%	Percent change in unrounded model ceiling, not a share-price return
Dated comparison quote	7,585.00	22 October 2026 regular close; later-session coverage unconfirmed
Analytical / Excel Base	68	68 remains unchanged; thesis quality is not a second economic probability
Multiple	20	20x fully expensed rolling four-quarter operating profit
Published Q3 2026 ceiling	1,687.59	Frozen as-then value; current reconstruction is separate

At the verified dated quotation, this model does not show positive upside. Coverage does not establish an owner holding. Amounts are TWD millions and shares are millions unless another unit is stated. Actuals, attributed expert claims and model assumptions remain distinct.

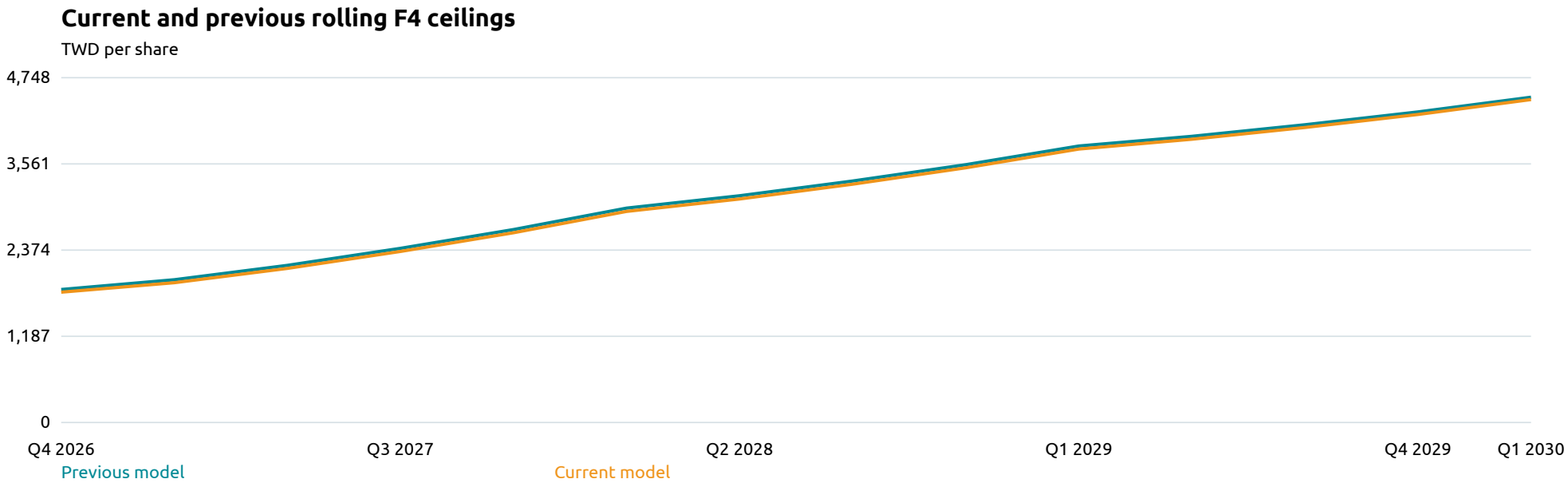


Figure 1. Same-quarter, same-method comparison. Each value uses four operating quarters and the last quarter's share denominator; the historical Q3 2026 display is excluded.

Business model, customer proof and competitive position

GUC supplies design implementation, verification, interface IP, package integration and production services. Non-recurring engineering (NRE) and IP income arise during design and milestone work; turnkey revenue follows manufactured-chip orders. Turnkey invoices include substantial foundry and packaging purchases. A larger or more advanced chip can expand booked revenue faster than retained profit.

Its TSMC relationship supports process access and implementation, but TSMC's profits and a hyperscaler's systems revenue do not belong to GUC. Customer-specific contract values, die prices, accepted unit volumes, rebates and gross-versus-net terms are not publicly mapped well enough to convert the expert's customer-unit claims directly into GUC sales.

Official solutions materials describe high-speed SerDes and Spec-in, RTL-in and GDSII-in service entry points. A blanket claim that GUC has no SerDes capability or only performs backend work would be inaccurate. Those portfolio statements do not establish ownership of a particular customer's latest-node architecture, a specific PCIe generation, IP licensing economics or an awarded TPU v 10 program.

Activity	Economic benefit	Constraint and competing route
Design / NRE / reusable IP	Engineering and interface capability can support qualification and switching costs.	Customer architecture teams, Broadcom, Marvell, Alchip and other ASIC-service providers offer alternative routes; contract scope matters.
Turnkey production	GUC can coordinate wafers, advanced packaging, test and delivery.	Wafer and packaging cost stay in COGS; foundry, OSAT and customer-funded purchases are not free inputs.
HBM / chiplets / advanced nodes	More difficult implementation can increase engineering content.	Tape-out or design readiness is not production qualification, usable capacity, yield or accepted shipment.
Optical / backend and systems collaboration	A bounded residual may capture additional GUC service revenue.	Partner optical-engine/rack revenue remains with partners; CPO and existing ASIC budgets overlap.

GUC's September HBM4E design-ready announcement identifies N2P and CoWoS-L use and customer adoption; it is a technical milestone, not a quantified production contract. Wiwynn collaboration and the named VSORA implementation demonstrate activity without disclosing a large contracted production value. No separate Tesla AI6 revenue or full TPU v 10 wallet is added.

Latest reported income, product mix and cash evidence

H1 2026 and Q2 2026 are the latest reported financial periods in the preserved source pack. The July-September monthly sales total is reported, but Q3 product mix, margins and profit remain forecasts pending the scheduled 29 October results. Statutory statements control exact figures where brief management releases round or contain editorial errors.

Measure	Q1 2026	Q2 2026	H1 2026	Q2 2025	H1 2025
revenue	11,447.812	13,896.557	25,344.369	6,104.577	13,128.221
COGS	8,334.505	10,902.540	19,237.045	4,070.659	9,121.688
GP	3,113.307	2,994.017	6,107.324	2,033.918	4,006.533
RD	1,013.423	1,046.188	2,059.611	836.987	1,636.655
selling	124.524	118.751	243.275	93.772	190.413
administration	151.182	152.732	303.914	159.497	287.189
total opex	1,289.129	1,317.671	2,606.800	1,090.256	1,968.234
OP	1,824.178	1,676.346	3,500.524	943.662	2,038.299
PBT	1,924.721	1,755.972	3,680.693	929.011	2,071.644
income tax	278.481	200.715	479.196	147.346	328.611
NI	1,646.240	1,555.257	3,201.497	781.665	1,743.033
basic EPS TWD	12.28	11.61	23.89	5.84	13.01

Q2 revenue increased from Q1 while gross profit and operating profit declined. The written management record attributes lower NRE and turnkey margins to product mix. This is direct counter-evidence to treating the whole production ramp as high-margin IP income. Current-period product margins remain undisclosed assumptions.

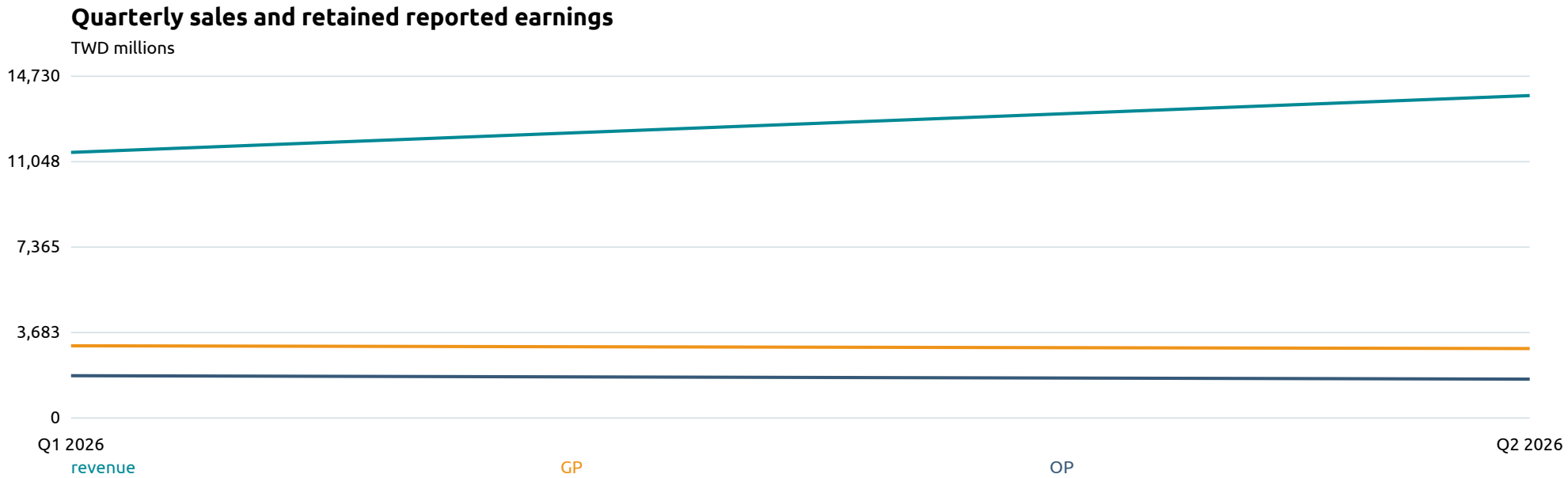


Figure 2. Reviewed statutory statements. H1 is not plotted as a third quarter or a further revenue tranche.

Measure	Q1 2026	Q2 2026	H1 2026	2025FY
NRE	1,642.739	2,310.470	3,953.209	8,405.177
goods turnkey	9,805.073	11,586.087	21,391.160	25,735.801

Measure	Value	Scope
cash equivalents	10,034.146	Source actual / dated statement
financial assets amortized cost	337.520	Source actual / dated statement
pledged time deposits separate from cash	53.219	Source actual / dated statement
trade AR including related	3,933.121	Source actual / dated statement
inventory	14,539.650	Source actual / dated statement
purchase prepayments	4,418.131	Source actual / dated statement
other current assets	606.198	Source actual / dated statement
AP including related	6,591.440	Source actual / dated statement
contract liabilities	13,214.722	Source actual / dated statement
accrued employee director compensation	1,829.919	Source actual / dated statement
PPE	2,216.193	Source actual / dated statement
ROU assets	248.463	Source actual / dated statement

Measure	Value	Scope
intangibles	566.870	Source actual / dated statement
prepaid equipment	21.303	Source actual / dated statement
refundable deposits	386.654	Source actual / dated statement
lease liabilities	271.581	Source actual / dated statement

Measure	Value	Date / unit
Issued ordinary shares	134,011,911	30 June 2026; shares, not millions

Measure	Value	Scope
CFO	5,223.546	Source actual / dated statement
depreciation addback PPE plus ROU	180.326	Source actual / dated statement
amortization addback	224.757	Source actual / dated statement
AR increase cash use	515.772	Source actual / dated statement
inventory increase cash use	4,391.627	Source actual / dated statement
purchase prepayment increase cash use	1,820.848	Source actual / dated statement
contract liability increase cash source	5,875.835	Source actual / dated statement
AP increase cash source	1,993.199	Source actual / dated statement
cash tax paid	203.952	Source actual / dated statement
cash PPE acquisitions	848.752	Source actual / dated statement
cash intangible acquisitions	301.138	Source actual / dated statement
cash dividends	2,680.238	Source actual / dated statement
cash lease principal	46.687	Source actual / dated statement
cash interest paid	3.266	Source actual / dated statement

Written management records, recordings and source completeness

The newest available written quarterly management records are Q2 (30 July 2026) and Q1 (30 April 2026), each five pages, together with their financial statements and decks. These are written records; they are not a complete spoken-call transcript.

Public issuer-linked recordings are available for Q2 (31 July, approximately 90 minutes) and FY2025 Q4 (30 January, approximately 100 minutes). No complete listening or transcription of those recordings has been performed for this review, and provided captions were unavailable. The FY2025 Q4 deck is available and was read separately. Audio availability does not establish spoken coverage.

The owner's anonymous expert material contains 15 supplied translated image pages and 30 questions. All supplied pages were reviewed in the source work. The exact underlying call day, expert identity, original-language recording and an authenticated full spoken transcript remain unavailable. The note has an AI-translation/inconsistency notice. It is attributed evidence, not issuer guidance.

Material	Coverage	What remains unknown
Latest issuer actuals	Reviewed Q1 and H1 / Q2 financial statements and written materials.	Q3 profit, latest capital after June and later-session quote coverage.
Issuer spoken calls	Public Q2 and FY25 Q4 recordings preserved / available.	Full spoken contents have not been listened to or transcribed.
Anonymous expert note	All 15 supplied pages / 30 question dispositions; private exhibits below.	Authenticity, original audio, exact day and customer contract mapping.
Official solutions correction	High-speed SerDes and multiple service-entry models are publicly described.	Particular contract/IP ownership, latest-node performance and specific PCIe generation.

All 30 expert claims: attribution, overlap and limits

The following table preserves every supplied question's claim and source boundary. Reader spacing is normalized; numbers and substantive meaning are retained. Question 19 and Question 21 must be read with the official-solutions correction: no blanket no-SerDes or backend-only statement is adopted as fact. The financial capture case is a model judgment about the reference margin level, not an announcement of a new fee cut.

Question / physical pages	Attributed claim	Boundary / overlap disposition
Q01 | pp 6	Google CPU shipments asserted above 1.2 million in 2026 and at least 60-70% growth in 2027; anonymous estimate, not GUC disclosed orders or revenue.	incremental customer detail; existing model already has strong cloud/advanced-node growth Source: anonymous translated expert; not issuer guidance.
Q02 | pp 6	Expert says 2H25 view of 2026 shipments became roughly 2.5 times earlier 2025 view after alleged TSMC allocation and hyperscaler investment. Historical forecast revisions are not an additional future revenue tranche.	potential overlap with already reported 2026 ramp Source: anonymous translated expert; not issuer guidance.
Q03 | pp 6, 7	2027 capacity not finalized; expert regards 5 million Google CPU volume as too optimistic, despite continuing growth.	capacity/acceptance limit; no firm capacity commitment Source: anonymous translated expert; not issuer guidance.
Q04 | pp 7	Microsoft Maia 200 asserted near 2 million 2027 units versus prior just-over 1 million expectation; called first mass-production year.	new named program estimate; unit/package and GUC accounting-capture gaps remain Source: anonymous translated expert; not issuer guidance.
Q05 | pp 7	Large customers mainly lower-margin TK3/SCM; expert identifies NRE and smaller-customer TK1 as important profit sources.	mix detail challenges treating all turnkey growth as equal-margin revenue Source: anonymous translated expert; not issuer guidance.
Q06 | pp 7	TK1 said double-digit and TK3 single-digit margins; denominator and gross-versus-operating definition are not supplied.	cannot insert as reported segment GM or OP margins Source: anonymous translated expert; not issuer guidance.
Q07 | pp 8	TK1 share could fall while its absolute revenue and NRE grow as hyperscaler production expands.	mix-versus-level distinction; overlaps broad volume growth Source: anonymous translated expert; not issuer guidance.
Q08 | pp 8	Small/startup and established system-house customers provide design/turnkey opportunities and ecosystem services.	qualitative business opportunity; not identified awards Source: anonymous translated expert; not issuer guidance.
Q09 | pp 8	Large direct-TSMC accounts may use GUC design service without TK1 manufacturing production route.	principal/agent and wallet exclusion lead, not contract-level accounting determination Source: anonymous translated expert; not issuer guidance.
Q10 | pp 8	Advanced packaging/SoIC/3D opportunity called TK2, distinct from TK3; expert places maturation in 2H2027-2028.	do not move TK2 future service economics into existing TK3 fees Source: anonymous translated expert; not issuer guidance.
Q11 | pp 9	Google CPU TK3 SCM profit and TSMC rebate both described as low-single-digit percentages.	two economic descriptions may overlap reported margin/cost; no audited supplier-rebate amount Source: anonymous translated expert; not issuer guidance.
Q12 | pp 9	Rebate tied to manufacturing scale/volume threshold; translated rate per total delivered units is dimensionally ambiguous.	no established dollar-per-chip or percentage-of-chip-price conversion Source: anonymous translated expert; not issuer guidance.
Q13 | pp 9	2026-2027 product-generation transition may affect wafer capacity/production scheduling.	timing qualification, no contracted delivery schedule Source: anonymous translated expert; not issuer guidance.
Q14 | pp 9, 10	Next-generation CPU package price/performance asserted around 2 times; two chips in package also mentioned.	package versus die units and GUC recognized revenue not determined Source: anonymous translated expert; not issuer guidance.

Question / physical pages	Attributed claim	Boundary / overlap disposition
Q15 | pp 10	Expert production chain covers wafer order, CP, advanced packaging, outsourced assembly/test and delivery.	procurement/acceptance chain; does not remove cash funding exposure Source: anonymous translated expert; not issuer guidance.
Q16 | pp 10	GUC characterized as design/service with third-party manufacturing and light own material content.	not proof customer-owned inventory or net reporting for every contract Source: anonymous translated expert; not issuer guidance.
Q17 | pp 10, 11	Design/verification occur before release to production; later volume work becomes SCM.	design-versus-production timing boundary Source: anonymous translated expert; not issuer guidance.
Q18 | pp 11	Possible Google TPU v 10 IP/3D integration discussed; service scope still negotiated and nothing settled.	not a confirmed design win; CPU and TPU programs must stay distinct Source: anonymous translated expert; not issuer guidance.
Q19 | pp 11, 12	Expert contrasts MediaTek broader RTL/architecture with GUC backend implementation and compares AVGO high-performance role.	comparative opinion, not verified supplier award or GUC revenue share Source: anonymous translated expert; not issuer guidance.
Q20 | pp 12	Google internal design capability may lead to targeted lower-cost outsourcing; expert expects more in-house work.	insourcing/capture risk alongside growth Source: anonymous translated expert; not issuer guidance.
Q21 | pp 12	HBM PHY viewed most valuable; UCIe/GLink and analog/external IP capabilities discussed.	technical opinions require issuer specification checks; not specific royalty contract Source: anonymous translated expert; not issuer guidance.
Q22 | pp 13	GLink described as chiplet/2.5D/3D interconnect; translated technical wording is imprecise.	do not treat as quantified external IP market-share or royalty forecast Source: anonymous translated expert; not issuer guidance.
Q23 | pp 13	Expert expects Microsoft/Google 2027 margins broadly stable and higher Tesla/Meta economics; Meta production ramp stated 2028.	question-answer 2028 timing is narrower than broad summary 2027 outlook Source: anonymous translated expert; not issuer guidance.
Q24 | pp 13	Illustrative Meta margin around 20%, mid-sized customer 25%, small customer above 30%.	undefined margin scope/contract terms; not current reported segment margins Source: anonymous translated expert; not issuer guidance.
Q25 | pp 14	Tesla described as NRE/design fees plus TSMC volume rebate while wafers ordered directly withTSMC.	exclude full wafer/chip wallet unless a distinct principal contract is evidenced Source: anonymous translated expert; not issuer guidance.
Q26 | pp 14	Higher Tesla chip selling price does not automatically raise GUC fee/rebate income.	explicit non-proportional revenue-capture limit Source: anonymous translated expert; not issuer guidance.
Q27 | pp 14	China exposure described broadly stable without exact current customer amounts.	qualitative regional claim, not source for numeric customer allocation Source: anonymous translated expert; not issuer guidance.
Q28 | pp 14	Top 3Google/Microsoft/Tesla around 70% ofUS revenue; US around 70% oftotal.	roughly 49% total only if scopes and periods align; not 70% ofglobal Source: anonymous translated expert; not issuer guidance.
Q29 | pp 14, 15	Tesla 1 million 2027 volume appears in question premise; response says GUC Tesla revenue below Microsoft because fee/rebate rather than fullchip sales.	premise is not independent expert confirmation of shipment volume Source: anonymous translated expert; not issuer guidance.
Q30 | pp 15	Horizon Robotics and unnamed Japanese company discussed as potential contributors; expert remains optimistic while noting Google insourcing.	unconfirmed qualitative pipeline, no new model revenue selected Source: anonymous translated expert; not issuer guidance.

Google CPU shipment estimates are distinct from possible TPU v 10 negotiations. Two dies per package or a package-price change is not a GUC revenue or margin multiplier. The top-three-customer assertion is about roughly 70% of US-region revenue, with the US region roughly 70% of total; the approximate 49% product is not a disclosed 70% global share. Tesla's one-million-unit 2027 number is a question premise. TK2 timing in H2 2027-2028 is distinct from TK3. Supplier rebate quantities and the margin denominator remain unclear.

Five events, conditional economics and joint probabilities

The model retains 60% design conversion, 35% optical production, 30% NRE pricing and 30% reference ramp delay. The added capture event has 40% probability that the existing reference margin budget is overstated on the selected pool. The separate 60% source-authenticity judgment is not multiplied into that probability again. Existing reference growth and overlap are explicitly retained; this is not a stack of five independent revenue increases.

Possible TPU v 10 backend or 3D work stays inside the existing design pool. A 30% preliminary identity attribution is a subset of the 60% design event (50% conditional on design and zero without it). It adds no separate contract dollars, full-platform wallet or sixth revenue event. Separately retained new-design economics are not proved immune to commercial mix risk; that scope remains uncertain.

Event	Probability	Conditional economics and overlap
New Design Conversion	60.0000%	CY2027/28/29/30 gross new NRE TWD 1,000/2,000/3,000/4,000m and associated turnkey sales TWD 12,000/22,000/30,000/40,000m; retain 40% NRE and 30% turnkey after overlap. Reference already includes existing 3 nm cloud ramps, advanced-node migration and most HBM-related growth. No Tesla AI6 revenue is assigned and no separate addition for the generic ASIC TAM. 60% NRE GP less 25% engineering expense; turnkey GP 15.5-17% less 2.5% engineering and selling cost. Weight the combined conditional economics once.
Optical Io Production	35.0000%	Gross optical-I/O-related GUC sales TWD 300/1,500/4,000/7,000m in CY2027-30; retain only 25% after overlap and partner invoice exclusions. 75% deduction covers advanced-node reference growth, the new-design branch and partner-owned hardware. No additive Wiwynn rack, Lightmatter optical-engine or Ayar revenue. 28% blended GP less 10% engineering and integration expense = 18% conditional OP.
New Nre Pricing	30.0000%	3% premium on 25% of reference NRE plus probability-weighted new NRE; remove 50% for mix and prior commercial overlap. No premium on fixed-price backlog or pass-through wafers. Existing NRE mix and rising engineering costs remain in the reference. 15% delivery cost and 20% operating support leave 65% of incremental price as OP. Funding stress states recompute eligibility using each new-design outcome.
Ramp Delay	30.0000%	15% of reference turnkey sales shift beyond their forecast year, plus TWD 250m annual extra qualification cost. One combined customer-schedule, packaging-yield and acceptance risk rather than separately stacking correlated bottlenecks. Lost turnkey gross profit offset by 2% avoidable operating cost; retain extra qualification spending.
Reference Margin Capture	40.0000%	35% referenceTK eligible; extra normalized COGS/GP loss 2/2.5/2/1.5/1 ppt inQ426/CY27/28/29/30, after the same-state ramp delay. No revenue increase/decrease and no assumed opex saving. Reference pool only; R/Mjoint handled by Frechet interpolation. Do not stack a second rebate or full customer chip wallet. GP loss=referenceTK*eligible*loss*(1-.15*R) for 2027+; OPsame loss; tax/cash recomputed. Q426 no prior ramp branch, Q326 historical no new loss.

Selected event probabilities

Fraction; selected judgments

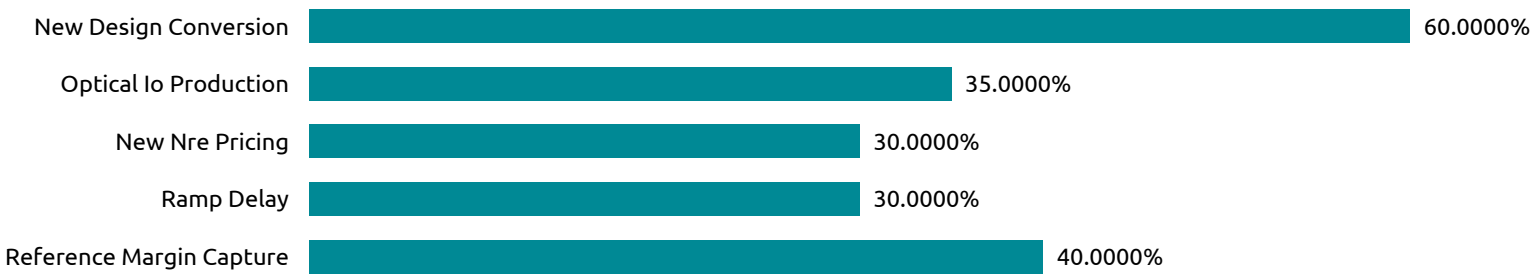


Figure 3. Five financial events. Authenticity and the analytical score are separate from these probabilities.

Ramp delay (R) and capture loss (M) are positively associated. Their four joint cells are 45% / 25% / 15% / 15% for 00 / 01 / 10 / 11. The association is a one-sixth interpolation from independence toward the Frechet upper bound. Design, optical and pricing combine with this R/M pair to form 32 persistent full-path states. Events are not redrawn independently every quarter.

R / M	Joint probability	Interpretation
00	45.0000%	No delay, no capture loss
01	25.0000%	Capture loss only
10	15.0000%	Delay only
11	15.0000%	Delay and capture loss

Joint ramp and capture probability cells

Fraction

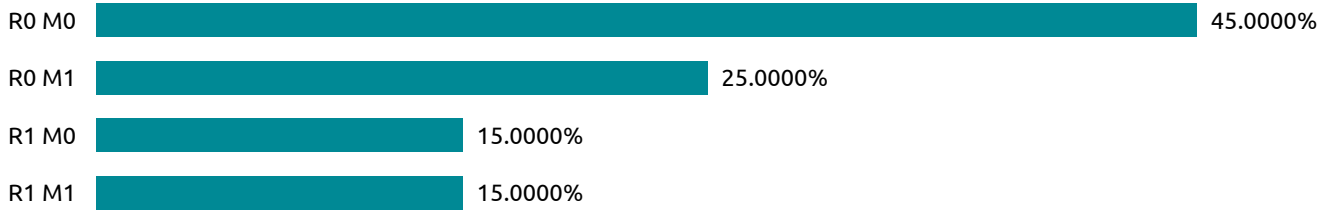


Figure 4. Nonnegative probability arithmetic only. MAX used to close tiny numerical probability residue is not a floor on the signed valuation.

Margin capture: normalized COGS, pool and duration

For each coherent state, capture GP loss = reference turnkey revenue x eligible share x selected percentage-point gap x (1 - 15% x ramp-delay bit) from 2027 onward. Q4 2026 has no prior ramp reduction and Q3 2026 has no new capture charge. Booked revenue is unchanged by capture; implied normalized COGS rises by the same amount, so OP and cash fall without an extra noncash addback.

The 35% eligible reference-turnkey pool is a low-confidence sizing assumption. It is not inferred from the source's approximate 49% regional concentration arithmetic or disclosed customer contracts. The selected 2/2.5/2/1.5/1 percentage-point gaps are a bounded baseline-level/mix risk. Source customer margins were described as stable for 2027; no reported new fee decrease is claimed. The 2028-2030 fade is a model assumption, not an expert forecast.

In the weighted operating bridge, expected loss = reference TK x eligible share x gap x (pM - 15% x P(R=1,M=1)) for 2027+. The coupled overlap removes capture loss from reference sales already delayed in the same state. Pricing uses (reference NRE + design-state new NRE) x 25% x 3% x 50%, then the pricing-state bit; averaging therefore retains the design-pricing cross-term. Neither supplier rebates nor the customer's full chip wallet are added again.

Period	Conditional GP gap	Eligible reference TK	Scope
Q4 2026	2.0000%	35.0000%	Extra normalized COGS; remaining same-state reference pool only
2027	2.5000%	35.0000%	Extra normalized COGS; remaining same-state reference pool only
2028	2.0000%	35.0000%	Extra normalized COGS; remaining same-state reference pool only
2029	1.5000%	35.0000%	Extra normalized COGS; remaining same-state reference pool only
2030	1.0000%	35.0000%	Extra normalized COGS; remaining same-state reference pool only

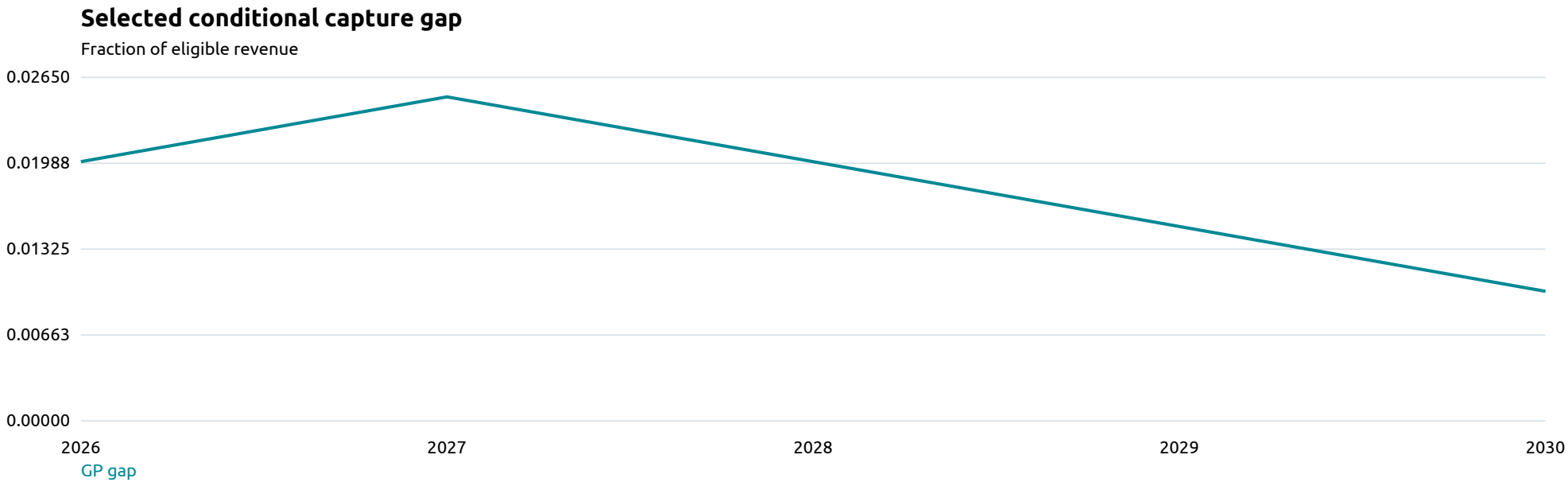


Figure 5. The gap is in percentage points of eligible-revenue gross margin. It is not a disclosed TK3 gross or operating margin.

Complete quarterly inputs and operating bridge

The reference already budgets substantial advanced-node growth. Calendar seasonality is 21% / 24% / 26% / 29%. Reference NRE and turnkey gross margins, program values, cash reserves and future costs are model-selected assumptions unless specifically tied to a dated actual below. Q3 2026 total revenue is monthly reported sales; its NRE split, margins and OP are current reconstructed estimates, not a replacement historical display.

Quarter	NRE / IP	Turnkey	nre gm	tk gm	Raw share control	multiple
Q3 2026	2,500.00	16,254.50	58.0000%	15.0000%	134.287911	20
Q4 2026	3,000.00	19,000.00	60.0000%	15.0000%	134.312911	20
Q1 2027	2,520.00	17,430.00	60.0000%	15.5000%	134.337911	20
Q2 2027	2,880.00	19,920.00	60.0000%	15.5000%	134.362911	20
Q3 2027	3,120.00	21,580.00	60.0000%	15.5000%	134.387911	20
Q4 2027	3,480.00	24,070.00	60.0000%	15.5000%	134.412911	20
Q1 2028	3,150.00	24,150.00	61.0000%	16.5000%	134.437911	20
Q2 2028	3,600.00	27,600.00	61.0000%	16.5000%	134.462911	20
Q3 2028	3,900.00	29,900.00	61.0000%	16.5000%	134.487911	20
Q4 2028	4,350.00	33,350.00	61.0000%	16.5000%	134.512911	20
Q1 2029	3,780.00	29,820.00	62.0000%	17.0000%	134.537911	20
Q2 2029	4,320.00	34,080.00	62.0000%	17.0000%	134.562911	20
Q3 2029	4,680.00	36,920.00	62.0000%	17.0000%	134.587911	20
Q4 2029	5,220.00	41,180.00	62.0000%	17.0000%	134.612911	20
Q1 2030	4,410.00	34,440.00	62.0000%	17.0000%	134.637911	20
Q2 2030	5,040.00	39,360.00	62.0000%	17.0000%	134.662911	20
Q3 2030	5,460.00	42,640.00	62.0000%	17.0000%	134.687911	20
Q4 2030	6,090.00	47,560.00	62.0000%	17.0000%	134.712911	20

Quarter	Research and development	Selling and administration	rent reserve	Non-ROU DA proxy	Dated financial claims	shares
Q3 2026	1,120.00	300.00	4.00	210.00	-5,367.35	134.287911
Q4 2026	1,200.00	320.00	4.00	210.00	-5,367.35	134.312911
Q1 2027	1,350.00	375.00	5.00	210.00	-5,367.35	134.337911
Q2 2027	1,350.00	375.00	5.00	210.00	-5,367.35	134.362911
Q3 2027	1,350.00	375.00	5.00	210.00	-5,367.35	134.387911
Q4 2027	1,350.00	375.00	5.00	210.00	-5,367.35	134.412911
Q1 2028	1,650.00	462.50	5.00	250.00	-5,367.35	134.437911
Q2 2028	1,650.00	462.50	5.00	250.00	-5,367.35	134.462911

Quarter	Research and development	Selling and administration	rent reserve	Non-ROU DA proxy	Dated financial claims	shares
Q3 2028	1,650.00	462.50	5.00	250.00	-5,367.35	134.487911
Q4 2028	1,650.00	462.50	5.00	250.00	-5,367.35	134.512911
Q1 2029	1,950.00	550.00	5.00	250.00	-5,367.35	134.537911
Q2 2029	1,950.00	550.00	5.00	250.00	-5,367.35	134.562911
Q3 2029	1,950.00	550.00	5.00	250.00	-5,367.35	134.587911
Q4 2029	1,950.00	550.00	5.00	250.00	-5,367.35	134.612911
Q1 2030	2,200.00	625.00	5.00	250.00	-5,367.35	134.637911
Q2 2030	2,200.00	625.00	5.00	250.00	-5,367.35	134.662911
Q3 2030	2,200.00	625.00	5.00	250.00	-5,367.35	134.687911
Q4 2030	2,200.00	625.00	5.00	250.00	-5,367.35	134.712911

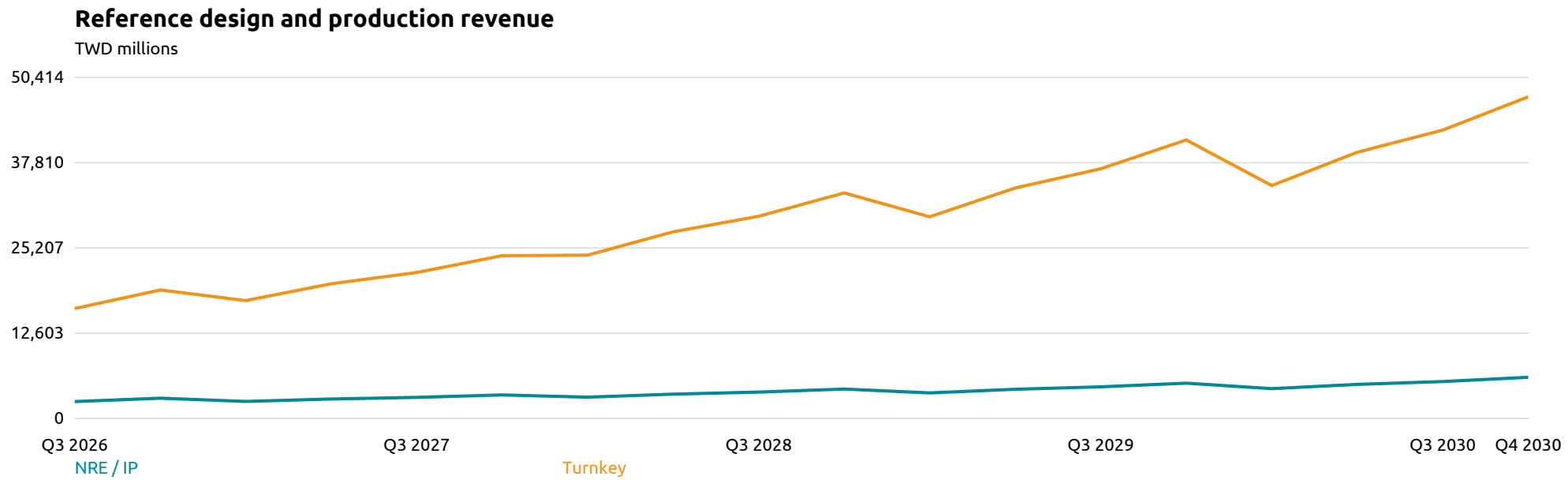


Figure 6. Separate revenue streams. Turnkey includes wafer and packaging pass-through costs; its gross margin is not an IP royalty margin.

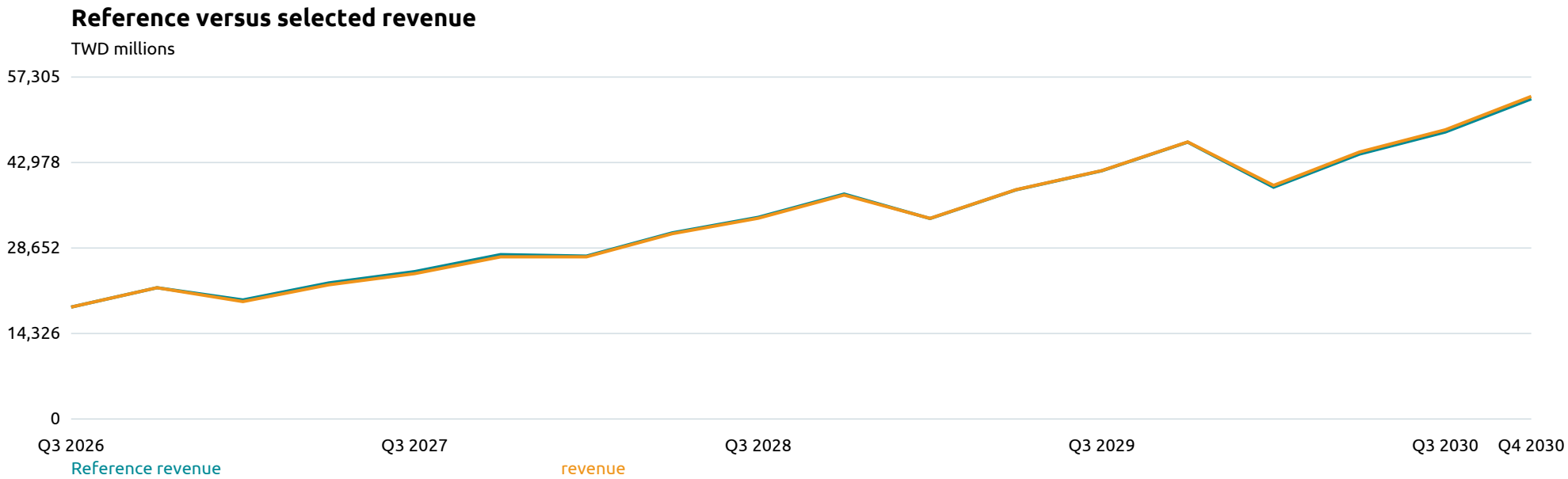


Figure 7. The added capture case changes normalized COGS and retained profit, not booked revenue. Existing delay and new-program branches still change revenue.

Quarter	Reference gross profit	revenue	Gross profit	opex	Operating profit	Expected capture GP loss
Q3 2026	3,888.17	18,754.50	3,888.17	1,424.00	2,464.17	0.00
Q4 2026	4,650.00	22,000.00	4,596.80	1,524.00	3,072.80	53.20
Q1 2027	4,213.65	19,678.05	4,139.05	1,755.13	2,383.92	57.57
Q2 2027	4,815.60	22,489.20	4,730.34	1,758.72	2,971.62	65.80
Q3 2027	5,216.90	24,363.31	5,124.54	1,761.12	3,363.42	71.28
Q4 2027	5,818.85	27,174.46	5,715.83	1,764.71	3,951.13	79.51
Q1 2028	5,906.25	27,176.87	5,871.64	2,160.99	3,710.65	63.82
Q2 2028	6,750.00	31,059.28	6,710.45	2,167.21	4,543.24	72.93
Q3 2028	7,312.50	33,647.55	7,269.65	2,171.35	5,098.30	79.01
Q4 2028	8,156.25	37,529.96	8,108.46	2,177.56	5,930.89	88.13
Q1 2029	7,413.00	33,621.22	7,433.62	2,568.30	4,865.32	59.10
Q2 2029	8,472.00	38,424.25	8,495.56	2,577.34	5,918.22	67.54
Q3 2029	9,178.00	41,626.28	9,203.53	2,583.37	6,620.16	73.17
Q4 2029	10,237.00	46,429.31	10,265.47	2,592.41	7,673.06	81.61
Q1 2030	8,589.00	39,147.61	8,698.45	2,916.85	5,781.60	45.50

Quarter	Reference gross profit	revenue	Gross profit	opex	Operating profit	Expected capture GP loss
Q2 2030	9,816.00	44,740.13	9,941.09	2,929.26	7,011.83	52.00
Q3 2030	10,634.00	48,468.47	10,769.52	2,937.53	7,831.98	56.34
Q4 2030	11,861.00	54,060.99	12,012.15	2,949.94	9,062.21	62.84

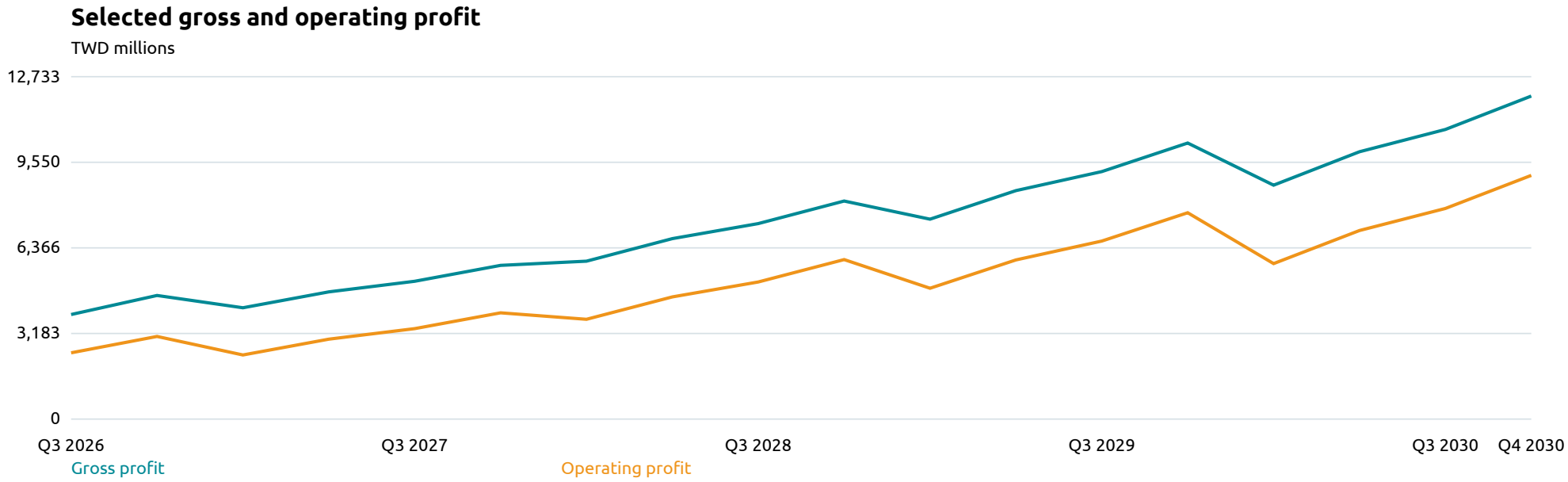


Figure 8. Fully expensed operating costs remain. Capture loss is extra normalized COGS; no noncash addback is created.

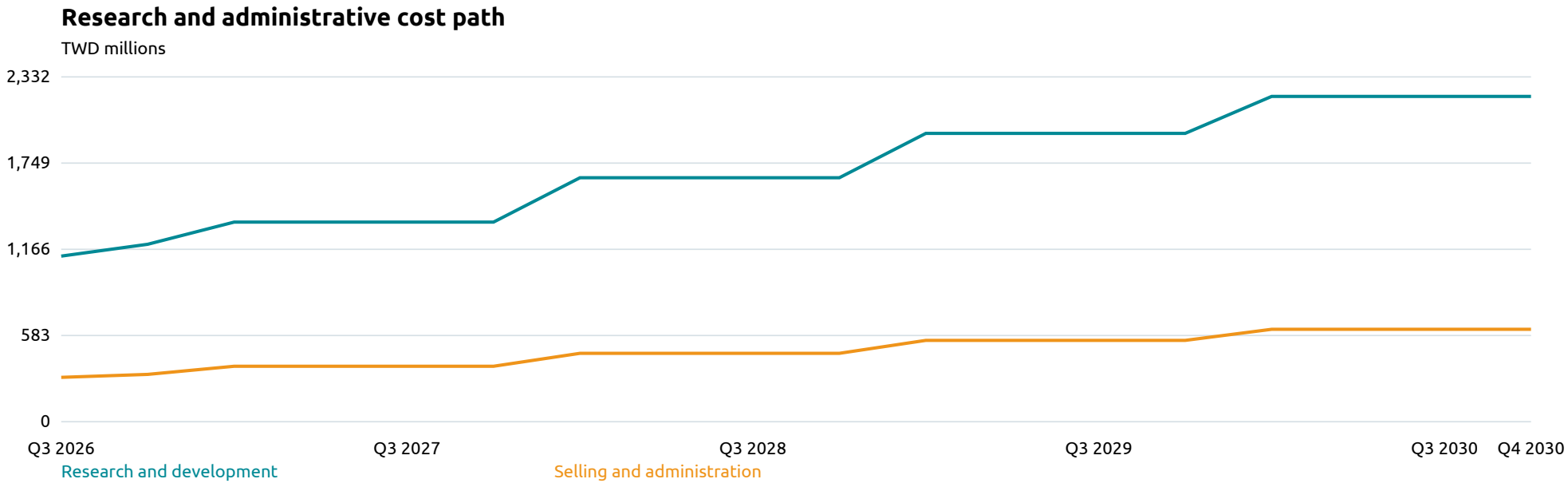


Figure 9. Engineering, staff and tools are paid for. Ordinary employee compensation and software costs are not removed to manufacture an adjusted profit.

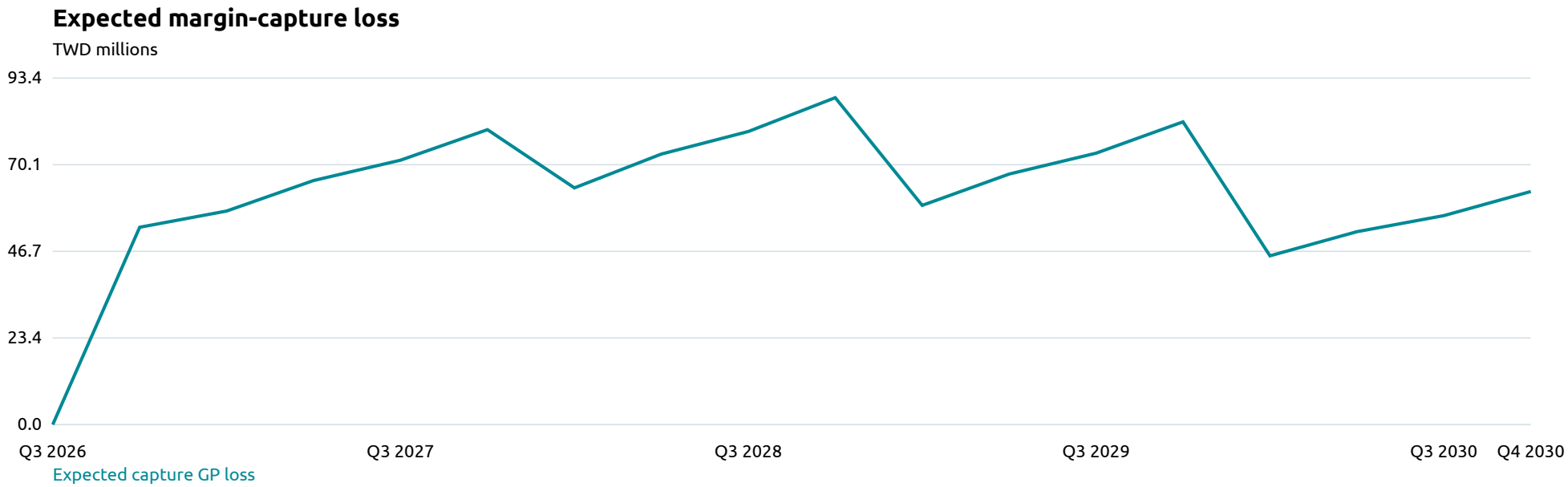


Figure 10. Reference turnkey remaining after the same-state ramp shortfall bears the selected loss. Q3 2026 reconstruction has no new capture charge.

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q3 2026	60.0000%	0.00	0.00	0.00
Q4 2026	60.0000%	0.00	0.00	0.00
Q1 2027	60.0000%	840.00	167.58	39.90
Q2 2027	60.0000%	960.00	191.52	45.60
Q3 2027	60.0000%	1,040.00	207.48	49.40
Q4 2027	60.0000%	1,160.00	231.42	55.10
Q1 2028	60.0000%	1,554.00	329.49	76.65
Q2 2028	60.0000%	1,776.00	376.56	87.60
Q3 2028	60.0000%	1,924.00	407.94	94.90
Q4 2028	60.0000%	2,146.00	455.01	105.85
Q1 2029	60.0000%	2,142.00	472.50	110.25
Q2 2029	60.0000%	2,448.00	540.00	126.00
Q3 2029	60.0000%	2,652.00	585.00	136.50
Q4 2029	60.0000%	2,958.00	652.50	152.25
Q1 2030	60.0000%	2,856.00	630.00	147.00

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q2 2030	60.0000%	3,264.00	720.00	168.00
Q3 2030	60.0000%	3,536.00	780.00	182.00
Q4 2030	60.0000%	3,944.00	870.00	203.00

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q3 2026	35.0000%	0.00	0.00	0.00
Q4 2026	35.0000%	0.00	0.00	0.00
Q1 2027	35.0000%	15.75	4.41	1.58
Q2 2027	35.0000%	18.00	5.04	1.80
Q3 2027	35.0000%	19.50	5.46	1.95
Q4 2027	35.0000%	21.75	6.09	2.18
Q1 2028	35.0000%	78.75	22.05	7.88
Q2 2028	35.0000%	90.00	25.20	9.00
Q3 2028	35.0000%	97.50	27.30	9.75
Q4 2028	35.0000%	108.75	30.45	10.88
Q1 2029	35.0000%	210.00	58.80	21.00
Q2 2029	35.0000%	240.00	67.20	24.00
Q3 2029	35.0000%	260.00	72.80	26.00
Q4 2029	35.0000%	290.00	81.20	29.00
Q1 2030	35.0000%	367.50	102.90	36.75
Q2 2030	35.0000%	420.00	117.60	42.00
Q3 2030	35.0000%	455.00	127.40	45.50
Q4 2030	35.0000%	507.50	142.10	50.75

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q3 2026	30.0000%	0.00	0.00	0.00
Q4 2026	30.0000%	0.00	0.00	0.00
Q1 2027	30.0000%	9.64	8.19	1.93
Q2 2027	30.0000%	11.02	9.36	2.20
Q3 2027	30.0000%	11.93	10.14	2.39
Q4 2027	30.0000%	13.31	11.31	2.66
Q1 2028	30.0000%	12.19	10.36	2.44
Q2 2028	30.0000%	13.93	11.84	2.79
Q3 2028	30.0000%	15.09	12.83	3.02

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q4 2028	30.0000%	16.83	14.31	3.37
Q1 2029	30.0000%	14.74	12.53	2.95
Q2 2029	30.0000%	16.85	14.32	3.37
Q3 2029	30.0000%	18.25	15.51	3.65
Q4 2029	30.0000%	20.36	17.30	4.07
Q1 2030	30.0000%	17.29	14.70	3.46
Q2 2030	30.0000%	19.76	16.80	3.95
Q3 2030	30.0000%	21.41	18.20	4.28
Q4 2030	30.0000%	23.88	20.30	4.78

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q3 2026	30.0000%	0.00	0.00	0.00
Q4 2026	30.0000%	0.00	0.00	0.00
Q1 2027	30.0000%	-2,614.50	-405.25	0.21
Q2 2027	30.0000%	-2,988.00	-463.14	0.24
Q3 2027	30.0000%	-3,237.00	-501.74	0.26
Q4 2027	30.0000%	-3,610.50	-559.63	0.29
Q1 2028	30.0000%	-3,622.50	-597.71	-19.95
Q2 2028	30.0000%	-4,140.00	-683.10	-22.80
Q3 2028	30.0000%	-4,485.00	-740.03	-24.70
Q4 2028	30.0000%	-5,002.50	-825.41	-27.55
Q1 2029	30.0000%	-4,473.00	-760.41	-36.96
Q2 2029	30.0000%	-5,112.00	-869.04	-42.24
Q3 2029	30.0000%	-5,538.00	-941.46	-45.76
Q4 2029	30.0000%	-6,177.00	-1,050.09	-51.04
Q1 2030	30.0000%	-5,166.00	-878.22	-50.82
Q2 2030	30.0000%	-5,904.00	-1,003.68	-58.08
Q3 2030	30.0000%	-6,396.00	-1,087.32	-62.92
Q4 2030	30.0000%	-7,134.00	-1,212.78	-70.18

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q3 2026	40.0000%	0.00	-0.00	0.00
Q4 2026	40.0000%	0.00	-133.00	0.00
Q1 2027	40.0000%	0.00	-143.93	0.00

Quarter	Probability	Conditional revenue	Conditional GP	Conditional opex
Q2 2027	40.0000%	0.00	-164.50	0.00
Q3 2027	40.0000%	0.00	-178.20	0.00
Q4 2027	40.0000%	0.00	-198.77	0.00
Q1 2028	40.0000%	0.00	-159.54	0.00
Q2 2028	40.0000%	0.00	-182.33	0.00
Q3 2028	40.0000%	0.00	-197.53	0.00
Q4 2028	40.0000%	0.00	-220.32	0.00
Q1 2029	40.0000%	0.00	-147.75	0.00
Q2 2029	40.0000%	0.00	-168.86	0.00
Q3 2029	40.0000%	0.00	-182.93	0.00
Q4 2029	40.0000%	0.00	-204.03	0.00
Q1 2030	40.0000%	0.00	-113.76	0.00
Q2 2030	40.0000%	0.00	-130.01	0.00
Q3 2030	40.0000%	0.00	-140.85	0.00
Q4 2030	40.0000%	0.00	-157.10	0.00

Year	Reference revenue	Selected revenue	Gross profit	Opex	OP	Expected capture loss
2027	95,000.00	93,705.02	19,709.77	7,039.68	12,670.09	274.16
2028	130,000.00	129,413.67	27,960.19	8,677.11	19,283.08	303.89
2029	160,000.00	160,101.06	35,398.17	10,321.41	25,076.76	281.43
2030	185,000.00	186,417.21	41,421.21	11,733.59	29,687.62	216.69

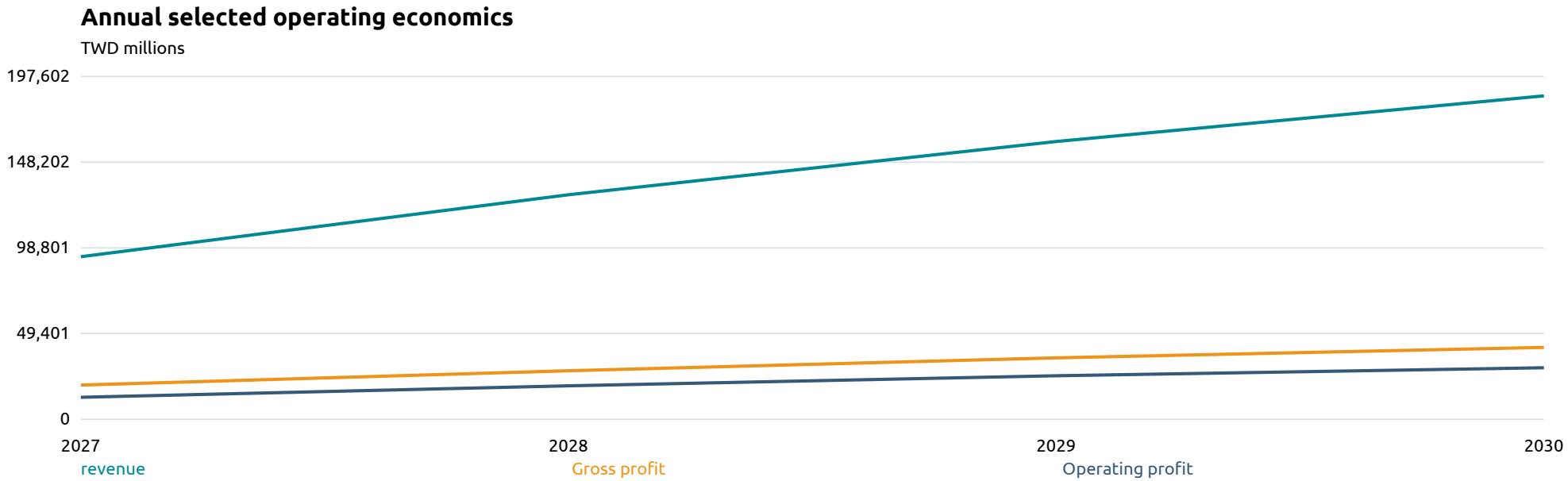


Figure 11. Calendar-year sums of the exact quarterly model, not issuer guidance.

Complete valuation method and all 14 forward windows

For starting quarter t, F4 OP is the sum of current and next three quarters. Price = (20 x F4 operating profit - dated end-window financial claims) / end-window diluted shares. This is a linear, signed valuation; no zero floor is added. Claims are dated, not future accumulated cash. Ordinary leases are normalized in operating costs and not deducted a second time.

Native valuation-control note. The historical OP-attribution reserve in D208 remains the frozen source value of zero. The live future controls D209:D225 are absolute quarterly TWD-million amounts, not fractions; all 17 are zero in the current snapshot. Valuation operating profit equals core group operating profit less that quarter's reserve. Reference operating profit uses the same subtraction, and each F4 window sums the adjusted operating profits before applying the unchanged 20x multiple, dated financial claims and end-window shares. All 32 group funding paths, cash taxes, working capital and dividends continue to use the unadjusted R3 core operating profit. This valuation-only reserve creates no cash expense or tax shield. At the current zero settings, every reported financial figure remains unchanged.

Start	OP t	OP t+1	OP t+2	OP t+3	F4 OP	End shares
Q4 2026	3,072.80	2,383.92	2,971.62	3,363.42	11,791.76	134.387911
Q1 2027	2,383.92	2,971.62	3,363.42	3,951.13	12,670.09	134.412911
Q2 2027	2,971.62	3,363.42	3,951.13	3,710.65	13,996.82	134.437911
Q3 2027	3,363.42	3,951.13	3,710.65	4,543.24	15,568.44	134.462911
Q4 2027	3,951.13	3,710.65	4,543.24	5,098.30	17,303.31	134.487911
Q1 2028	3,710.65	4,543.24	5,098.30	5,930.89	19,283.08	134.512911
Q2 2028	4,543.24	5,098.30	5,930.89	4,865.32	20,437.76	134.537911
Q3 2028	5,098.30	5,930.89	4,865.32	5,918.22	21,812.74	134.562911
Q4 2028	5,930.89	4,865.32	5,918.22	6,620.16	23,334.60	134.587911
Q1 2029	4,865.32	5,918.22	6,620.16	7,673.06	25,076.76	134.612911
Q2 2029	5,918.22	6,620.16	7,673.06	5,781.60	25,993.04	134.637911
Q3 2029	6,620.16	7,673.06	5,781.60	7,011.83	27,086.65	134.662911
Q4 2029	7,673.06	5,781.60	7,011.83	7,831.98	28,298.47	134.687911
Q1 2030	5,781.60	7,011.83	7,831.98	9,062.21	29,687.62	134.712911

Start	Multiple	Dated claim	Previous ceiling	New ceiling	Revision (%)
Q4 2026	20	-5,367.35	1,831.71	1,794.82	-2.0138%
Q1 2027	20	-5,367.35	1,965.97	1,925.18	-2.0750%
Q2 2027	20	-5,367.35	2,163.91	2,122.20	-1.9277%
Q3 2027	20	-5,367.35	2,398.33	2,355.56	-1.7832%
Q4 2027	20	-5,367.35	2,657.03	2,613.12	-1.6526%
Q1 2028	20	-5,367.35	2,952.18	2,907.00	-1.5305%
Q2 2028	20	-5,367.35	3,122.58	3,078.11	-1.4243%

Start	Multiple	Dated claim	Previous ceiling	New ceiling	Revision (%)
Q3 2028	20	-5,367.35	3,325.57	3,281.90	-1.3130%
Q4 2028	20	-5,367.35	3,550.23	3,507.44	-1.2052%
Q1 2029	20	-5,367.35	3,807.44	3,765.63	-1.0982%
Q2 2029	20	-5,367.35	3,940.83	3,901.04	-1.0096%
Q3 2029	20	-5,367.35	4,100.21	4,062.74	-0.9139%
Q4 2029	20	-5,367.35	4,276.89	4,241.93	-0.8175%
Q1 2030	20	-5,367.35	4,479.55	4,447.38	-0.7181%

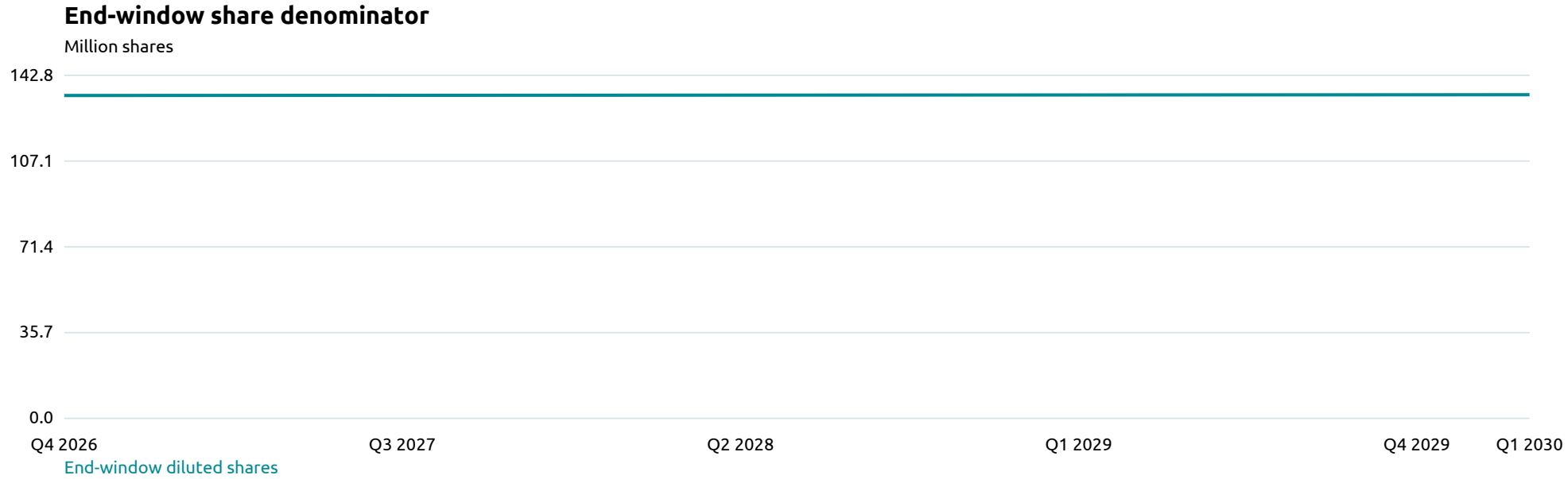


Figure 12. Common shares are dated 30 June; potential employee shares and the quarterly allowance are explicit model assumptions. No post-June issued-count date is inferred.

The published Q3 2026 ceiling of TWD 1,687.5869822439317 remains a frozen as-then display. The current Q3 reconstruction uses today's forward operating assumptions and therefore has a different F4 value. Neither is silently substituted for the other. The remaining tail quarters support the last forward windows and are retained in full.

Item	Value	Interpretation
Published Q3 2026 ceiling	1,687.59	Immutable historical display
Current reconstructed Q3 F4 value	1,661.30	Current analytical bridge, not published history
Published historical atoms	156	156 values / formula vintages are preserved under the native contract

Working capital, advances, leases, DA, SBC and FX

Dated claims = operating cash reserve + pension claim - cash equivalents - unrestricted time deposits. The TWD 5,000 million operational reserve remains outside excess cash. Future cash accumulation is not capitalized again. Forecast shares add the dated common count, potential employee shares and a quarterly allowance; they do not generate funding cash.

Funding starts from cash plus unrestricted deposits, less the remaining USD 25 million packaging deposit at the selected TWD 32 per USD assumption, license cash due and the construction commitment. The USD 37.5 million contract deposit / USD 12.5 million paid disclosure supports the remaining-25 arithmetic; 32 is a model FX assumption, not a current market quotation. Deposit timing is charged upfront in the funding test.

Opening net operating working capital is AR + inventory + purchase prepayments - AP - customer advances. Each state's future working capital is 30% of its own revenue; the cash bridge charges the change from the prior state. Customer advances fund operations but are liabilities, not profit or free distributable excess cash.

Quarter cash flow = OP - positive-OP cash tax + non-ROU DA proxy - working-capital increase - capex/software cash. Cash dividends use 40% of positive after-tax OP from 2027. Capture loss reduces OP and cash; it is not added back as noncash. There is no automatic equity raise or new bank facility in this bridge. The advance-loss stress is separate from the desired operating reserve.

Ordinary leases are normalized to full cash rent. H1 source arithmetic gives a TWD 2.018 million illustrative rent normalization with a TWD 0.051 million timing residual. Future quarterly rent reserves of 4 / 5 and non-ROU DA of 210 / 250 are low-confidence model proxies, not certified functional factory DA allocations. Do not deduct total lease cash again, add back ROU depreciation on top of full rent, or add a second lease claim.

Reported compensation accruals and potential employee dilution do not imply that IFRS 2 expense is zero. The model's separate SBC-adjustment field is zero because fully expensed OP is not adjusted again; ordinary staff, software and compensation costs remain inside operating expenses. No reported IFRS 2 zero or cost-free employee award is inferred.

Measure	Value	Scope
nre residual	40.0000%	Model-selected assumption / proxy; not issuer guidance
tk residual	30.0000%	Model-selected assumption / proxy; not issuer guidance
optical residual	25.0000%	Model-selected assumption / proxy; not issuer guidance
design nre gp	60.0000%	Model-selected assumption / proxy; not issuer guidance
design nre opex	25.0000%	Model-selected assumption / proxy; not issuer guidance
design tk opex	2.5000%	Model-selected assumption / proxy; not issuer guidance
optical gp	28.0000%	Model-selected assumption / proxy; not issuer guidance
optical opex	10.0000%	Model-selected assumption / proxy; not issuer guidance
price eligible	25.0000%	Model-selected assumption / proxy; not issuer guidance
price premium	3.0000%	Model-selected assumption / proxy; not issuer guidance
price overlap retention	50.0000%	Model-selected assumption / proxy; not issuer guidance
price gp	85.0000%	Model-selected assumption / proxy; not issuer guidance

Measure	Value	Scope
price opex	20.0000%	Model-selected assumption / proxy; not issuer guidance
delay fraction	15.0000%	Model-selected assumption / proxy; not issuer guidance
delay avoidable opex	2.0000%	Model-selected assumption / proxy; not issuer guidance
annual qualification cost	250.00	Model-selected assumption / proxy; not issuer guidance
capture eligible reference tk	35.0000%	Model-selected assumption / proxy; not issuer guidance
capture delay association	0.166666666667	Model-selected assumption / proxy; not issuer guidance
common shares	134.011911	Dated June actual / source anchor; common shares in millions; million shares
potential employee shares	0.251000	Model-selected assumption / proxy; not issuer guidance; million shares
quarterly share allowance	0.025000	Model-selected assumption / proxy; not issuer guidance; million shares
cash	10,034.146	Dated June actual / source anchor; common shares in millions
unrestricted time deposits	337.520	Dated June actual / source anchor; common shares in millions
operating reserve	5,000.00	Model-selected assumption / proxy; not issuer guidance
pension claim	4.321	Dated June actual / source anchor; common shares in millions
opening ar	3,933.121	Dated June actual / source anchor; common shares in millions
opening inventory	14,539.650	Dated June actual / source anchor; common shares in millions
opening purchase prepay	4,418.131	Dated June actual / source anchor; common shares in millions
opening ap	6,591.440	Dated June actual / source anchor; common shares in millions
opening customer advances	13,214.722	Dated June actual / source anchor; common shares in millions
deposit usdm remaining	25.00	Derived remaining USD million deposit (37.5 less 12.5 paid); timing selected separately
deposit fx assumption	32	Model FX assumption: TWD per USD; not a current FX quotation
license cash due	402.792	Dated disclosed cash commitment; upfront timing is a model convention
construction commitment	78.900	Dated disclosed cash commitment; upfront timing is a model convention
working capital revenue ratio	30.0000%	Model-selected assumption / proxy; not issuer guidance
tax rate	22.0000%	Model-selected assumption / proxy; not issuer guidance
dividend after tax ratio	40.0000%	Model-selected assumption / proxy; not issuer guidance
advance loss stress	25.0000%	Model-selected assumption / proxy; not issuer guidance

Period	COGS	Operating expense	Total
Q1 2026	4.104	48.363	52.467
Q2 2026	4.107	75.766	79.873
H1 2026	8.211	124.129	132.340

Period	COGS	Operating expense	Total
Q1 2026	1.061	23.671	24.732

Period	COGS	Operating expense	Total
Q2 2026	1.036	22.218	23.254
H1 2026	2.097	45.889	47.986

Period	COGS	Operating expense	Total
Q1 2026	4.832	105.036	109.868
Q2 2026	4.761	110.128	114.889
H1 2026	9.593	215.164	224.757

Measure	Value	Scope
Q1 2026	465.305	Source actual / dated statement
Q2 2026	424.875	Source actual / dated statement
H1 2026	890.180	Source actual / dated statement

Measure	Value	Scope
Q2 2026	10.814	Source actual / dated statement
H1 2026	21.564	Source actual / dated statement

Lease measure	Q1 2026	Q2 2026	H1 2026	2025FY
cash interest paid	1.652	1.614	3.266	4.171
cash lease principal	23.047	23.640	46.687	78.252
cash residual beyond principal and interest	1.219	12.623	13.842	6.493
cash total leases	25.918	37.877	63.795	88.916
low value expense	0.074	0.113	0.187	0.207
short term expense	1.094	1.011	2.105	6.168
variable expense not in lease liability	Not separately reported	11.499	11.499	Not separately reported

The exact full economic law and all numeric controls are reproduced in the derivation appendix. The cash schedules below retain every state and every quarter; a source fact, a current reconstruction and a future proxy are not interchangeable.

All 32 persistent cash outcomes and financing limits

Bits are ordered Design / Optical / NRE price / Ramp delay / Margin capture. Each probability is a coherent full-path weight, not five independent quarterly redraws. The joint R/M cells enforce the selected association. The default cash test is not proof of committed future financing or guaranteed customer advance availability.

State DOPRM	Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
00000	8.1900%	8,080.42	4,776.74	32,924.93	223.26	0.00
00001	4.5500%	8,080.42	4,776.74	31,487.04	223.26	0.00
00010	2.7300%	8,080.42	4,776.74	29,417.77	223.26	0.00
00011	2.7300%	8,080.42	4,776.74	28,180.01	223.26	0.00
00100	3.5100%	8,080.42	4,776.74	32,993.37	223.26	0.00
00101	1.9500%	8,080.42	4,776.74	31,555.48	223.26	0.00
00110	1.1700%	8,080.42	4,776.74	29,486.21	223.26	0.00
00111	1.1700%	8,080.42	4,776.74	28,248.44	223.26	0.00
01000	4.4100%	8,080.42	4,776.74	33,042.25	223.26	0.00
01001	2.4500%	8,080.42	4,776.74	31,604.36	223.26	0.00
01010	1.4700%	8,080.42	4,776.74	29,535.09	223.26	0.00
01011	1.4700%	8,080.42	4,776.74	28,297.32	223.26	0.00
01100	1.8900%	8,080.42	4,776.74	33,110.69	223.26	0.00
01101	1.0500%	8,080.42	4,776.74	31,672.80	223.26	0.00
01110	0.6300%	8,080.42	4,776.74	29,603.53	223.26	0.00
01111	0.6300%	8,080.42	4,776.74	28,365.76	223.26	0.00
10000	12.2850%	8,080.42	4,776.74	34,473.45	223.26	0.00
10001	6.8250%	8,080.42	4,776.74	33,035.56	223.26	0.00
10010	4.0950%	8,080.42	4,776.74	30,966.29	223.26	0.00
10011	4.0950%	8,080.42	4,776.74	29,728.52	223.26	0.00
10100	5.2650%	8,080.42	4,776.74	34,545.93	223.26	0.00
10101	2.9250%	8,080.42	4,776.74	33,108.03	223.26	0.00
10110	1.7550%	8,080.42	4,776.74	31,038.77	223.26	0.00
10111	1.7550%	8,080.42	4,776.74	29,801.00	223.26	0.00
11000	6.6150%	8,080.42	4,776.74	34,590.76	223.26	0.00
11001	3.6750%	8,080.42	4,776.74	33,152.87	223.26	0.00
11010	2.2050%	8,080.42	4,776.74	31,083.61	223.26	0.00
11011	2.2050%	8,080.42	4,776.74	29,845.84	223.26	0.00
11100	2.8350%	8,080.42	4,776.74	34,663.24	223.26	0.00

State DOPRM	Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
11101	1.5750%	8,080.42	4,776.74	33,225.35	223.26	0.00
11110	0.9450%	8,080.42	4,776.74	31,156.09	223.26	0.00
11111	0.9450%	8,080.42	4,776.74	29,918.32	223.26	0.00

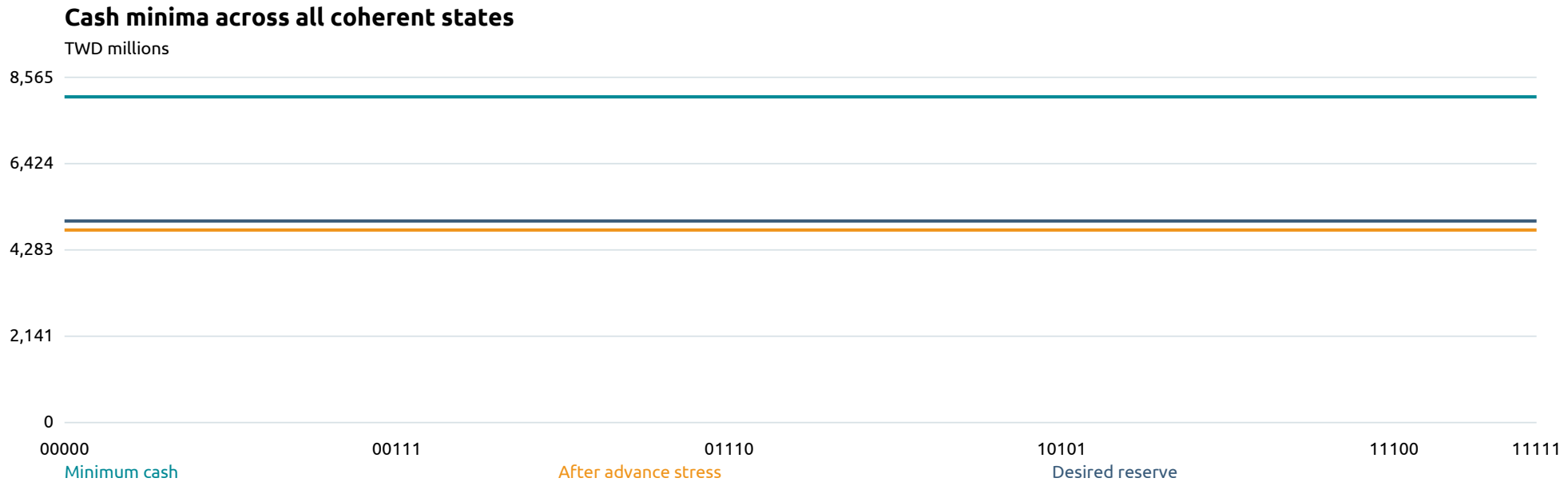


Figure 13. The adverse advance shock can breach the desired buffer without exhausting cash. The minimum after stress is about TWD 4,777 million; the desired-buffer gap is about TWD 223 million, and the hard cash gap is zero under selected defaults.

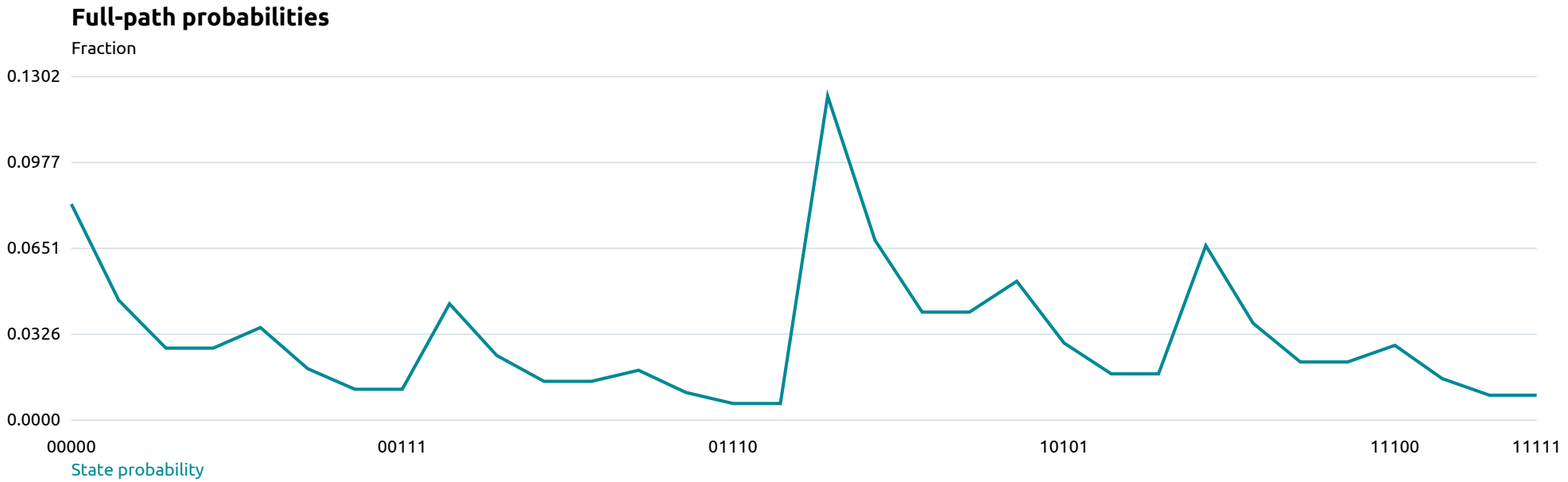


Figure 14. Every state appears in the adjacent table. Display rounding does not replace the exact typed probability or its conservation check.

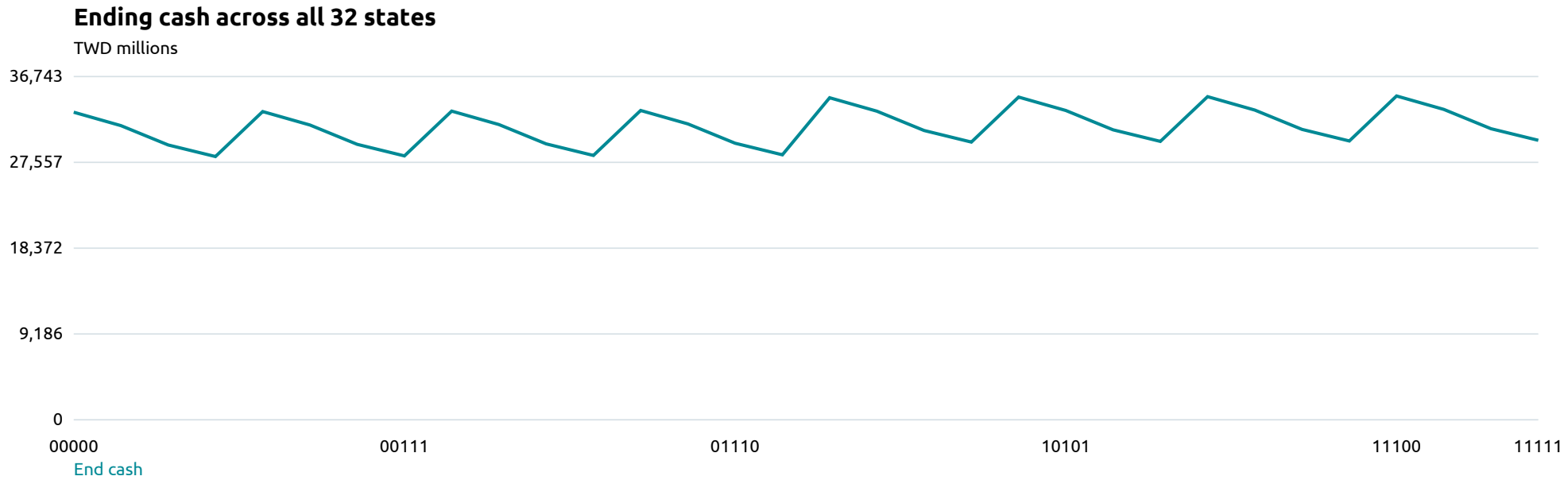


Figure 15. Ending cash is a funding diagnostic and is not added again to the dated valuation claim.

Scorecard, falsification tests and remaining uncertainty

Factor	Selected points	Reason
Core	16	Reusable interface IP, customer-specific engineering and TSMC integration create qualification and switching costs. GUC still pays the foundry and competes with large ASIC vendors and in-house teams.
Cycle	9	Nine-month sales and 3nm turnkey expansion are already real; booked sales do not establish subsequent margins.
TAM	9	More advanced-node and HBM-rich custom silicon increases design difficulty and production content, with substantial wafer pass-through.
Catalysts	4	October 29 results, NRE milestones, HBM4E silicon validation, production qualifications and accepted turnkey shipments.
Macro	7	AI infrastructure investment is supportive, offset by Taiwan concentration, export rules and customer concentration.
Alpha	2	Revenue versus retained profit and advance-funded working capital are more useful than naming an unconfirmed hyperscaler.
Risk	-6	Six deductions: concentration, production pass-through margins, customer qualification delays, packaging utilization commitments, working-capital exposure and demanding quotation.

The raw score is 41 / 60, rounded to an analytical Base of 68; the protected Excel Base also remains 68. Existing risk deductions already recognize pass-through margins and concentration. The financial capture downside is not mechanically converted into a second score deduction or multiplied into the operating-state probabilities.

Test	Evidence that matters	Model implication
29 October results	Actual Q3 mix, margins, OP and cash conversion.	Update the current reconstruction and forward inputs with comparable reported figures; the published historical Q3 2026 Excel value stays as-then.
Qualified production	Named accepted programs, usable packaging capacity, yield and delivery.	Revise program conversion and delay together with full costs.
Commercial retention	Contract gross/net treatment, customer-owned inputs, scope and quantified rebates.	Test the reference-level capture budget; no invented fee-cut fact.
Design and IP ownership	Verified service entry, IP ownership and project-specific technical scope.	Separate customer chip wallets from GUC design/production receipts.
Funding and advances	Realized collections, customer advances, deposit usage and working-capital change.	Test cash and desired reserve; do not assume automatic capital.
Source coverage	Authenticated expert identity/original audio and full issuer spoken coverage.	Update evidence confidence without multiplying probabilities twice.

The issuer does not disclose customer-specific contract values, ASIC unit shipments, die ASPs or NRE/turnkey gross margins. Segment margins and the program-value proxy are explicit assumptions.

No primary confirmation of a Tesla AI6 award, exclusivity, chip count or economics was found. This is an unresolved customer-identification claim, not a separate modeled win.

Q3 total revenue is known from monthly disclosures; Q3 product split and earnings are not yet reported. The next scheduled results are October 29.

HBM4E is design-ready and taped out; the September release does not establish completed production qualification or yield.

Security-deposit refund thresholds, packaging quantities and utilization commitments are not disclosed. Third-party 60,000-wafer claims are not used.

Cash is June 30 reported cash plus unrestricted time deposits. Customer advances make much of it operationally necessary; TWD 5 bn is reserved rather than treating the whole balance as excess.

Potential employee-bonus shares are estimated using H1 EPS dilution plus 25,000 quarterly future shares. Bonuses remain fully expensed, whether ultimately cash or stock; no share buyback is assumed.

Public issuer recordings 31Jul/30Jan are available (5410.389/6018.798s); no full spoken listening/transcription yet. Written summaries are not complete spokenQ&A substitutes.

TK3 single-digit margin denominator is undefined; no contract-level gross/net mapping, absolute supplier rebate, customer units/prices or allocation confirmed.

Anonymous expert authenticity, translated technical wording and customer identification remain uncertain. Current index review is not an exhaustive regulatory-event search.

PostJune actual capital count is not established; declared 134.011911m common shares dated 30June plus explicit employee-share allowance retained.

Historical source genericJSON floor differed from unfloored native method; defaults positive, literal native linear convention retained.

Source probability/pool/capture-gap/tail assumptions are bounded analytical judgments; no official-confirmation gate or minimum-publication threshold used.

Persistent cash path 00000 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
8.1900%	8,080.42	4,776.74	32,924.93	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	19,950.00	4,213.65	1,730.00	2,483.65	0.00	5,985.00
Q2 2027	22,800.00	4,815.60	1,730.00	3,085.60	0.00	6,840.00
Q3 2027	24,700.00	5,216.90	1,730.00	3,486.90	0.00	7,410.00
Q4 2027	27,550.00	5,818.85	1,730.00	4,088.85	0.00	8,265.00
Q1 2028	27,300.00	5,906.25	2,117.50	3,788.75	0.00	8,190.00
Q2 2028	31,200.00	6,750.00	2,117.50	4,632.50	0.00	9,360.00
Q3 2028	33,800.00	7,312.50	2,117.50	5,195.00	0.00	10,140.00
Q4 2028	37,700.00	8,156.25	2,117.50	6,038.75	0.00	11,310.00
Q1 2029	33,600.00	7,413.00	2,505.00	4,908.00	0.00	10,080.00
Q2 2029	38,400.00	8,472.00	2,505.00	5,967.00	0.00	11,520.00
Q3 2029	41,600.00	9,178.00	2,505.00	6,673.00	0.00	12,480.00
Q4 2029	46,400.00	10,237.00	2,505.00	7,732.00	0.00	13,920.00
Q1 2030	38,850.00	8,589.00	2,830.00	5,759.00	0.00	11,655.00
Q2 2030	44,400.00	9,816.00	2,830.00	6,986.00	0.00	13,320.00
Q3 2030	48,100.00	10,634.00	2,830.00	7,804.00	0.00	14,430.00
Q4 2030	53,650.00	11,861.00	2,830.00	9,031.00	0.00	16,095.00

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-615.00	546.40	210.00	600.00	774.90	2,162.25	10,542.40
Q2 2027	855.00	678.83	210.00	600.00	962.71	1,161.77	10,741.46
Q3 2027	570.00	767.12	210.00	600.00	1,087.91	1,759.78	11,413.33
Q4 2027	855.00	899.55	210.00	600.00	1,275.72	1,944.30	12,081.91
Q1 2028	-75.00	833.52	250.00	700.00	1,182.09	2,580.22	13,480.05
Q2 2028	1,170.00	1,019.15	250.00	700.00	1,445.34	1,993.35	14,028.06
Q3 2028	780.00	1,142.90	250.00	700.00	1,620.84	2,822.10	15,229.32
Q4 2028	1,170.00	1,328.53	250.00	700.00	1,884.09	3,090.23	16,435.45
Q1 2029	-1,230.00	1,079.76	250.00	800.00	1,531.30	4,508.24	19,412.39
Q2 2029	1,440.00	1,312.74	250.00	800.00	1,861.70	2,664.26	20,214.95
Q3 2029	960.00	1,468.06	250.00	800.00	2,081.98	3,694.94	21,827.91
Q4 2029	1,440.00	1,701.04	250.00	800.00	2,412.38	4,040.96	23,456.49
Q1 2030	-2,265.00	1,266.98	250.00	800.00	1,796.81	6,207.02	27,866.70
Q2 2030	1,665.00	1,536.92	250.00	800.00	2,179.63	3,234.08	28,921.15
Q3 2030	1,110.00	1,716.88	250.00	800.00	2,434.85	4,427.12	30,913.42
Q4 2030	1,665.00	1,986.82	250.00	800.00	2,817.67	4,829.18	32,924.93

Persistent cash path 00001 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
4.5500%	8,080.42	4,776.74	31,487.04	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	19,950.00	4,061.14	1,730.00	2,331.14	152.51	5,985.00
Q2 2027	22,800.00	4,641.30	1,730.00	2,911.30	174.30	6,840.00
Q3 2027	24,700.00	5,028.07	1,730.00	3,298.07	188.82	7,410.00
Q4 2027	27,550.00	5,608.24	1,730.00	3,878.24	210.61	8,265.00
Q1 2028	27,300.00	5,737.20	2,117.50	3,619.70	169.05	8,190.00
Q2 2028	31,200.00	6,556.80	2,117.50	4,439.30	193.20	9,360.00
Q3 2028	33,800.00	7,103.20	2,117.50	4,985.70	209.30	10,140.00
Q4 2028	37,700.00	7,922.80	2,117.50	5,805.30	233.45	11,310.00
Q1 2029	33,600.00	7,256.44	2,505.00	4,751.44	156.56	10,080.00
Q2 2029	38,400.00	8,293.08	2,505.00	5,788.08	178.92	11,520.00
Q3 2029	41,600.00	8,984.17	2,505.00	6,479.17	193.83	12,480.00
Q4 2029	46,400.00	10,020.81	2,505.00	7,515.81	216.19	13,920.00
Q1 2030	38,850.00	8,468.46	2,830.00	5,638.46	120.54	11,655.00
Q2 2030	44,400.00	9,678.24	2,830.00	6,848.24	137.76	13,320.00
Q3 2030	48,100.00	10,484.76	2,830.00	7,654.76	149.24	14,430.00
Q4 2030	53,650.00	11,694.54	2,830.00	8,864.54	166.46	16,095.00

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-615.00	512.85	210.00	600.00	727.31	2,043.29	10,367.28
Q2 2027	855.00	640.49	210.00	600.00	908.33	1,025.81	10,484.77
Q3 2027	570.00	725.58	210.00	600.00	1,029.00	1,612.50	11,068.27
Q4 2027	855.00	853.21	210.00	600.00	1,210.01	1,780.03	11,638.29
Q1 2028	-75.00	796.33	250.00	700.00	1,129.35	2,448.37	12,957.31
Q2 2028	1,170.00	976.65	250.00	700.00	1,385.06	1,842.65	13,414.90
Q3 2028	780.00	1,096.85	250.00	700.00	1,555.54	2,658.85	14,518.21
Q4 2028	1,170.00	1,277.17	250.00	700.00	1,811.25	2,908.13	15,615.09
Q1 2029	-1,230.00	1,045.32	250.00	800.00	1,482.45	4,386.13	18,518.76
Q2 2029	1,440.00	1,273.38	250.00	800.00	1,805.88	2,524.70	19,237.58
Q3 2029	960.00	1,425.42	250.00	800.00	2,021.50	3,543.75	20,759.83
Q4 2029	1,440.00	1,653.48	250.00	800.00	2,344.93	3,872.33	22,287.23
Q1 2030	-2,265.00	1,240.46	250.00	800.00	1,759.20	6,113.00	26,641.03
Q2 2030	1,665.00	1,506.61	250.00	800.00	2,136.65	3,126.63	27,631.01
Q3 2030	1,110.00	1,684.05	250.00	800.00	2,388.29	4,310.71	29,553.43
Q4 2030	1,665.00	1,950.20	250.00	800.00	2,765.74	4,699.34	31,487.04

Persistent cash path 00010 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
2.7300%	8,080.42	4,776.74	29,417.77	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	17,335.50	3,808.40	1,730.21	2,078.19	0.00	5,200.65
Q2 2027	19,812.00	4,352.46	1,730.24	2,622.22	0.00	5,943.60
Q3 2027	21,463.00	4,715.16	1,730.26	2,984.90	0.00	6,438.90
Q4 2027	23,939.50	5,259.22	1,730.29	3,528.93	0.00	7,181.85
Q1 2028	23,677.50	5,308.54	2,097.55	3,210.99	0.00	7,103.25
Q2 2028	27,060.00	6,066.90	2,094.70	3,972.20	0.00	8,118.00
Q3 2028	29,315.00	6,572.48	2,092.80	4,479.68	0.00	8,794.50
Q4 2028	32,697.50	7,330.84	2,089.95	5,240.89	0.00	9,809.25
Q1 2029	29,127.00	6,652.59	2,468.04	4,184.55	0.00	8,738.10
Q2 2029	33,288.00	7,602.96	2,462.76	5,140.20	0.00	9,986.40
Q3 2029	36,062.00	8,236.54	2,459.24	5,777.30	0.00	10,818.60
Q4 2029	40,223.00	9,186.91	2,453.96	6,732.95	0.00	12,066.90
Q1 2030	33,684.00	7,710.78	2,779.18	4,931.60	0.00	10,105.20
Q2 2030	38,496.00	8,812.32	2,771.92	6,040.40	0.00	11,548.80
Q3 2030	41,704.00	9,546.68	2,767.08	6,779.60	0.00	12,511.20
Q4 2030	46,516.00	10,648.22	2,759.82	7,888.40	0.00	13,954.80

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,399.35	457.20	210.00	600.00	648.40	2,630.34	11,136.99
Q2 2027	742.95	576.89	210.00	600.00	818.13	912.38	11,231.24
Q3 2027	495.30	656.68	210.00	600.00	931.29	1,442.93	11,742.88
Q4 2027	742.95	776.37	210.00	600.00	1,101.03	1,619.62	12,261.47
Q1 2028	-78.60	706.42	250.00	700.00	1,001.83	2,133.17	13,392.81
Q2 2028	1,014.75	873.88	250.00	700.00	1,239.33	1,633.57	13,787.05
Q3 2028	676.50	985.53	250.00	700.00	1,397.66	2,367.65	14,757.04
Q4 2028	1,014.75	1,153.00	250.00	700.00	1,635.16	2,623.14	15,745.02
Q1 2029	-1,071.15	920.60	250.00	800.00	1,305.58	3,785.10	18,224.54
Q2 2029	1,248.30	1,130.84	250.00	800.00	1,603.74	2,211.06	18,831.86
Q3 2029	832.20	1,271.01	250.00	800.00	1,802.52	3,124.09	20,153.43
Q4 2029	1,248.30	1,481.25	250.00	800.00	2,100.68	3,453.40	21,506.15
Q1 2030	-1,961.70	1,084.95	250.00	800.00	1,538.66	5,258.35	25,225.84
Q2 2030	1,443.60	1,328.89	250.00	800.00	1,884.60	2,717.91	26,059.15
Q3 2030	962.40	1,491.51	250.00	800.00	2,115.24	3,775.69	27,719.60
Q4 2030	1,443.60	1,735.45	250.00	800.00	2,461.18	4,159.35	29,417.77

Persistent cash path 00011 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
2.7300%	8,080.42	4,776.74	28,180.01	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	17,335.50	3,678.77	1,730.21	1,948.56	129.64	5,200.65
Q2 2027	19,812.00	4,204.31	1,730.24	2,474.07	148.16	5,943.60
Q3 2027	21,463.00	4,554.66	1,730.26	2,824.40	160.50	6,438.90
Q4 2027	23,939.50	5,080.20	1,730.29	3,349.91	179.02	7,181.85
Q1 2028	23,677.50	5,164.85	2,097.55	3,067.30	143.69	7,103.25
Q2 2028	27,060.00	5,902.68	2,094.70	3,807.98	164.22	8,118.00
Q3 2028	29,315.00	6,394.57	2,092.80	4,301.77	177.91	8,794.50
Q4 2028	32,697.50	7,132.40	2,089.95	5,042.45	198.43	9,809.25
Q1 2029	29,127.00	6,519.52	2,468.04	4,051.48	133.07	8,738.10
Q2 2029	33,288.00	7,450.88	2,462.76	4,988.12	152.08	9,986.40
Q3 2029	36,062.00	8,071.78	2,459.24	5,612.54	164.76	10,818.60
Q4 2029	40,223.00	9,003.14	2,453.96	6,549.18	183.77	12,066.90
Q1 2030	33,684.00	7,608.32	2,779.18	4,829.14	102.46	10,105.20
Q2 2030	38,496.00	8,695.22	2,771.92	5,923.30	117.10	11,548.80
Q3 2030	41,704.00	9,419.83	2,767.08	6,652.75	126.85	12,511.20
Q4 2030	46,516.00	10,506.73	2,759.82	7,746.91	141.49	13,954.80

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,399.35	428.68	210.00	600.00	607.95	2,529.22	10,972.59
Q2 2027	742.95	544.29	210.00	600.00	771.91	796.82	10,997.50
Q3 2027	495.30	621.37	210.00	600.00	881.21	1,317.73	11,434.02
Q4 2027	742.95	736.98	210.00	600.00	1,045.17	1,479.98	11,868.83
Q1 2028	-78.60	674.80	250.00	700.00	957.00	2,021.09	12,932.92
Q2 2028	1,014.75	837.76	250.00	700.00	1,188.09	1,505.47	13,250.31
Q3 2028	676.50	946.39	250.00	700.00	1,342.15	2,228.88	14,137.03
Q4 2028	1,014.75	1,109.34	250.00	700.00	1,573.25	2,468.36	15,032.15
Q1 2029	-1,071.15	891.33	250.00	800.00	1,264.06	3,681.30	17,449.40
Q2 2029	1,248.30	1,097.39	250.00	800.00	1,556.29	2,092.43	17,985.53
Q3 2029	832.20	1,234.76	250.00	800.00	1,751.11	2,995.58	19,230.01
Q4 2029	1,248.30	1,440.82	250.00	800.00	2,043.35	3,310.06	20,496.72
Q1 2030	-1,961.70	1,062.41	250.00	800.00	1,506.69	5,178.43	24,168.46
Q2 2030	1,443.60	1,303.13	250.00	800.00	1,848.07	2,626.58	24,946.97
Q3 2030	962.40	1,463.60	250.00	800.00	2,075.66	3,676.74	26,548.05
Q4 2030	1,443.60	1,704.32	250.00	800.00	2,417.04	4,048.99	28,180.01

Persistent cash path 00100 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
3.5100%	8,080.42	4,776.74	32,993.37	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	19,959.45	4,221.68	1,731.89	2,489.79	0.00	5,987.84
Q2 2027	22,810.80	4,824.78	1,732.16	3,092.62	0.00	6,843.24
Q3 2027	24,711.70	5,226.84	1,732.34	3,494.50	0.00	7,413.51
Q4 2027	27,563.05	5,829.94	1,732.61	4,097.33	0.00	8,268.91
Q1 2028	27,311.81	5,916.29	2,119.86	3,796.43	0.00	8,193.54
Q2 2028	31,213.50	6,761.48	2,120.20	4,641.28	0.00	9,364.05
Q3 2028	33,814.62	7,324.93	2,120.43	5,204.51	0.00	10,144.39
Q4 2028	37,716.31	8,170.12	2,120.76	6,049.35	0.00	11,314.89
Q1 2029	33,614.18	7,425.05	2,507.84	4,917.21	0.00	10,084.25
Q2 2029	38,416.20	8,485.77	2,508.24	5,977.53	0.00	11,524.86
Q3 2029	41,617.55	9,192.92	2,508.51	6,684.41	0.00	12,485.27
Q4 2029	46,419.57	10,253.64	2,508.91	7,744.72	0.00	13,925.87
Q1 2030	38,866.54	8,603.06	2,833.31	5,769.75	0.00	11,659.96
Q2 2030	44,418.90	9,832.07	2,833.78	6,998.28	0.00	13,325.67
Q3 2030	48,120.47	10,651.40	2,834.09	7,817.31	0.00	14,436.14
Q4 2030	53,672.84	11,880.41	2,834.57	9,045.84	0.00	16,101.85

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-612.16	547.75	210.00	600.00	776.82	2,164.20	10,542.44
Q2 2027	855.40	680.38	210.00	600.00	964.90	1,166.84	10,744.38
Q3 2027	570.27	768.79	210.00	600.00	1,090.29	1,765.44	11,419.54
Q4 2027	855.40	901.41	210.00	600.00	1,278.37	1,950.51	12,091.68
Q1 2028	-75.37	835.21	250.00	700.00	1,184.49	2,586.59	13,493.78
Q2 2028	1,170.51	1,021.08	250.00	700.00	1,448.08	1,999.69	14,045.39
Q3 2028	780.34	1,144.99	250.00	700.00	1,623.81	2,829.18	15,250.77
Q4 2028	1,170.51	1,330.86	250.00	700.00	1,887.40	3,097.99	16,461.36
Q1 2029	-1,230.64	1,081.79	250.00	800.00	1,534.17	4,516.07	19,443.25
Q2 2029	1,440.61	1,315.06	250.00	800.00	1,864.99	2,671.87	20,250.13
Q3 2029	960.41	1,470.57	250.00	800.00	2,085.54	3,703.43	21,868.03
Q4 2029	1,440.61	1,703.84	250.00	800.00	2,416.35	4,050.28	23,501.95
Q1 2030	-2,265.91	1,269.34	250.00	800.00	1,800.16	6,216.32	27,918.11
Q2 2030	1,665.71	1,539.62	250.00	800.00	2,183.46	3,242.95	28,977.59
Q3 2030	1,110.47	1,719.81	250.00	800.00	2,439.00	4,437.03	30,975.62
Q4 2030	1,665.71	1,990.09	250.00	800.00	2,822.30	4,840.05	32,993.37

Persistent cash path 00101 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.9500%	8,080.42	4,776.74	31,555.48	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	19,959.45	4,069.17	1,731.89	2,337.28	152.51	5,987.84
Q2 2027	22,810.80	4,650.48	1,732.16	2,918.32	174.30	6,843.24
Q3 2027	24,711.70	5,038.02	1,732.34	3,305.68	188.82	7,413.51
Q4 2027	27,563.05	5,619.33	1,732.61	3,886.72	210.61	8,268.91
Q1 2028	27,311.81	5,747.24	2,119.86	3,627.38	169.05	8,193.54
Q2 2028	31,213.50	6,568.28	2,120.20	4,448.08	193.20	9,364.05
Q3 2028	33,814.62	7,115.63	2,120.43	4,995.21	209.30	10,144.39
Q4 2028	37,716.31	7,936.67	2,120.76	5,815.90	233.45	11,314.89
Q1 2029	33,614.18	7,268.49	2,507.84	4,760.66	156.56	10,084.25
Q2 2029	38,416.20	8,306.85	2,508.24	5,798.61	178.92	11,524.86
Q3 2029	41,617.55	8,999.09	2,508.51	6,490.58	193.83	12,485.27
Q4 2029	46,419.57	10,037.44	2,508.91	7,528.53	216.19	13,925.87
Q1 2030	38,866.54	8,482.52	2,833.31	5,649.21	120.54	11,659.96
Q2 2030	44,418.90	9,694.31	2,833.78	6,860.52	137.76	13,325.67
Q3 2030	48,120.47	10,502.16	2,834.09	7,668.07	149.24	14,436.14
Q4 2030	53,672.84	11,713.95	2,834.57	8,879.38	166.46	16,101.85

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-612.16	514.20	210.00	600.00	729.23	2,045.24	10,367.32
Q2 2027	855.40	642.03	210.00	600.00	910.52	1,030.88	10,487.69
Q3 2027	570.27	727.25	210.00	600.00	1,031.37	1,618.16	11,074.48
Q4 2027	855.40	855.08	210.00	600.00	1,212.66	1,786.24	11,648.06
Q1 2028	-75.37	798.02	250.00	700.00	1,131.74	2,454.73	12,971.04
Q2 2028	1,170.51	978.58	250.00	700.00	1,387.80	1,848.99	13,432.24
Q3 2028	780.34	1,098.95	250.00	700.00	1,558.50	2,665.92	14,539.66
Q4 2028	1,170.51	1,279.50	250.00	700.00	1,814.56	2,915.90	15,640.99
Q1 2029	-1,230.64	1,047.34	250.00	800.00	1,485.33	4,393.96	18,549.62
Q2 2029	1,440.61	1,275.69	250.00	800.00	1,809.17	2,532.31	19,272.76
Q3 2029	960.41	1,427.93	250.00	800.00	2,025.06	3,552.25	20,799.95
Q4 2029	1,440.61	1,656.28	250.00	800.00	2,348.90	3,881.64	22,332.69
Q1 2030	-2,265.91	1,242.83	250.00	800.00	1,762.55	6,122.29	26,692.43
Q2 2030	1,665.71	1,509.32	250.00	800.00	2,140.48	3,135.50	27,687.45
Q3 2030	1,110.47	1,686.98	250.00	800.00	2,392.44	4,320.62	29,615.63
Q4 2030	1,665.71	1,953.46	250.00	800.00	2,770.37	4,710.21	31,555.48

Persistent cash path 00110 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.1700%	8,080.42	4,776.74	29,486.21	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	17,344.95	3,816.43	1,732.10	2,084.34	0.00	5,203.48
Q2 2027	19,822.80	4,361.64	1,732.40	2,629.24	0.00	5,946.84
Q3 2027	21,474.70	4,725.11	1,732.60	2,992.51	0.00	6,442.41
Q4 2027	23,952.55	5,270.31	1,732.90	3,537.41	0.00	7,185.76
Q1 2028	23,689.31	5,318.58	2,099.91	3,218.67	0.00	7,106.79
Q2 2028	27,073.50	6,078.38	2,097.40	3,980.98	0.00	8,122.05
Q3 2028	29,329.62	6,584.91	2,095.73	4,489.18	0.00	8,798.89
Q4 2028	32,713.81	7,344.70	2,093.21	5,251.49	0.00	9,814.14
Q1 2029	29,141.18	6,664.64	2,470.88	4,193.76	0.00	8,742.35
Q2 2029	33,304.20	7,616.73	2,466.00	5,150.73	0.00	9,991.26
Q3 2029	36,079.55	8,251.46	2,462.75	5,788.71	0.00	10,823.86
Q4 2029	40,242.57	9,203.55	2,457.88	6,745.67	0.00	12,072.77
Q1 2030	33,700.54	7,724.84	2,782.49	4,942.35	0.00	10,110.16
Q2 2030	38,514.90	8,828.39	2,775.70	6,052.68	0.00	11,554.47
Q3 2030	41,724.47	9,564.08	2,771.17	6,792.91	0.00	12,517.34
Q4 2030	46,538.84	10,667.63	2,764.39	7,903.24	0.00	13,961.65

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,396.52	458.55	210.00	600.00	650.31	2,632.30	11,137.03
Q2 2027	743.35	578.43	210.00	600.00	820.32	917.45	11,234.16
Q3 2027	495.57	658.35	210.00	600.00	933.66	1,448.59	11,749.09
Q4 2027	743.35	778.23	210.00	600.00	1,103.67	1,625.83	12,271.24
Q1 2028	-78.97	708.11	250.00	700.00	1,004.22	2,139.53	13,406.55
Q2 2028	1,015.26	875.81	250.00	700.00	1,242.06	1,639.90	13,804.39
Q3 2028	676.84	987.62	250.00	700.00	1,400.62	2,374.72	14,778.49
Q4 2028	1,015.26	1,155.33	250.00	700.00	1,638.47	2,630.91	15,770.93
Q1 2029	-1,071.79	922.63	250.00	800.00	1,308.45	3,792.93	18,255.40
Q2 2029	1,248.91	1,133.16	250.00	800.00	1,607.03	2,218.66	18,867.04
Q3 2029	832.61	1,273.52	250.00	800.00	1,806.08	3,132.59	20,193.55
Q4 2029	1,248.91	1,484.05	250.00	800.00	2,104.65	3,462.72	21,551.62
Q1 2030	-1,962.61	1,087.32	250.00	800.00	1,542.01	5,267.64	25,277.25
Q2 2030	1,444.31	1,331.59	250.00	800.00	1,888.44	2,726.79	26,115.59
Q3 2030	962.87	1,494.44	250.00	800.00	2,119.39	3,785.60	27,781.80
Q4 2030	1,444.31	1,738.71	250.00	800.00	2,465.81	4,170.22	29,486.21

Persistent cash path 00111 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.1700%	8,080.42	4,776.74	28,248.44	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	17,344.95	3,686.80	1,732.10	1,954.70	129.64	5,203.48
Q2 2027	19,822.80	4,213.49	1,732.40	2,481.09	148.16	5,946.84
Q3 2027	21,474.70	4,564.61	1,732.60	2,832.01	160.50	6,442.41
Q4 2027	23,952.55	5,091.29	1,732.90	3,358.39	179.02	7,185.76
Q1 2028	23,689.31	5,174.89	2,099.91	3,074.97	143.69	7,106.79
Q2 2028	27,073.50	5,914.15	2,097.40	3,816.76	164.22	8,122.05
Q3 2028	29,329.62	6,407.00	2,095.73	4,311.28	177.91	8,798.89
Q4 2028	32,713.81	7,146.27	2,093.21	5,053.06	198.43	9,814.14
Q1 2029	29,141.18	6,531.57	2,470.88	4,060.69	133.07	8,742.35
Q2 2029	33,304.20	7,464.65	2,466.00	4,998.65	152.08	9,991.26
Q3 2029	36,079.55	8,086.70	2,462.75	5,623.95	164.76	10,823.86
Q4 2029	40,242.57	9,019.78	2,457.88	6,561.91	183.77	12,072.77
Q1 2030	33,700.54	7,622.38	2,782.49	4,839.89	102.46	10,110.16
Q2 2030	38,514.90	8,711.29	2,775.70	5,935.59	117.10	11,554.47
Q3 2030	41,724.47	9,437.23	2,771.17	6,666.05	126.85	12,517.34
Q4 2030	46,538.84	10,526.14	2,764.39	7,761.75	141.49	13,961.65

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,396.52	430.03	210.00	600.00	609.87	2,531.18	10,972.62
Q2 2027	743.35	545.84	210.00	600.00	774.10	801.89	11,000.42
Q3 2027	495.57	623.04	210.00	600.00	883.59	1,323.40	11,440.23
Q4 2027	743.35	738.85	210.00	600.00	1,047.82	1,486.19	11,878.60
Q1 2028	-78.97	676.49	250.00	700.00	959.39	2,027.45	12,946.66
Q2 2028	1,015.26	839.69	250.00	700.00	1,190.83	1,511.81	13,267.65
Q3 2028	676.84	948.48	250.00	700.00	1,345.12	2,235.96	14,158.48
Q4 2028	1,015.26	1,111.67	250.00	700.00	1,576.55	2,476.13	15,058.06
Q1 2029	-1,071.79	893.35	250.00	800.00	1,266.94	3,689.13	17,480.25
Q2 2029	1,248.91	1,099.70	250.00	800.00	1,559.58	2,100.04	18,020.71
Q3 2029	832.61	1,237.27	250.00	800.00	1,754.67	3,004.08	19,270.12
Q4 2029	1,248.91	1,443.62	250.00	800.00	2,047.32	3,319.38	20,542.18
Q1 2030	-1,962.61	1,064.78	250.00	800.00	1,510.05	5,187.73	24,219.86
Q2 2030	1,444.31	1,305.83	250.00	800.00	1,851.90	2,635.45	25,003.41
Q3 2030	962.87	1,466.53	250.00	800.00	2,079.81	3,686.65	26,610.25
Q4 2030	1,444.31	1,707.59	250.00	800.00	2,421.67	4,059.86	28,248.44

Persistent cash path 01000 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
4.4100%	8,080.42	4,776.74	33,042.25	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	19,965.75	4,218.06	1,731.58	2,486.48	0.00	5,989.72
Q2 2027	22,818.00	4,820.64	1,731.80	3,088.84	0.00	6,845.40
Q3 2027	24,719.50	5,222.36	1,731.95	3,490.41	0.00	7,415.85
Q4 2027	27,571.75	5,824.94	1,732.17	4,092.76	0.00	8,271.52
Q1 2028	27,378.75	5,928.30	2,125.38	3,802.93	0.00	8,213.62
Q2 2028	31,290.00	6,775.20	2,126.50	4,648.70	0.00	9,387.00
Q3 2028	33,897.50	7,339.80	2,127.25	5,212.55	0.00	10,169.25
Q4 2028	37,808.75	8,186.70	2,128.38	6,058.32	0.00	11,342.62
Q1 2029	33,810.00	7,471.80	2,526.00	4,945.80	0.00	10,143.00
Q2 2029	38,640.00	8,539.20	2,529.00	6,010.20	0.00	11,592.00
Q3 2029	41,860.00	9,250.80	2,531.00	6,719.80	0.00	12,558.00
Q4 2029	46,690.00	10,318.20	2,534.00	7,784.20	0.00	14,007.00
Q1 2030	39,217.50	8,691.90	2,866.75	5,825.15	0.00	11,765.25
Q2 2030	44,820.00	9,933.60	2,872.00	7,061.60	0.00	13,446.00
Q3 2030	48,555.00	10,761.40	2,875.50	7,885.90	0.00	14,566.50
Q4 2030	54,157.50	12,003.10	2,880.75	9,122.35	0.00	16,247.25

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-610.28	547.03	210.00	600.00	775.78	2,159.73	10,539.00
Q2 2027	855.68	679.54	210.00	600.00	963.72	1,163.62	10,738.90
Q3 2027	570.45	767.89	210.00	600.00	1,089.01	1,762.07	11,411.96
Q4 2027	855.68	900.41	210.00	600.00	1,276.94	1,946.68	12,081.70
Q1 2028	-57.90	836.64	250.00	700.00	1,186.51	2,574.18	13,469.37
Q2 2028	1,173.38	1,022.71	250.00	700.00	1,450.39	2,002.61	14,021.59
Q3 2028	782.25	1,146.76	250.00	700.00	1,626.32	2,833.54	15,228.81
Q4 2028	1,173.38	1,332.83	250.00	700.00	1,890.20	3,102.12	16,440.73
Q1 2029	-1,199.62	1,088.08	250.00	800.00	1,543.09	4,507.35	19,404.99
Q2 2029	1,449.00	1,322.24	250.00	800.00	1,875.18	2,688.96	20,218.77
Q3 2029	966.00	1,478.36	250.00	800.00	2,096.58	3,725.44	21,847.63
Q4 2029	1,449.00	1,712.52	250.00	800.00	2,428.67	4,072.68	23,491.64
Q1 2030	-2,241.75	1,281.53	250.00	800.00	1,817.45	6,235.37	27,909.56
Q2 2030	1,680.75	1,553.55	250.00	800.00	2,203.22	3,277.30	28,983.64
Q3 2030	1,120.50	1,734.90	250.00	800.00	2,460.40	4,480.50	31,003.74
Q4 2030	1,680.75	2,006.92	250.00	800.00	2,846.17	4,884.68	33,042.25

Persistent cash path 01001 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
2.4500%	8,080.42	4,776.74	31,604.36	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	19,965.75	4,065.55	1,731.58	2,333.97	152.51	5,989.72
Q2 2027	22,818.00	4,646.34	1,731.80	2,914.54	174.30	6,845.40
Q3 2027	24,719.50	5,033.53	1,731.95	3,301.59	188.82	7,415.85
Q4 2027	27,571.75	5,614.33	1,732.17	3,882.15	210.61	8,271.52
Q1 2028	27,378.75	5,759.25	2,125.38	3,633.88	169.05	8,213.62
Q2 2028	31,290.00	6,582.00	2,126.50	4,455.50	193.20	9,387.00
Q3 2028	33,897.50	7,130.50	2,127.25	5,003.25	209.30	10,169.25
Q4 2028	37,808.75	7,953.25	2,128.38	5,824.88	233.45	11,342.62
Q1 2029	33,810.00	7,315.24	2,526.00	4,789.24	156.56	10,143.00
Q2 2029	38,640.00	8,360.28	2,529.00	5,831.28	178.92	11,592.00
Q3 2029	41,860.00	9,056.97	2,531.00	6,525.97	193.83	12,558.00
Q4 2029	46,690.00	10,102.01	2,534.00	7,568.01	216.19	14,007.00
Q1 2030	39,217.50	8,571.36	2,866.75	5,704.61	120.54	11,765.25
Q2 2030	44,820.00	9,795.84	2,872.00	6,923.84	137.76	13,446.00
Q3 2030	48,555.00	10,612.16	2,875.50	7,736.66	149.24	14,566.50
Q4 2030	54,157.50	11,836.64	2,880.75	8,955.89	166.46	16,247.25

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-610.28	513.47	210.00	600.00	728.20	2,040.77	10,363.88
Q2 2027	855.68	641.20	210.00	600.00	909.34	1,027.67	10,482.21
Q3 2027	570.45	726.35	210.00	600.00	1,030.09	1,614.79	11,066.91
Q4 2027	855.68	854.07	210.00	600.00	1,211.23	1,782.40	11,638.08
Q1 2028	-57.90	799.45	250.00	700.00	1,133.77	2,442.32	12,946.63
Q2 2028	1,173.38	980.21	250.00	700.00	1,390.12	1,851.91	13,408.43
Q3 2028	782.25	1,100.71	250.00	700.00	1,561.01	2,670.28	14,517.70
Q4 2028	1,173.38	1,281.47	250.00	700.00	1,817.36	2,920.03	15,620.37
Q1 2029	-1,199.62	1,053.63	250.00	800.00	1,494.24	4,385.24	18,511.36
Q2 2029	1,449.00	1,282.88	250.00	800.00	1,819.36	2,549.40	19,241.40
Q3 2029	966.00	1,435.71	250.00	800.00	2,036.10	3,574.26	20,779.55
Q4 2029	1,449.00	1,664.96	250.00	800.00	2,361.22	3,904.04	22,322.38
Q1 2030	-2,241.75	1,255.01	250.00	800.00	1,779.84	6,141.35	26,683.89
Q2 2030	1,680.75	1,523.24	250.00	800.00	2,160.24	3,169.85	27,693.49
Q3 2030	1,120.50	1,702.07	250.00	800.00	2,413.84	4,364.09	29,643.75
Q4 2030	1,680.75	1,970.30	250.00	800.00	2,794.24	4,754.84	31,604.36

Persistent cash path 01010 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.4700%	8,080.42	4,776.74	29,535.09	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	17,351.25	3,812.81	1,731.79	2,081.03	0.00	5,205.38
Q2 2027	19,830.00	4,357.50	1,732.04	2,625.46	0.00	5,949.00
Q3 2027	21,482.50	4,720.62	1,732.21	2,988.41	0.00	6,444.75
Q4 2027	23,961.25	5,265.31	1,732.46	3,532.85	0.00	7,188.38
Q1 2028	23,756.25	5,330.59	2,105.43	3,225.16	0.00	7,126.88
Q2 2028	27,150.00	6,092.10	2,103.70	3,988.40	0.00	8,145.00
Q3 2028	29,412.50	6,599.77	2,102.55	4,497.22	0.00	8,823.75
Q4 2028	32,806.25	7,361.29	2,100.82	5,260.46	0.00	9,841.88
Q1 2029	29,337.00	6,711.39	2,489.04	4,222.35	0.00	8,801.10
Q2 2029	33,528.00	7,670.16	2,486.76	5,183.40	0.00	10,058.40
Q3 2029	36,322.00	8,309.34	2,485.24	5,824.10	0.00	10,896.60
Q4 2029	40,513.00	9,268.11	2,482.96	6,785.15	0.00	12,153.90
Q1 2030	34,051.50	7,813.68	2,815.93	4,997.75	0.00	10,215.45
Q2 2030	38,916.00	8,929.92	2,813.92	6,116.00	0.00	11,674.80
Q3 2030	42,159.00	9,674.08	2,812.58	6,861.50	0.00	12,647.70
Q4 2030	47,023.50	10,790.32	2,810.57	7,979.75	0.00	14,107.05

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,394.62	457.83	210.00	600.00	649.28	2,627.83	11,133.60
Q2 2027	743.62	577.60	210.00	600.00	819.14	914.23	11,228.69
Q3 2027	495.75	657.45	210.00	600.00	932.39	1,445.21	11,741.51
Q4 2027	743.62	777.23	210.00	600.00	1,102.25	1,622.00	12,261.26
Q1 2028	-61.50	709.54	250.00	700.00	1,006.25	2,127.13	13,382.14
Q2 2028	1,018.12	877.45	250.00	700.00	1,244.38	1,642.83	13,780.58
Q3 2028	678.75	989.39	250.00	700.00	1,403.13	2,379.09	14,756.54
Q4 2028	1,018.12	1,157.30	250.00	700.00	1,641.26	2,635.04	15,750.31
Q1 2029	-1,040.77	928.92	250.00	800.00	1,317.37	3,784.21	18,217.14
Q2 2029	1,257.30	1,140.35	250.00	800.00	1,617.22	2,235.75	18,835.67
Q3 2029	838.20	1,281.30	250.00	800.00	1,817.12	3,154.60	20,173.15
Q4 2029	1,257.30	1,492.73	250.00	800.00	2,116.97	3,485.12	21,541.30
Q1 2030	-1,938.45	1,099.51	250.00	800.00	1,559.30	5,286.70	25,268.70
Q2 2030	1,459.35	1,345.52	250.00	800.00	1,908.19	2,761.13	26,121.64
Q3 2030	972.90	1,509.53	250.00	800.00	2,140.79	3,829.07	27,809.92
Q4 2030	1,459.35	1,755.55	250.00	800.00	2,489.68	4,214.85	29,535.09

Persistent cash path 01011 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.4700%	8,080.42	4,776.74	28,297.32	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	17,351.25	3,683.18	1,731.79	1,951.39	129.64	5,205.38
Q2 2027	19,830.00	4,209.35	1,732.04	2,477.31	148.16	5,949.00
Q3 2027	21,482.50	4,560.12	1,732.21	2,827.91	160.50	6,444.75
Q4 2027	23,961.25	5,086.29	1,732.46	3,353.83	179.02	7,188.38
Q1 2028	23,756.25	5,186.90	2,105.43	3,081.47	143.69	7,126.88
Q2 2028	27,150.00	5,927.88	2,103.70	3,824.18	164.22	8,145.00
Q3 2028	29,412.50	6,421.87	2,102.55	4,319.32	177.91	8,823.75
Q4 2028	32,806.25	7,162.85	2,100.82	5,062.03	198.43	9,841.88
Q1 2029	29,337.00	6,578.32	2,489.04	4,089.28	133.07	8,801.10
Q2 2029	33,528.00	7,518.08	2,486.76	5,031.32	152.08	10,058.40
Q3 2029	36,322.00	8,144.58	2,485.24	5,659.34	164.76	10,896.60
Q4 2029	40,513.00	9,084.34	2,482.96	6,601.38	183.77	12,153.90
Q1 2030	34,051.50	7,711.22	2,815.93	4,895.29	102.46	10,215.45
Q2 2030	38,916.00	8,812.82	2,813.92	5,998.90	117.10	11,674.80
Q3 2030	42,159.00	9,547.23	2,812.58	6,734.65	126.85	12,647.70
Q4 2030	47,023.50	10,648.83	2,810.57	7,838.26	141.49	14,107.05

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,394.62	429.31	210.00	600.00	608.83	2,526.71	10,969.19
Q2 2027	743.62	545.01	210.00	600.00	772.92	798.67	10,994.94
Q3 2027	495.75	622.14	210.00	600.00	882.31	1,320.02	11,432.65
Q4 2027	743.62	737.84	210.00	600.00	1,046.39	1,482.36	11,868.62
Q1 2028	-61.50	677.92	250.00	700.00	961.42	2,015.05	12,922.25
Q2 2028	1,018.12	841.32	250.00	700.00	1,193.14	1,514.74	13,243.84
Q3 2028	678.75	950.25	250.00	700.00	1,347.63	2,240.32	14,136.53
Q4 2028	1,018.12	1,113.65	250.00	700.00	1,579.35	2,480.26	15,037.44
Q1 2029	-1,040.77	899.64	250.00	800.00	1,275.85	3,680.41	17,441.99
Q2 2029	1,257.30	1,106.89	250.00	800.00	1,569.77	2,117.13	17,989.35
Q3 2029	838.20	1,245.06	250.00	800.00	1,765.72	3,026.09	19,249.72
Q4 2029	1,257.30	1,452.30	250.00	800.00	2,059.63	3,341.78	20,531.87
Q1 2030	-1,938.45	1,076.96	250.00	800.00	1,527.33	5,206.78	24,211.32
Q2 2030	1,459.35	1,319.76	250.00	800.00	1,871.66	2,669.80	25,009.45
Q3 2030	972.90	1,481.62	250.00	800.00	2,101.21	3,730.12	26,638.37
Q4 2030	1,459.35	1,724.42	250.00	800.00	2,445.54	4,104.49	28,297.32

Persistent cash path 01100 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.8900%	8,080.42	4,776.74	33,110.69	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	19,975.20	4,226.09	1,733.47	2,492.63	0.00	5,992.56
Q2 2027	22,828.80	4,829.82	1,733.96	3,095.86	0.00	6,848.64
Q3 2027	24,731.20	5,232.30	1,734.29	3,498.01	0.00	7,419.36
Q4 2027	27,584.80	5,836.03	1,734.78	4,101.25	0.00	8,275.44
Q1 2028	27,390.56	5,938.34	2,127.74	3,810.60	0.00	8,217.17
Q2 2028	31,303.50	6,786.68	2,129.20	4,657.48	0.00	9,391.05
Q3 2028	33,912.12	7,352.23	2,130.18	5,222.06	0.00	10,173.64
Q4 2028	37,825.06	8,200.57	2,131.64	6,068.93	0.00	11,347.52
Q1 2029	33,824.18	7,483.85	2,528.84	4,955.01	0.00	10,147.25
Q2 2029	38,656.20	8,552.97	2,532.24	6,020.73	0.00	11,596.86
Q3 2029	41,877.55	9,265.72	2,534.51	6,731.21	0.00	12,563.27
Q4 2029	46,709.57	10,334.84	2,537.91	7,796.92	0.00	14,012.87
Q1 2030	39,234.04	8,705.96	2,870.06	5,835.90	0.00	11,770.21
Q2 2030	44,838.90	9,949.67	2,875.78	7,073.89	0.00	13,451.67
Q3 2030	48,575.47	10,778.80	2,879.59	7,899.21	0.00	14,572.64
Q4 2030	54,180.34	12,022.51	2,885.32	9,137.19	0.00	16,254.10

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-607.44	548.38	210.00	600.00	777.70	2,161.69	10,539.04
Q2 2027	856.08	681.09	210.00	600.00	965.91	1,168.69	10,741.82
Q3 2027	570.72	769.56	210.00	600.00	1,091.38	1,767.73	11,418.17
Q4 2027	856.08	902.27	210.00	600.00	1,279.59	1,952.89	12,091.48
Q1 2028	-58.27	838.33	250.00	700.00	1,188.91	2,580.54	13,483.11
Q2 2028	1,173.88	1,024.64	250.00	700.00	1,453.13	2,008.95	14,038.93
Q3 2028	782.59	1,148.85	250.00	700.00	1,629.28	2,840.62	15,250.26
Q4 2028	1,173.88	1,335.16	250.00	700.00	1,893.51	3,109.88	16,466.64
Q1 2029	-1,200.27	1,090.10	250.00	800.00	1,545.96	4,515.18	19,435.85
Q2 2029	1,449.61	1,324.56	250.00	800.00	1,878.47	2,696.56	20,253.95
Q3 2029	966.41	1,480.87	250.00	800.00	2,100.14	3,733.94	21,887.75
Q4 2029	1,449.61	1,715.32	250.00	800.00	2,432.64	4,081.99	23,537.10
Q1 2030	-2,242.66	1,283.90	250.00	800.00	1,820.80	6,244.66	27,960.96
Q2 2030	1,681.46	1,556.25	250.00	800.00	2,207.05	3,286.17	29,040.08
Q3 2030	1,120.97	1,737.83	250.00	800.00	2,464.55	4,490.41	31,065.94
Q4 2030	1,681.46	2,010.18	250.00	800.00	2,850.80	4,895.55	33,110.69

Persistent cash path 01101 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.0500%	8,080.42	4,776.74	31,672.80	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	19,975.20	4,073.58	1,733.47	2,340.11	152.51	5,992.56
Q2 2027	22,828.80	4,655.52	1,733.96	2,921.56	174.30	6,848.64
Q3 2027	24,731.20	5,043.48	1,734.29	3,309.19	188.82	7,419.36
Q4 2027	27,584.80	5,625.42	1,734.78	3,890.63	210.61	8,275.44
Q1 2028	27,390.56	5,769.29	2,127.74	3,641.55	169.05	8,217.17
Q2 2028	31,303.50	6,593.48	2,129.20	4,464.28	193.20	9,391.05
Q3 2028	33,912.12	7,142.93	2,130.18	5,012.76	209.30	10,173.64
Q4 2028	37,825.06	7,967.12	2,131.64	5,835.48	233.45	11,347.52
Q1 2029	33,824.18	7,327.29	2,528.84	4,798.46	156.56	10,147.25
Q2 2029	38,656.20	8,374.05	2,532.24	5,841.81	178.92	11,596.86
Q3 2029	41,877.55	9,071.89	2,534.51	6,537.38	193.83	12,563.27
Q4 2029	46,709.57	10,118.64	2,537.91	7,580.73	216.19	14,012.87
Q1 2030	39,234.04	8,585.42	2,870.06	5,715.36	120.54	11,770.21
Q2 2030	44,838.90	9,811.91	2,875.78	6,936.12	137.76	13,451.67
Q3 2030	48,575.47	10,629.56	2,879.59	7,749.97	149.24	14,572.64
Q4 2030	54,180.34	11,856.05	2,885.32	8,970.73	166.46	16,254.10

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-607.44	514.83	210.00	600.00	730.12	2,042.73	10,363.92
Q2 2027	856.08	642.74	210.00	600.00	911.53	1,032.74	10,485.13
Q3 2027	570.72	728.02	210.00	600.00	1,032.47	1,620.45	11,073.12
Q4 2027	856.08	855.94	210.00	600.00	1,213.88	1,788.62	11,647.85
Q1 2028	-58.27	801.14	250.00	700.00	1,136.16	2,448.68	12,960.37
Q2 2028	1,173.88	982.14	250.00	700.00	1,392.85	1,858.25	13,425.77
Q3 2028	782.59	1,102.81	250.00	700.00	1,563.98	2,677.36	14,539.15
Q4 2028	1,173.88	1,283.81	250.00	700.00	1,820.67	2,927.79	15,646.28
Q1 2029	-1,200.27	1,055.66	250.00	800.00	1,497.12	4,393.06	18,542.22
Q2 2029	1,449.61	1,285.20	250.00	800.00	1,822.64	2,557.00	19,276.58
Q3 2029	966.41	1,438.22	250.00	800.00	2,039.66	3,582.75	20,819.67
Q4 2029	1,449.61	1,667.76	250.00	800.00	2,365.19	3,913.36	22,367.84
Q1 2030	-2,242.66	1,257.38	250.00	800.00	1,783.19	6,150.64	26,735.29
Q2 2030	1,681.46	1,525.95	250.00	800.00	2,164.07	3,178.72	27,749.94
Q3 2030	1,120.97	1,704.99	250.00	800.00	2,417.99	4,374.00	29,705.95
Q4 2030	1,681.46	1,973.56	250.00	800.00	2,798.87	4,765.71	31,672.80

Persistent cash path 01110 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
0.6300%	8,080.42	4,776.74	29,603.53	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	17,360.70	3,820.84	1,733.68	2,087.17	0.00	5,208.21
Q2 2027	19,840.80	4,366.68	1,734.20	2,632.48	0.00	5,952.24
Q3 2027	21,494.20	4,730.57	1,734.55	2,996.02	0.00	6,448.26
Q4 2027	23,974.30	5,276.40	1,735.07	3,541.33	0.00	7,192.29
Q1 2028	23,768.06	5,340.63	2,107.79	3,232.84	0.00	7,130.42
Q2 2028	27,163.50	6,103.57	2,106.40	3,997.18	0.00	8,149.05
Q3 2028	29,427.12	6,612.21	2,105.48	4,506.73	0.00	8,828.14
Q4 2028	32,822.56	7,375.15	2,104.09	5,271.07	0.00	9,846.77
Q1 2029	29,351.18	6,723.44	2,491.88	4,231.56	0.00	8,805.35
Q2 2029	33,544.20	7,683.93	2,490.00	5,193.93	0.00	10,063.26
Q3 2029	36,339.55	8,324.26	2,488.75	5,835.51	0.00	10,901.86
Q4 2029	40,532.57	9,284.75	2,486.88	6,797.87	0.00	12,159.77
Q1 2030	34,068.04	7,827.74	2,819.24	5,008.50	0.00	10,220.41
Q2 2030	38,934.90	8,945.99	2,817.70	6,128.28	0.00	11,680.47
Q3 2030	42,179.47	9,691.48	2,816.67	6,874.81	0.00	12,653.84
Q4 2030	47,046.34	10,809.73	2,815.14	7,994.59	0.00	14,113.90

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,391.79	459.18	210.00	600.00	651.20	2,629.78	11,133.64
Q2 2027	744.03	579.15	210.00	600.00	821.33	919.30	11,231.61
Q3 2027	496.02	659.12	210.00	600.00	934.76	1,450.88	11,747.72
Q4 2027	744.03	779.09	210.00	600.00	1,104.89	1,628.21	12,271.04
Q1 2028	-61.87	711.22	250.00	700.00	1,008.65	2,133.49	13,395.88
Q2 2028	1,018.63	879.38	250.00	700.00	1,247.12	1,649.17	13,797.92
Q3 2028	679.09	991.48	250.00	700.00	1,406.10	2,386.16	14,777.99
Q4 2028	1,018.63	1,159.63	250.00	700.00	1,644.57	2,642.80	15,776.21
Q1 2029	-1,041.42	930.94	250.00	800.00	1,320.25	3,792.04	18,248.00
Q2 2029	1,257.91	1,142.66	250.00	800.00	1,620.51	2,243.36	18,870.85
Q3 2029	838.61	1,283.81	250.00	800.00	1,820.68	3,163.09	20,213.27
Q4 2029	1,257.91	1,495.53	250.00	800.00	2,120.94	3,494.43	21,586.76
Q1 2030	-1,939.36	1,101.87	250.00	800.00	1,562.65	5,295.99	25,320.10
Q2 2030	1,460.06	1,348.22	250.00	800.00	1,912.02	2,770.00	26,178.08
Q3 2030	973.37	1,512.46	250.00	800.00	2,144.94	3,838.98	27,872.12
Q4 2030	1,460.06	1,758.81	250.00	800.00	2,494.31	4,225.72	29,603.53

Persistent cash path 01111 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
0.6300%	8,080.42	4,776.74	28,365.76	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	17,360.70	3,691.21	1,733.68	1,957.53	129.64	5,208.21
Q2 2027	19,840.80	4,218.53	1,734.20	2,484.33	148.16	5,952.24
Q3 2027	21,494.20	4,570.07	1,734.55	2,835.52	160.50	6,448.26
Q4 2027	23,974.30	5,097.38	1,735.07	3,362.31	179.02	7,192.29
Q1 2028	23,768.06	5,196.94	2,107.79	3,089.15	143.69	7,130.42
Q2 2028	27,163.50	5,939.35	2,106.40	3,832.95	164.22	8,149.05
Q3 2028	29,427.12	6,434.30	2,105.48	4,328.83	177.91	8,828.14
Q4 2028	32,822.56	7,176.72	2,104.09	5,072.63	198.43	9,846.77
Q1 2029	29,351.18	6,590.37	2,491.88	4,098.49	133.07	8,805.35
Q2 2029	33,544.20	7,531.85	2,490.00	5,041.85	152.08	10,063.26
Q3 2029	36,339.55	8,159.50	2,488.75	5,670.75	164.76	10,901.86
Q4 2029	40,532.57	9,100.98	2,486.88	6,614.11	183.77	12,159.77
Q1 2030	34,068.04	7,725.28	2,819.24	4,906.04	102.46	10,220.41
Q2 2030	38,934.90	8,828.89	2,817.70	6,011.19	117.10	11,680.47
Q3 2030	42,179.47	9,564.63	2,816.67	6,747.95	126.85	12,653.84
Q4 2030	47,046.34	10,668.24	2,815.14	7,853.10	141.49	14,113.90

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,391.79	430.66	210.00	600.00	610.75	2,528.67	10,969.23
Q2 2027	744.03	546.55	210.00	600.00	775.11	803.74	10,997.86
Q3 2027	496.02	623.81	210.00	600.00	884.68	1,325.68	11,438.86
Q4 2027	744.03	739.71	210.00	600.00	1,049.04	1,488.57	11,878.39
Q1 2028	-61.87	679.61	250.00	700.00	963.81	2,021.41	12,935.99
Q2 2028	1,018.63	843.25	250.00	700.00	1,195.88	1,521.07	13,261.18
Q3 2028	679.09	952.34	250.00	700.00	1,350.59	2,247.40	14,157.98
Q4 2028	1,018.63	1,115.98	250.00	700.00	1,582.66	2,488.02	15,063.34
Q1 2029	-1,041.42	901.67	250.00	800.00	1,278.73	3,688.24	17,472.85
Q2 2029	1,257.91	1,109.21	250.00	800.00	1,573.06	2,124.73	18,024.53
Q3 2029	838.61	1,247.57	250.00	800.00	1,769.27	3,034.58	19,289.84
Q4 2029	1,257.91	1,455.10	250.00	800.00	2,063.60	3,351.10	20,577.33
Q1 2030	-1,939.36	1,079.33	250.00	800.00	1,530.68	5,216.07	24,262.72
Q2 2030	1,460.06	1,322.46	250.00	800.00	1,875.49	2,678.67	25,065.90
Q3 2030	973.37	1,484.55	250.00	800.00	2,105.36	3,740.03	26,700.57
Q4 2030	1,460.06	1,727.68	250.00	800.00	2,450.17	4,115.36	28,365.76

Persistent cash path 10000 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
12.2850%	8,080.42	4,776.74	34,473.45	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	20,790.00	4,381.23	1,769.90	2,611.33	0.00	6,237.00
Q2 2027	23,760.00	5,007.12	1,775.60	3,231.52	0.00	7,128.00
Q3 2027	25,740.00	5,424.38	1,779.40	3,644.98	0.00	7,722.00
Q4 2027	28,710.00	6,050.27	1,785.10	4,265.17	0.00	8,613.00
Q1 2028	28,854.00	6,235.74	2,194.15	4,041.59	0.00	8,656.20
Q2 2028	32,976.00	7,126.56	2,205.10	4,921.46	0.00	9,892.80
Q3 2028	35,724.00	7,720.44	2,212.40	5,508.04	0.00	10,717.20
Q4 2028	39,846.00	8,611.26	2,223.35	6,387.91	0.00	11,953.80
Q1 2029	35,742.00	7,885.50	2,615.25	5,270.25	0.00	10,722.60
Q2 2029	40,848.00	9,012.00	2,631.00	6,381.00	0.00	12,254.40
Q3 2029	44,252.00	9,763.00	2,641.50	7,121.50	0.00	13,275.60
Q4 2029	49,358.00	10,889.50	2,657.25	8,232.25	0.00	14,807.40
Q1 2030	41,706.00	9,219.00	2,977.00	6,242.00	0.00	12,511.80
Q2 2030	47,664.00	10,536.00	2,998.00	7,538.00	0.00	14,299.20
Q3 2030	51,636.00	11,414.00	3,012.00	8,402.00	0.00	15,490.80
Q4 2030	57,594.00	12,731.00	3,033.00	9,698.00	0.00	17,278.20

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-363.00	574.49	210.00	600.00	814.73	2,009.84	10,350.15
Q2 2027	891.00	710.93	210.00	600.00	1,008.23	1,239.59	10,581.50
Q3 2027	594.00	801.90	210.00	600.00	1,137.23	1,859.08	11,303.35
Q4 2027	891.00	938.34	210.00	600.00	1,330.73	2,045.83	12,018.45
Q1 2028	43.20	889.15	250.00	700.00	1,260.98	2,659.24	13,416.72
Q2 2028	1,236.60	1,082.72	250.00	700.00	1,535.50	2,152.14	14,033.36
Q3 2028	824.40	1,211.77	250.00	700.00	1,718.51	3,021.87	15,336.72
Q4 2028	1,236.60	1,405.34	250.00	700.00	1,993.03	3,295.97	16,639.67
Q1 2029	-1,231.20	1,159.45	250.00	800.00	1,644.32	4,791.99	19,787.34
Q2 2029	1,531.80	1,403.82	250.00	800.00	1,990.87	2,895.38	20,691.85
Q3 2029	1,021.20	1,566.73	250.00	800.00	2,221.91	3,983.57	22,453.51
Q4 2029	1,531.80	1,811.10	250.00	800.00	2,568.46	4,339.36	24,224.41
Q1 2030	-2,295.60	1,373.24	250.00	800.00	1,947.50	6,614.36	28,891.26
Q2 2030	1,787.40	1,658.36	250.00	800.00	2,351.86	3,542.24	30,081.65
Q3 2030	1,191.60	1,848.44	250.00	800.00	2,621.42	4,811.96	32,272.18
Q4 2030	1,787.40	2,133.56	250.00	800.00	3,025.78	5,227.04	34,473.45

Persistent cash path 10001 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
6.8250%	8,080.42	4,776.74	33,035.56	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	20,790.00	4,228.72	1,769.90	2,458.82	152.51	6,237.00
Q2 2027	23,760.00	4,832.82	1,775.60	3,057.22	174.30	7,128.00
Q3 2027	25,740.00	5,235.55	1,779.40	3,456.15	188.82	7,722.00
Q4 2027	28,710.00	5,839.66	1,785.10	4,054.56	210.61	8,613.00
Q1 2028	28,854.00	6,066.69	2,194.15	3,872.54	169.05	8,656.20
Q2 2028	32,976.00	6,933.36	2,205.10	4,728.26	193.20	9,892.80
Q3 2028	35,724.00	7,511.14	2,212.40	5,298.74	209.30	10,717.20
Q4 2028	39,846.00	8,377.81	2,223.35	6,154.46	233.45	11,953.80
Q1 2029	35,742.00	7,728.94	2,615.25	5,113.69	156.56	10,722.60
Q2 2029	40,848.00	8,833.08	2,631.00	6,202.08	178.92	12,254.40
Q3 2029	44,252.00	9,569.17	2,641.50	6,927.67	193.83	13,275.60
Q4 2029	49,358.00	10,673.31	2,657.25	8,016.06	216.19	14,807.40
Q1 2030	41,706.00	9,098.46	2,977.00	6,121.46	120.54	12,511.80
Q2 2030	47,664.00	10,398.24	2,998.00	7,400.24	137.76	14,299.20
Q3 2030	51,636.00	11,264.76	3,012.00	8,252.76	149.24	15,490.80
Q4 2030	57,594.00	12,564.54	3,033.00	9,531.54	166.46	17,278.20

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-363.00	540.94	210.00	600.00	767.15	1,890.88	10,175.04
Q2 2027	891.00	672.59	210.00	600.00	953.85	1,103.63	10,324.82
Q3 2027	594.00	760.35	210.00	600.00	1,078.32	1,711.80	10,958.30
Q4 2027	891.00	892.00	210.00	600.00	1,265.02	1,881.55	11,574.83
Q1 2028	43.20	851.96	250.00	700.00	1,208.23	2,527.38	12,893.98
Q2 2028	1,236.60	1,040.22	250.00	700.00	1,475.22	2,001.44	13,420.20
Q3 2028	824.40	1,165.72	250.00	700.00	1,653.21	2,858.62	14,625.61
Q4 2028	1,236.60	1,353.98	250.00	700.00	1,920.19	3,113.88	15,819.30
Q1 2029	-1,231.20	1,125.01	250.00	800.00	1,595.47	4,669.88	18,893.71
Q2 2029	1,531.80	1,364.46	250.00	800.00	1,935.05	2,755.82	19,714.48
Q3 2029	1,021.20	1,524.09	250.00	800.00	2,161.43	3,832.38	21,385.43
Q4 2029	1,531.80	1,763.53	250.00	800.00	2,501.01	4,170.72	23,055.15
Q1 2030	-2,295.60	1,346.72	250.00	800.00	1,909.90	6,520.34	27,665.59
Q2 2030	1,787.40	1,628.05	250.00	800.00	2,308.87	3,434.79	28,791.50
Q3 2030	1,191.60	1,815.61	250.00	800.00	2,574.86	4,695.55	30,912.19
Q4 2030	1,787.40	2,096.94	250.00	800.00	2,973.84	5,097.20	33,035.56

Persistent cash path 10010 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
4.0950%	8,080.42	4,776.74	30,966.29	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	18,175.50	3,975.98	1,770.11	2,205.87	0.00	5,452.65
Q2 2027	20,772.00	4,543.98	1,775.84	2,768.14	0.00	6,231.60
Q3 2027	22,503.00	4,922.64	1,779.66	3,142.98	0.00	6,750.90
Q4 2027	25,099.50	5,490.64	1,785.39	3,705.25	0.00	7,529.85
Q1 2028	25,231.50	5,638.03	2,174.20	3,463.83	0.00	7,569.45
Q2 2028	28,836.00	6,443.46	2,182.30	4,261.16	0.00	8,650.80
Q3 2028	31,239.00	6,980.41	2,187.70	4,792.71	0.00	9,371.70
Q4 2028	34,843.50	7,785.85	2,195.80	5,590.05	0.00	10,453.05
Q1 2029	31,269.00	7,125.09	2,578.29	4,546.80	0.00	9,380.70
Q2 2029	35,736.00	8,142.96	2,588.76	5,554.20	0.00	10,720.80
Q3 2029	38,714.00	8,821.54	2,595.74	6,225.80	0.00	11,614.20
Q4 2029	43,181.00	9,839.41	2,606.21	7,233.20	0.00	12,954.30
Q1 2030	36,540.00	8,340.78	2,926.18	5,414.60	0.00	10,962.00
Q2 2030	41,760.00	9,532.32	2,939.92	6,592.40	0.00	12,528.00
Q3 2030	45,240.00	10,326.68	2,949.08	7,377.60	0.00	13,572.00
Q4 2030	50,460.00	11,518.22	2,962.82	8,555.40	0.00	15,138.00

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,147.35	485.29	210.00	600.00	688.23	2,477.93	10,944.75
Q2 2027	778.95	608.99	210.00	600.00	863.66	990.20	11,071.29
Q3 2027	519.30	691.46	210.00	600.00	980.61	1,542.23	11,632.91
Q4 2027	778.95	815.16	210.00	600.00	1,156.04	1,721.15	12,198.01
Q1 2028	39.60	762.04	250.00	700.00	1,080.71	2,212.19	13,329.48
Q2 2028	1,081.35	937.46	250.00	700.00	1,329.48	1,792.35	13,792.36
Q3 2028	720.90	1,054.40	250.00	700.00	1,495.33	2,567.42	14,864.45
Q4 2028	1,081.35	1,229.81	250.00	700.00	1,744.09	2,828.89	15,949.24
Q1 2029	-1,072.35	1,000.30	250.00	800.00	1,418.60	4,068.85	18,599.49
Q2 2029	1,340.10	1,221.92	250.00	800.00	1,732.91	2,442.18	19,308.76
Q3 2029	893.40	1,369.68	250.00	800.00	1,942.45	3,412.72	20,779.03
Q4 2029	1,340.10	1,591.30	250.00	800.00	2,256.76	3,751.80	22,274.07
Q1 2030	-1,992.30	1,191.21	250.00	800.00	1,689.36	5,665.69	26,250.40
Q2 2030	1,566.00	1,450.33	250.00	800.00	2,056.83	3,026.07	27,219.65
Q3 2030	1,044.00	1,623.07	250.00	800.00	2,301.81	4,160.53	29,078.36
Q4 2030	1,566.00	1,882.19	250.00	800.00	2,669.28	4,557.21	30,966.29

Persistent cash path 10011 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
4.0950%	8,080.42	4,776.74	29,728.52	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	18,175.50	3,846.35	1,770.11	2,076.24	129.64	5,452.65
Q2 2027	20,772.00	4,395.83	1,775.84	2,619.99	148.16	6,231.60
Q3 2027	22,503.00	4,762.14	1,779.66	2,982.48	160.50	6,750.90
Q4 2027	25,099.50	5,311.62	1,785.39	3,526.23	179.02	7,529.85
Q1 2028	25,231.50	5,494.34	2,174.20	3,320.13	143.69	7,569.45
Q2 2028	28,836.00	6,279.24	2,182.30	4,096.94	164.22	8,650.80
Q3 2028	31,239.00	6,802.51	2,187.70	4,614.81	177.91	9,371.70
Q4 2028	34,843.50	7,587.41	2,195.80	5,391.61	198.43	10,453.05
Q1 2029	31,269.00	6,992.02	2,578.29	4,413.73	133.07	9,380.70
Q2 2029	35,736.00	7,990.88	2,588.76	5,402.12	152.08	10,720.80
Q3 2029	38,714.00	8,656.78	2,595.74	6,061.04	164.76	11,614.20
Q4 2029	43,181.00	9,655.64	2,606.21	7,049.43	183.77	12,954.30
Q1 2030	36,540.00	8,238.32	2,926.18	5,312.14	102.46	10,962.00
Q2 2030	41,760.00	9,415.22	2,939.92	6,475.30	117.10	12,528.00
Q3 2030	45,240.00	10,199.83	2,949.08	7,250.75	126.85	13,572.00
Q4 2030	50,460.00	11,376.73	2,962.82	8,413.91	141.49	15,138.00

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,147.35	456.77	210.00	600.00	647.79	2,376.81	10,780.34
Q2 2027	778.95	576.40	210.00	600.00	817.44	874.64	10,837.54
Q3 2027	519.30	656.15	210.00	600.00	930.53	1,417.04	11,324.04
Q4 2027	778.95	775.77	210.00	600.00	1,100.18	1,581.51	11,805.37
Q1 2028	39.60	730.43	250.00	700.00	1,035.88	2,100.11	12,869.59
Q2 2028	1,081.35	901.33	250.00	700.00	1,278.25	1,664.26	13,255.61
Q3 2028	720.90	1,015.26	250.00	700.00	1,439.82	2,428.65	14,244.44
Q4 2028	1,081.35	1,186.16	250.00	700.00	1,682.18	2,674.11	15,236.37
Q1 2029	-1,072.35	971.02	250.00	800.00	1,377.08	3,965.06	17,824.34
Q2 2029	1,340.10	1,188.47	250.00	800.00	1,685.46	2,323.55	18,462.44
Q3 2029	893.40	1,333.43	250.00	800.00	1,891.05	3,284.21	19,855.60
Q4 2029	1,340.10	1,550.88	250.00	800.00	2,199.42	3,608.46	21,264.64
Q1 2030	-1,992.30	1,168.67	250.00	800.00	1,657.39	5,585.77	25,193.02
Q2 2030	1,566.00	1,424.57	250.00	800.00	2,020.29	2,934.74	26,107.46
Q3 2030	1,044.00	1,595.16	250.00	800.00	2,262.23	4,061.58	27,906.81
Q4 2030	1,566.00	1,851.06	250.00	800.00	2,625.14	4,446.85	29,728.52

Persistent cash path 10100 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
5.2650%	8,080.42	4,776.74	34,545.93	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	20,799.76	4,389.53	1,771.85	2,617.68	0.00	6,239.93
Q2 2027	23,771.16	5,016.61	1,777.83	3,238.77	0.00	7,131.35
Q3 2027	25,752.09	5,434.66	1,781.82	3,652.84	0.00	7,725.63
Q4 2027	28,723.49	6,061.73	1,787.80	4,273.94	0.00	8,617.05
Q1 2028	28,866.44	6,246.32	2,196.64	4,049.68	0.00	8,659.93
Q2 2028	32,990.22	7,138.65	2,207.94	4,930.70	0.00	9,897.07
Q3 2028	35,739.40	7,733.53	2,215.48	5,518.05	0.00	10,721.82
Q4 2028	39,863.18	8,625.87	2,226.79	6,399.08	0.00	11,958.95
Q1 2029	35,757.12	7,898.35	2,618.27	5,280.08	0.00	10,727.14
Q2 2029	40,865.28	9,026.69	2,634.46	6,392.23	0.00	12,259.58
Q3 2029	44,270.72	9,778.91	2,645.24	7,133.67	0.00	13,281.22
Q4 2029	49,378.88	10,907.25	2,661.43	8,245.82	0.00	14,813.66
Q1 2030	41,723.80	9,234.13	2,980.56	6,253.57	0.00	12,517.14
Q2 2030	47,684.34	10,553.29	3,002.07	7,551.22	0.00	14,305.30
Q3 2030	51,658.04	11,432.73	3,016.41	8,416.32	0.00	15,497.41
Q4 2030	57,618.58	12,751.89	3,037.92	9,713.98	0.00	17,285.57

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-360.07	575.89	210.00	600.00	816.72	2,011.86	10,350.19
Q2 2027	891.42	712.53	210.00	600.00	1,010.50	1,244.83	10,584.52
Q3 2027	594.28	803.62	210.00	600.00	1,139.69	1,864.94	11,309.77
Q4 2027	891.42	940.27	210.00	600.00	1,333.47	2,052.25	12,028.55
Q1 2028	42.89	890.93	250.00	700.00	1,263.50	2,665.86	13,430.92
Q2 2028	1,237.13	1,084.75	250.00	700.00	1,538.38	2,158.82	14,051.35
Q3 2028	824.76	1,213.97	250.00	700.00	1,721.63	3,029.33	15,359.05
Q4 2028	1,237.13	1,407.80	250.00	700.00	1,996.51	3,304.15	16,666.68
Q1 2029	-1,231.82	1,161.62	250.00	800.00	1,647.38	4,800.28	19,819.58
Q2 2029	1,532.45	1,406.29	250.00	800.00	1,994.38	2,903.49	20,728.69
Q3 2029	1,021.63	1,569.41	250.00	800.00	2,225.70	3,992.63	22,495.62
Q4 2029	1,532.45	1,814.08	250.00	800.00	2,572.70	4,349.29	24,272.21
Q1 2030	-2,296.52	1,375.79	250.00	800.00	1,951.11	6,624.31	28,945.41
Q2 2030	1,788.16	1,661.27	250.00	800.00	2,355.98	3,551.79	30,141.22
Q3 2030	1,192.11	1,851.59	250.00	800.00	2,625.89	4,822.62	32,337.95
Q4 2030	1,788.16	2,137.07	250.00	800.00	3,030.76	5,238.74	34,545.93

Persistent cash path 10101 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
2.9250%	8,080.42	4,776.74	33,108.03	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	20,799.76	4,237.02	1,771.85	2,465.16	152.51	6,239.93
Q2 2027	23,771.16	4,842.31	1,777.83	3,064.47	174.30	7,131.35
Q3 2027	25,752.09	5,245.83	1,781.82	3,464.01	188.82	7,725.63
Q4 2027	28,723.49	5,851.12	1,787.80	4,063.32	210.61	8,617.05
Q1 2028	28,866.44	6,077.27	2,196.64	3,880.63	169.05	8,659.93
Q2 2028	32,990.22	6,945.45	2,207.94	4,737.50	193.20	9,897.07
Q3 2028	35,739.40	7,524.23	2,215.48	5,308.75	209.30	10,721.82
Q4 2028	39,863.18	8,392.42	2,226.79	6,165.63	233.45	11,958.95
Q1 2029	35,757.12	7,741.80	2,618.27	5,123.52	156.56	10,727.14
Q2 2029	40,865.28	8,847.77	2,634.46	6,213.31	178.92	12,259.58
Q3 2029	44,270.72	9,585.08	2,645.24	6,939.84	193.83	13,281.22
Q4 2029	49,378.88	10,691.05	2,661.43	8,029.63	216.19	14,813.66
Q1 2030	41,723.80	9,113.59	2,980.56	6,133.03	120.54	12,517.14
Q2 2030	47,684.34	10,415.53	3,002.07	7,413.46	137.76	14,305.30
Q3 2030	51,658.04	11,283.49	3,016.41	8,267.08	149.24	15,497.41
Q4 2030	57,618.58	12,585.43	3,037.92	9,547.52	166.46	17,285.57

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-360.07	542.34	210.00	600.00	769.13	1,892.90	10,175.08
Q2 2027	891.42	674.18	210.00	600.00	956.12	1,108.87	10,327.83
Q3 2027	594.28	762.08	210.00	600.00	1,080.77	1,717.65	10,964.71
Q4 2027	891.42	893.93	210.00	600.00	1,267.76	1,887.97	11,584.93
Q1 2028	42.89	853.74	250.00	700.00	1,210.76	2,534.00	12,908.18
Q2 2028	1,237.13	1,042.25	250.00	700.00	1,478.10	2,008.12	13,438.19
Q3 2028	824.76	1,167.93	250.00	700.00	1,656.33	2,866.07	14,647.93
Q4 2028	1,237.13	1,356.44	250.00	700.00	1,923.68	3,122.06	15,846.32
Q1 2029	-1,231.82	1,127.18	250.00	800.00	1,598.54	4,678.17	18,925.94
Q2 2029	1,532.45	1,366.93	250.00	800.00	1,938.55	2,763.94	19,751.33
Q3 2029	1,021.63	1,526.76	250.00	800.00	2,165.23	3,841.44	21,427.54
Q4 2029	1,532.45	1,766.52	250.00	800.00	2,505.24	4,180.66	23,102.96
Q1 2030	-2,296.52	1,349.27	250.00	800.00	1,913.50	6,530.29	27,719.74
Q2 2030	1,788.16	1,630.96	250.00	800.00	2,313.00	3,444.34	28,851.07
Q3 2030	1,192.11	1,818.76	250.00	800.00	2,579.33	4,706.22	30,977.96
Q4 2030	1,788.16	2,100.45	250.00	800.00	2,978.82	5,108.90	33,108.03

Persistent cash path 10110 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.7550%	8,080.42	4,776.74	31,038.77	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	18,185.26	3,984.28	1,772.06	2,212.22	0.00	5,455.58
Q2 2027	20,783.16	4,553.47	1,778.07	2,775.39	0.00	6,234.95
Q3 2027	22,515.09	4,932.92	1,782.08	3,150.84	0.00	6,754.53
Q4 2027	25,112.99	5,502.10	1,788.09	3,714.02	0.00	7,533.90
Q1 2028	25,243.94	5,648.60	2,176.69	3,471.92	0.00	7,573.18
Q2 2028	28,850.22	6,455.55	2,185.14	4,270.40	0.00	8,655.07
Q3 2028	31,254.40	6,993.51	2,190.78	4,802.73	0.00	9,376.32
Q4 2028	34,860.68	7,800.45	2,199.24	5,601.22	0.00	10,458.20
Q1 2029	31,284.12	7,137.94	2,581.31	4,556.63	0.00	9,385.24
Q2 2029	35,753.28	8,157.65	2,592.22	5,565.43	0.00	10,725.98
Q3 2029	38,732.72	8,837.45	2,599.48	6,237.97	0.00	11,619.82
Q4 2029	43,201.88	9,857.16	2,610.39	7,246.77	0.00	12,960.56
Q1 2030	36,557.80	8,355.91	2,929.74	5,426.17	0.00	10,967.34
Q2 2030	41,780.34	9,549.61	2,943.99	6,605.62	0.00	12,534.10
Q3 2030	45,262.04	10,345.41	2,953.49	7,391.92	0.00	13,578.61
Q4 2030	50,484.58	11,539.11	2,967.74	8,571.38	0.00	15,145.37

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,144.42	486.69	210.00	600.00	690.21	2,479.95	10,944.79
Q2 2027	779.37	610.59	210.00	600.00	865.92	995.44	11,074.31
Q3 2027	519.58	693.19	210.00	600.00	983.06	1,548.08	11,639.32
Q4 2027	779.37	817.08	210.00	600.00	1,158.77	1,727.57	12,208.11
Q1 2028	39.29	763.82	250.00	700.00	1,083.24	2,218.81	13,343.68
Q2 2028	1,081.88	939.49	250.00	700.00	1,332.37	1,799.03	13,810.35
Q3 2028	721.26	1,056.60	250.00	700.00	1,498.45	2,574.87	14,886.77
Q4 2028	1,081.88	1,232.27	250.00	700.00	1,747.58	2,837.07	15,976.25
Q1 2029	-1,072.97	1,002.46	250.00	800.00	1,421.67	4,077.14	18,631.73
Q2 2029	1,340.75	1,224.40	250.00	800.00	1,736.41	2,450.29	19,345.60
Q3 2029	893.83	1,372.35	250.00	800.00	1,946.25	3,421.78	20,821.14
Q4 2029	1,340.75	1,594.29	250.00	800.00	2,260.99	3,761.73	22,321.88
Q1 2030	-1,993.22	1,193.76	250.00	800.00	1,692.96	5,675.64	26,304.55
Q2 2030	1,566.76	1,453.24	250.00	800.00	2,060.95	3,035.62	27,279.22
Q3 2030	1,044.51	1,626.22	250.00	800.00	2,306.28	4,171.19	29,144.13
Q4 2030	1,566.76	1,885.70	250.00	800.00	2,674.27	4,568.91	31,038.77

Persistent cash path 10111 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.7550%	8,080.42	4,776.74	29,801.00	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	18,185.26	3,854.65	1,772.06	2,082.58	129.64	5,455.58
Q2 2027	20,783.16	4,405.31	1,778.07	2,627.24	148.16	6,234.95
Q3 2027	22,515.09	4,772.42	1,782.08	2,990.34	160.50	6,754.53
Q4 2027	25,112.99	5,323.08	1,788.09	3,535.00	179.02	7,533.90
Q1 2028	25,243.94	5,504.91	2,176.69	3,328.22	143.69	7,573.18
Q2 2028	28,850.22	6,291.33	2,185.14	4,106.18	164.22	8,655.07
Q3 2028	31,254.40	6,815.60	2,190.78	4,624.82	177.91	9,376.32
Q4 2028	34,860.68	7,602.02	2,199.24	5,402.78	198.43	10,458.20
Q1 2029	31,284.12	7,004.87	2,581.31	4,423.56	133.07	9,385.24
Q2 2029	35,753.28	8,005.57	2,592.22	5,413.35	152.08	10,725.98
Q3 2029	38,732.72	8,672.70	2,599.48	6,073.21	164.76	11,619.82
Q4 2029	43,201.88	9,673.39	2,610.39	7,063.01	183.77	12,960.56
Q1 2030	36,557.80	8,253.45	2,929.74	5,323.71	102.46	10,967.34
Q2 2030	41,780.34	9,432.51	2,943.99	6,488.53	117.10	12,534.10
Q3 2030	45,262.04	10,218.56	2,953.49	7,265.07	126.85	13,578.61
Q4 2030	50,484.58	11,397.62	2,967.74	8,429.88	141.49	15,145.37

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,144.42	458.17	210.00	600.00	649.77	2,378.84	10,780.38
Q2 2027	779.37	577.99	210.00	600.00	819.70	879.88	10,840.56
Q3 2027	519.58	657.88	210.00	600.00	932.99	1,422.89	11,330.46
Q4 2027	779.37	777.70	210.00	600.00	1,102.92	1,587.93	11,815.47
Q1 2028	39.29	732.21	250.00	700.00	1,038.41	2,106.73	12,883.79
Q2 2028	1,081.88	903.36	250.00	700.00	1,281.13	1,670.94	13,273.60
Q3 2028	721.26	1,017.46	250.00	700.00	1,442.94	2,436.11	14,266.76
Q4 2028	1,081.88	1,188.61	250.00	700.00	1,685.67	2,682.29	15,263.38
Q1 2029	-1,072.97	973.18	250.00	800.00	1,380.15	3,973.34	17,856.58
Q2 2029	1,340.75	1,190.94	250.00	800.00	1,688.97	2,331.67	18,499.28
Q3 2029	893.83	1,336.11	250.00	800.00	1,894.84	3,293.27	19,897.71
Q4 2029	1,340.75	1,553.86	250.00	800.00	2,203.66	3,618.40	21,312.45
Q1 2030	-1,993.22	1,171.22	250.00	800.00	1,661.00	5,595.72	25,247.17
Q2 2030	1,566.76	1,427.48	250.00	800.00	2,024.42	2,944.29	26,167.03
Q3 2030	1,044.51	1,598.32	250.00	800.00	2,266.70	4,072.25	27,972.58
Q4 2030	1,566.76	1,854.57	250.00	800.00	2,630.12	4,458.55	29,801.00

Persistent cash path 11000 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
6.6150%	8,080.42	4,776.74	34,590.76	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	20,805.75	4,385.64	1,771.48	2,614.16	0.00	6,241.72
Q2 2027	23,778.00	5,012.16	1,777.40	3,234.76	0.00	7,133.40
Q3 2027	25,759.50	5,429.84	1,781.35	3,648.49	0.00	7,727.85
Q4 2027	28,731.75	6,056.36	1,787.27	4,269.09	0.00	8,619.52
Q1 2028	28,932.75	6,257.79	2,202.03	4,055.76	0.00	8,679.82
Q2 2028	33,066.00	7,151.76	2,214.10	4,937.66	0.00	9,919.80
Q3 2028	35,821.50	7,747.74	2,222.15	5,525.59	0.00	10,746.45
Q4 2028	39,954.75	8,641.71	2,234.22	6,407.49	0.00	11,986.42
Q1 2029	35,952.00	7,944.30	2,636.25	5,308.05	0.00	10,785.60
Q2 2029	41,088.00	9,079.20	2,655.00	6,424.20	0.00	12,326.40
Q3 2029	44,512.00	9,835.80	2,667.50	7,168.30	0.00	13,353.60
Q4 2029	49,648.00	10,970.70	2,686.25	8,284.45	0.00	14,894.40
Q1 2030	42,073.50	9,321.90	3,013.75	6,308.15	0.00	12,622.05
Q2 2030	48,084.00	10,653.60	3,040.00	7,613.60	0.00	14,425.20
Q3 2030	52,091.00	11,541.40	3,057.50	8,483.90	0.00	15,627.30
Q4 2030	58,101.50	12,873.10	3,083.75	9,789.35	0.00	17,430.45

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-358.28	575.12	210.00	600.00	815.62	2,007.32	10,346.75
Q2 2027	891.68	711.65	210.00	600.00	1,009.25	1,241.44	10,578.95
Q3 2027	594.45	802.67	210.00	600.00	1,138.33	1,861.37	11,301.99
Q4 2027	891.68	939.20	210.00	600.00	1,331.95	2,048.21	12,018.25
Q1 2028	60.30	892.27	250.00	700.00	1,265.40	2,653.20	13,406.05
Q2 2028	1,239.98	1,086.29	250.00	700.00	1,540.55	2,161.40	14,026.90
Q3 2028	826.65	1,215.63	250.00	700.00	1,723.98	3,033.31	15,336.22
Q4 2028	1,239.98	1,409.65	250.00	700.00	1,999.14	3,307.86	16,644.95
Q1 2029	-1,200.82	1,167.77	250.00	800.00	1,656.11	4,791.10	19,779.94
Q2 2029	1,540.80	1,413.32	250.00	800.00	2,004.35	2,920.08	20,695.67
Q3 2029	1,027.20	1,577.03	250.00	800.00	2,236.51	4,014.07	22,473.23
Q4 2029	1,540.80	1,822.58	250.00	800.00	2,584.75	4,371.07	24,259.55
Q1 2030	-2,272.35	1,387.79	250.00	800.00	1,968.14	6,642.71	28,934.12
Q2 2030	1,803.15	1,674.99	250.00	800.00	2,375.44	3,585.46	30,144.13
Q3 2030	1,202.10	1,866.46	250.00	800.00	2,646.98	4,865.34	32,362.50
Q4 2030	1,803.15	2,153.66	250.00	800.00	3,054.28	5,282.54	34,590.76

Persistent cash path 11001 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
3.6750%	8,080.42	4,776.74	33,152.87	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	20,805.75	4,233.13	1,771.48	2,461.65	152.51	6,241.72
Q2 2027	23,778.00	4,837.86	1,777.40	3,060.46	174.30	7,133.40
Q3 2027	25,759.50	5,241.01	1,781.35	3,459.66	188.82	7,727.85
Q4 2027	28,731.75	5,845.75	1,787.27	4,058.47	210.61	8,619.52
Q1 2028	28,932.75	6,088.74	2,202.03	3,886.71	169.05	8,679.82
Q2 2028	33,066.00	6,958.56	2,214.10	4,744.46	193.20	9,919.80
Q3 2028	35,821.50	7,538.44	2,222.15	5,316.29	209.30	10,746.45
Q4 2028	39,954.75	8,408.26	2,234.22	6,174.03	233.45	11,986.42
Q1 2029	35,952.00	7,787.74	2,636.25	5,151.49	156.56	10,785.60
Q2 2029	41,088.00	8,900.28	2,655.00	6,245.28	178.92	12,326.40
Q3 2029	44,512.00	9,641.97	2,667.50	6,974.47	193.83	13,353.60
Q4 2029	49,648.00	10,754.51	2,686.25	8,068.26	216.19	14,894.40
Q1 2030	42,073.50	9,201.36	3,013.75	6,187.61	120.54	12,622.05
Q2 2030	48,084.00	10,515.84	3,040.00	7,475.84	137.76	14,425.20
Q3 2030	52,091.00	11,392.16	3,057.50	8,334.66	149.24	15,627.30
Q4 2030	58,101.50	12,706.64	3,083.75	9,622.89	166.46	17,430.45

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-358.28	541.56	210.00	600.00	768.04	1,888.36	10,171.64
Q2 2027	891.68	673.30	210.00	600.00	954.86	1,105.48	10,322.26
Q3 2027	594.45	761.13	210.00	600.00	1,079.42	1,714.09	10,956.93
Q4 2027	891.68	892.86	210.00	600.00	1,266.24	1,883.93	11,574.62
Q1 2028	60.30	855.08	250.00	700.00	1,212.66	2,521.34	12,883.31
Q2 2028	1,239.98	1,043.78	250.00	700.00	1,480.27	2,010.70	13,413.74
Q3 2028	826.65	1,169.58	250.00	700.00	1,658.68	2,870.06	14,625.11
Q4 2028	1,239.98	1,358.29	250.00	700.00	1,926.30	3,125.77	15,824.58
Q1 2029	-1,200.82	1,133.33	250.00	800.00	1,607.27	4,668.99	18,886.31
Q2 2029	1,540.80	1,373.96	250.00	800.00	1,948.53	2,780.52	19,718.30
Q3 2029	1,027.20	1,534.38	250.00	800.00	2,176.03	3,862.89	21,405.15
Q4 2029	1,540.80	1,775.02	250.00	800.00	2,517.30	4,202.44	23,090.30
Q1 2030	-2,272.35	1,361.27	250.00	800.00	1,930.53	6,548.69	27,708.45
Q2 2030	1,803.15	1,644.68	250.00	800.00	2,332.46	3,478.01	28,853.99
Q3 2030	1,202.10	1,833.63	250.00	800.00	2,600.41	4,748.93	31,002.51
Q4 2030	1,803.15	2,117.04	250.00	800.00	3,002.34	5,152.70	33,152.87

Persistent cash path 11010 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
2.2050%	8,080.42	4,776.74	31,083.61	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	18,191.25	3,980.39	1,771.69	2,208.71	0.00	5,457.38
Q2 2027	20,790.00	4,549.02	1,777.64	2,771.38	0.00	6,237.00
Q3 2027	22,522.50	4,928.10	1,781.61	3,146.49	0.00	6,756.75
Q4 2027	25,121.25	5,496.73	1,787.56	3,709.17	0.00	7,536.38
Q1 2028	25,310.25	5,660.08	2,182.08	3,478.00	0.00	7,593.07
Q2 2028	28,926.00	6,468.66	2,191.30	4,277.36	0.00	8,677.80
Q3 2028	31,336.50	7,007.72	2,197.45	4,810.26	0.00	9,400.95
Q4 2028	34,952.25	7,816.30	2,206.67	5,609.62	0.00	10,485.67
Q1 2029	31,479.00	7,183.89	2,599.29	4,584.60	0.00	9,443.70
Q2 2029	35,976.00	8,210.16	2,612.76	5,597.40	0.00	10,792.80
Q3 2029	38,974.00	8,894.34	2,621.74	6,272.60	0.00	11,692.20
Q4 2029	43,471.00	9,920.61	2,635.21	7,285.40	0.00	13,041.30
Q1 2030	36,907.50	8,443.68	2,962.93	5,480.75	0.00	11,072.25
Q2 2030	42,180.00	9,649.92	2,981.92	6,668.00	0.00	12,654.00
Q3 2030	45,695.00	10,454.08	2,994.58	7,459.50	0.00	13,708.50
Q4 2030	50,967.50	11,660.32	3,013.57	8,646.75	0.00	15,290.25

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,142.62	485.92	210.00	600.00	689.12	2,475.42	10,941.35
Q2 2027	779.62	609.70	210.00	600.00	864.67	992.05	11,068.73
Q3 2027	519.75	692.23	210.00	600.00	981.71	1,544.52	11,631.54
Q4 2027	779.62	816.02	210.00	600.00	1,157.26	1,723.53	12,197.81
Q1 2028	56.70	765.16	250.00	700.00	1,085.14	2,206.14	13,318.81
Q2 2028	1,084.72	941.02	250.00	700.00	1,334.54	1,801.62	13,785.89
Q3 2028	723.15	1,058.26	250.00	700.00	1,500.80	2,578.86	14,863.95
Q4 2028	1,084.73	1,234.12	250.00	700.00	1,750.20	2,840.78	15,954.52
Q1 2029	-1,041.98	1,008.61	250.00	800.00	1,430.40	4,067.96	18,592.09
Q2 2029	1,349.10	1,231.43	250.00	800.00	1,746.39	2,466.87	19,312.57
Q3 2029	899.40	1,379.97	250.00	800.00	1,957.05	3,443.23	20,798.75
Q4 2029	1,349.10	1,602.79	250.00	800.00	2,273.04	3,783.51	22,309.22
Q1 2030	-1,969.05	1,205.77	250.00	800.00	1,709.99	5,694.03	26,293.26
Q2 2030	1,581.75	1,466.96	250.00	800.00	2,080.42	3,069.29	27,282.13
Q3 2030	1,054.50	1,641.09	250.00	800.00	2,327.36	4,213.91	29,168.68
Q4 2030	1,581.75	1,902.29	250.00	800.00	2,697.79	4,612.72	31,083.61

Persistent cash path 11011 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
2.2050%	8,080.42	4,776.74	29,845.84	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	18,191.25	3,850.76	1,771.69	2,079.07	129.64	5,457.38
Q2 2027	20,790.00	4,400.87	1,777.64	2,623.23	148.16	6,237.00
Q3 2027	22,522.50	4,767.60	1,781.61	2,985.99	160.50	6,756.75
Q4 2027	25,121.25	5,317.71	1,787.56	3,530.15	179.02	7,536.38
Q1 2028	25,310.25	5,516.39	2,182.08	3,334.31	143.69	7,593.07
Q2 2028	28,926.00	6,304.44	2,191.30	4,113.14	164.22	8,677.80
Q3 2028	31,336.50	6,829.81	2,197.45	4,632.36	177.91	9,400.95
Q4 2028	34,952.25	7,617.87	2,206.67	5,411.19	198.43	10,485.67
Q1 2029	31,479.00	7,050.82	2,599.29	4,451.53	133.07	9,443.70
Q2 2029	35,976.00	8,058.08	2,612.76	5,445.32	152.08	10,792.80
Q3 2029	38,974.00	8,729.58	2,621.74	6,107.84	164.76	11,692.20
Q4 2029	43,471.00	9,736.84	2,635.21	7,101.63	183.77	13,041.30
Q1 2030	36,907.50	8,341.22	2,962.93	5,378.29	102.46	11,072.25
Q2 2030	42,180.00	9,532.82	2,981.92	6,550.90	117.10	12,654.00
Q3 2030	45,695.00	10,327.23	2,994.58	7,332.65	126.85	13,708.50
Q4 2030	50,967.50	11,518.83	3,013.57	8,505.26	141.49	15,290.25

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,142.62	457.40	210.00	600.00	648.67	2,374.30	10,776.94
Q2 2027	779.62	577.11	210.00	600.00	818.45	876.49	10,834.99
Q3 2027	519.75	656.92	210.00	600.00	931.63	1,419.33	11,322.68
Q4 2027	779.62	776.63	210.00	600.00	1,101.41	1,583.89	11,805.16
Q1 2028	56.70	733.55	250.00	700.00	1,040.30	2,094.06	12,858.92
Q2 2028	1,084.72	904.89	250.00	700.00	1,283.30	1,673.52	13,249.15
Q3 2028	723.15	1,019.12	250.00	700.00	1,445.30	2,440.09	14,243.94
Q4 2028	1,084.73	1,190.46	250.00	700.00	1,688.29	2,686.00	15,241.65
Q1 2029	-1,041.98	979.34	250.00	800.00	1,388.88	3,964.17	17,816.94
Q2 2029	1,349.10	1,197.97	250.00	800.00	1,698.94	2,348.25	18,466.25
Q3 2029	899.40	1,343.73	250.00	800.00	1,905.65	3,314.72	19,875.32
Q4 2029	1,349.10	1,562.36	250.00	800.00	2,215.71	3,640.17	21,299.79
Q1 2030	-1,969.05	1,183.22	250.00	800.00	1,678.03	5,614.12	25,235.88
Q2 2030	1,581.75	1,441.20	250.00	800.00	2,043.88	2,977.96	26,169.95
Q3 2030	1,054.50	1,613.18	250.00	800.00	2,287.79	4,114.96	27,997.13
Q4 2030	1,581.75	1,871.16	250.00	800.00	2,653.64	4,502.35	29,845.84

Persistent cash path 11100 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
2.8350%	8,080.42	4,776.74	34,663.24	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	20,815.51	4,393.94	1,773.43	2,620.51	0.00	6,244.65
Q2 2027	23,789.16	5,021.65	1,779.63	3,242.01	0.00	7,136.75
Q3 2027	25,771.59	5,440.12	1,783.77	3,656.35	0.00	7,731.48
Q4 2027	28,745.24	6,067.82	1,789.97	4,277.85	0.00	8,623.57
Q1 2028	28,945.19	6,268.37	2,204.51	4,063.85	0.00	8,683.56
Q2 2028	33,080.22	7,163.85	2,216.94	4,946.90	0.00	9,924.07
Q3 2028	35,836.90	7,760.83	2,225.23	5,535.60	0.00	10,751.07
Q4 2028	39,971.93	8,656.32	2,237.66	6,418.65	0.00	11,991.58
Q1 2029	35,967.12	7,957.15	2,639.27	5,317.88	0.00	10,790.14
Q2 2029	41,105.28	9,093.89	2,658.46	6,435.43	0.00	12,331.58
Q3 2029	44,530.72	9,851.71	2,671.24	7,180.47	0.00	13,359.22
Q4 2029	49,668.88	10,988.45	2,690.43	8,298.02	0.00	14,900.66
Q1 2030	42,091.30	9,337.03	3,017.31	6,319.72	0.00	12,627.39
Q2 2030	48,104.34	10,670.89	3,044.07	7,626.82	0.00	14,431.30
Q3 2030	52,113.04	11,560.13	3,061.91	8,498.22	0.00	15,633.91
Q4 2030	58,126.08	12,893.99	3,088.67	9,805.33	0.00	17,437.82

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-355.35	576.51	210.00	600.00	817.60	2,009.35	10,346.80
Q2 2027	892.09	713.24	210.00	600.00	1,011.51	1,246.68	10,581.96
Q3 2027	594.73	804.40	210.00	600.00	1,140.78	1,867.22	11,308.41
Q4 2027	892.09	941.13	210.00	600.00	1,334.69	2,054.63	12,028.35
Q1 2028	59.99	894.05	250.00	700.00	1,267.92	2,659.82	13,420.24
Q2 2028	1,240.51	1,088.32	250.00	700.00	1,543.43	2,168.08	14,044.89
Q3 2028	827.01	1,217.83	250.00	700.00	1,727.11	3,040.77	15,358.54
Q4 2028	1,240.51	1,412.10	250.00	700.00	2,002.62	3,316.04	16,671.96
Q1 2029	-1,201.44	1,169.93	250.00	800.00	1,659.18	4,799.39	19,812.17
Q2 2029	1,541.45	1,415.80	250.00	800.00	2,007.85	2,928.19	20,732.51
Q3 2029	1,027.63	1,579.70	250.00	800.00	2,240.31	4,023.13	22,515.34
Q4 2029	1,541.45	1,825.56	250.00	800.00	2,588.98	4,381.01	24,307.36
Q1 2030	-2,273.27	1,390.34	250.00	800.00	1,971.75	6,652.66	28,988.26
Q2 2030	1,803.91	1,677.90	250.00	800.00	2,379.57	3,595.01	30,203.70
Q3 2030	1,202.61	1,869.61	250.00	800.00	2,651.45	4,876.01	32,428.26
Q4 2030	1,803.91	2,157.17	250.00	800.00	3,059.26	5,294.24	34,663.24

Persistent cash path 11101 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
1.5750%	8,080.42	4,776.74	33,225.35	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	20,815.51	4,241.43	1,773.43	2,468.00	152.51	6,244.65
Q2 2027	23,789.16	4,847.35	1,779.63	3,067.71	174.30	7,136.75
Q3 2027	25,771.59	5,251.29	1,783.77	3,467.52	188.82	7,731.48
Q4 2027	28,745.24	5,857.21	1,789.97	4,067.24	210.61	8,623.57
Q1 2028	28,945.19	6,099.32	2,204.51	3,894.80	169.05	8,683.56
Q2 2028	33,080.22	6,970.65	2,216.94	4,753.70	193.20	9,924.07
Q3 2028	35,836.90	7,551.53	2,225.23	5,326.30	209.30	10,751.07
Q4 2028	39,971.93	8,422.87	2,237.66	6,185.20	233.45	11,991.58
Q1 2029	35,967.12	7,800.60	2,639.27	5,161.32	156.56	10,790.14
Q2 2029	41,105.28	8,914.97	2,658.46	6,256.51	178.92	12,331.58
Q3 2029	44,530.72	9,657.88	2,671.24	6,986.64	193.83	13,359.22
Q4 2029	49,668.88	10,772.25	2,690.43	8,081.83	216.19	14,900.66
Q1 2030	42,091.30	9,216.49	3,017.31	6,199.18	120.54	12,627.39
Q2 2030	48,104.34	10,533.13	3,044.07	7,489.06	137.76	14,431.30
Q3 2030	52,113.04	11,410.89	3,061.91	8,348.98	149.24	15,633.91
Q4 2030	58,126.08	12,727.53	3,088.67	9,638.87	166.46	17,437.82

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-355.35	542.96	210.00	600.00	770.02	1,890.39	10,171.68
Q2 2027	892.09	674.90	210.00	600.00	957.13	1,110.72	10,325.28
Q3 2027	594.73	762.86	210.00	600.00	1,081.87	1,719.94	10,963.35
Q4 2027	892.09	894.79	210.00	600.00	1,268.98	1,890.35	11,584.72
Q1 2028	59.99	856.86	250.00	700.00	1,215.18	2,527.96	12,897.50
Q2 2028	1,240.51	1,045.81	250.00	700.00	1,483.16	2,017.38	13,431.73
Q3 2028	827.01	1,171.79	250.00	700.00	1,661.81	2,877.51	14,647.43
Q4 2028	1,240.51	1,360.74	250.00	700.00	1,929.78	3,133.95	15,851.60
Q1 2029	-1,201.44	1,135.49	250.00	800.00	1,610.33	4,677.28	18,918.54
Q2 2029	1,541.45	1,376.43	250.00	800.00	1,952.03	2,788.63	19,755.14
Q3 2029	1,027.63	1,537.06	250.00	800.00	2,179.83	3,871.95	21,447.26
Q4 2029	1,541.45	1,778.00	250.00	800.00	2,521.53	4,212.38	23,138.10
Q1 2030	-2,273.27	1,363.82	250.00	800.00	1,934.14	6,558.63	27,762.59
Q2 2030	1,803.91	1,647.59	250.00	800.00	2,336.59	3,487.55	28,913.56
Q3 2030	1,202.61	1,836.78	250.00	800.00	2,604.88	4,759.60	31,068.28
Q4 2030	1,803.91	2,120.55	250.00	800.00	3,007.33	5,164.40	33,225.35

Persistent cash path 11110 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
0.9450%	8,080.42	4,776.74	31,156.09	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,650.00	1,524.00	3,126.00	0.00	6,600.00
Q1 2027	18,201.01	3,988.69	1,773.64	2,215.05	0.00	5,460.30
Q2 2027	20,801.16	4,558.51	1,779.87	2,778.63	0.00	6,240.35
Q3 2027	22,534.59	4,938.38	1,784.03	3,154.35	0.00	6,760.38
Q4 2027	25,134.74	5,508.19	1,790.26	3,717.93	0.00	7,540.42
Q1 2028	25,322.69	5,670.65	2,184.56	3,486.09	0.00	7,596.81
Q2 2028	28,940.22	6,480.75	2,194.14	4,286.60	0.00	8,682.07
Q3 2028	31,351.90	7,020.81	2,200.53	4,820.28	0.00	9,405.57
Q4 2028	34,969.43	7,830.90	2,210.11	5,620.79	0.00	10,490.83
Q1 2029	31,494.12	7,196.74	2,602.31	4,594.43	0.00	9,448.24
Q2 2029	35,993.28	8,224.85	2,616.22	5,608.63	0.00	10,797.98
Q3 2029	38,992.72	8,910.25	2,625.48	6,284.77	0.00	11,697.82
Q4 2029	43,491.88	9,938.36	2,639.39	7,298.97	0.00	13,047.56
Q1 2030	36,925.30	8,458.81	2,966.49	5,492.32	0.00	11,077.59
Q2 2030	42,200.34	9,667.21	2,985.99	6,681.22	0.00	12,660.10
Q3 2030	45,717.04	10,472.81	2,998.99	7,473.82	0.00	13,715.11
Q4 2030	50,992.08	11,681.21	3,018.49	8,662.73	0.00	15,297.62

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	687.72	210.00	600.00	0.00	1,074.63	9,155.05
Q1 2027	-1,139.70	487.31	210.00	600.00	691.10	2,477.44	10,941.39
Q2 2027	780.04	611.30	210.00	600.00	866.93	997.29	11,071.75
Q3 2027	520.03	693.96	210.00	600.00	984.16	1,550.37	11,637.96
Q4 2027	780.04	817.95	210.00	600.00	1,160.00	1,729.94	12,207.91
Q1 2028	56.39	766.94	250.00	700.00	1,087.66	2,212.76	13,333.01
Q2 2028	1,085.26	943.05	250.00	700.00	1,337.42	1,808.29	13,803.88
Q3 2028	723.51	1,060.46	250.00	700.00	1,503.93	2,586.31	14,886.27
Q4 2028	1,085.26	1,236.57	250.00	700.00	1,753.69	2,848.96	15,981.54
Q1 2029	-1,042.59	1,010.77	250.00	800.00	1,433.46	4,076.25	18,624.32
Q2 2029	1,349.75	1,233.90	250.00	800.00	1,749.89	2,474.98	19,349.42
Q3 2029	899.83	1,382.65	250.00	800.00	1,960.85	3,452.29	20,840.86
Q4 2029	1,349.75	1,605.77	250.00	800.00	2,277.28	3,793.45	22,357.03
Q1 2030	-1,969.97	1,208.31	250.00	800.00	1,713.60	5,703.98	26,347.41
Q2 2030	1,582.51	1,469.87	250.00	800.00	2,084.54	3,078.84	27,341.70
Q3 2030	1,055.01	1,644.24	250.00	800.00	2,331.83	4,224.57	29,234.45
Q4 2030	1,582.51	1,905.80	250.00	800.00	2,702.77	4,624.41	31,156.09

Persistent cash path 11111 | complete 18-quarter schedule

Order: Design / Optical / NRE price / Ramp / Capture. One coherent outcome persists through the entire path. Values are TWD millions. All original source controls and state values are consumed directly; no independent report-side financial engine is used.

Probability	Minimum cash	After advance stress	End cash	Desired-buffer gap	Hard cash gap
0.9450%	8,080.42	4,776.74	29,918.32	223.26	0.00

Quarter	revenue	Gross profit	opex	Operating profit	capture gp loss	Net operating working capital
Q3 2026	18,754.50	3,888.17	1,424.00	2,464.17	0.00	5,626.35
Q4 2026	22,000.00	4,517.00	1,524.00	2,993.00	133.00	6,600.00
Q1 2027	18,201.01	3,859.06	1,773.64	2,085.42	129.64	5,460.30
Q2 2027	20,801.16	4,410.35	1,779.87	2,630.48	148.16	6,240.35
Q3 2027	22,534.59	4,777.88	1,784.03	2,993.85	160.50	6,760.38
Q4 2027	25,134.74	5,329.17	1,790.26	3,538.91	179.02	7,540.42
Q1 2028	25,322.69	5,526.96	2,184.56	3,342.40	143.69	7,596.81
Q2 2028	28,940.22	6,316.53	2,194.14	4,122.38	164.22	8,682.07
Q3 2028	31,351.90	6,842.90	2,200.53	4,642.37	177.91	9,405.57
Q4 2028	34,969.43	7,632.47	2,210.11	5,422.36	198.43	10,490.83
Q1 2029	31,494.12	7,063.67	2,602.31	4,461.36	133.07	9,448.24
Q2 2029	35,993.28	8,072.77	2,616.22	5,456.55	152.08	10,797.98
Q3 2029	38,992.72	8,745.50	2,625.48	6,120.01	164.76	11,697.82
Q4 2029	43,491.88	9,754.59	2,639.39	7,115.21	183.77	13,047.56
Q1 2030	36,925.30	8,356.35	2,966.49	5,389.86	102.46	11,077.59
Q2 2030	42,200.34	9,550.11	2,985.99	6,564.13	117.10	12,660.10
Q3 2030	45,717.04	10,345.96	2,998.99	7,346.97	126.85	13,715.11
Q4 2030	50,992.08	11,539.72	3,018.49	8,521.23	141.49	15,297.62

Quarter	Working-capital cash use	Cash tax	non ROU da proxy	Capex and software cash	dividend	Free cash before dividends	Closing cash
Q3 2026	2,541.61	542.12	210.00	600.00	0.00	-1,009.55	8,080.42
Q4 2026	973.65	658.46	210.00	600.00	0.00	970.89	9,051.31
Q1 2027	-1,139.70	458.79	210.00	600.00	650.65	2,376.32	10,776.98
Q2 2027	780.04	578.71	210.00	600.00	820.71	881.73	10,838.00
Q3 2027	520.03	658.65	210.00	600.00	934.08	1,425.18	11,329.10
Q4 2027	780.04	778.56	210.00	600.00	1,104.14	1,590.31	11,815.26
Q1 2028	56.39	735.33	250.00	700.00	1,042.83	2,100.68	12,873.12
Q2 2028	1,085.26	906.92	250.00	700.00	1,286.18	1,680.20	13,267.14
Q3 2028	723.51	1,021.32	250.00	700.00	1,448.42	2,447.55	14,266.26
Q4 2028	1,085.26	1,192.92	250.00	700.00	1,691.78	2,694.18	15,268.67
Q1 2029	-1,042.59	981.50	250.00	800.00	1,391.94	3,972.45	17,849.18
Q2 2029	1,349.75	1,200.44	250.00	800.00	1,702.44	2,356.36	18,503.09
Q3 2029	899.83	1,346.40	250.00	800.00	1,909.44	3,323.78	19,917.43
Q4 2029	1,349.75	1,565.35	250.00	800.00	2,219.94	3,650.11	21,347.59
Q1 2030	-1,969.97	1,185.77	250.00	800.00	1,681.64	5,624.07	25,290.02
Q2 2030	1,582.51	1,444.11	250.00	800.00	2,048.01	2,987.50	26,229.52
Q3 2030	1,055.01	1,616.33	250.00	800.00	2,292.25	4,125.63	28,062.89
Q4 2030	1,582.51	1,874.67	250.00	800.00	2,658.63	4,514.05	29,918.32

Source register and exact financial provenance

All charts and tables project the approved financial source and preserved factual pack. The linked standalone ArtifactTool workbook passed saved reimport and all 24 selected perturbation/restoration tests; the portable validation workbook passed its 30 selected tests. These are saved-engine checks for the bound workbooks. Full native Microsoft Excel compatibility remains unverified.

Item	SHA-256	Scope
Financial source model	39ec619ad10d9105d824662a43c16d41e20fca249ef6f17ca5c0df505c464630	Immutable approved source; current monetary objects
Financial law	b6e935588b9453c9c80cb39c8ef8a1bed999798aa0336576160f1fd249068b7c	Complete financial rules and selected parameters
Financial authoring approval	b9a33dd41c8a31fe1bc8b96bc64619fa68ef1a8d8622fb88319f142a5e2485e4	Root financial selection; production release is separate
Independent financial review	6d1277a2ea9c4a34fa702b09d136763cf1d4ec2fc720a300744eef8c9907958a	32 states and paired stresses; external review receipt
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Expert claim map	6d3a01c9d18abec324593c00ab68f3531b6d43db0afd2bcf79506d5603957e86	All 30 attributed claims; no issuer-guidance conversion
Native OP-reserve preservation adapter	ec8035258ee41710ef576a56cfb48b4407466e6653707ea03c8ed44da5e7e5e7	Preserved absolute TWD-million valuation-only control; zero current snapshot; no group cash expense
Canonical linked model	9f8d0c20ca416f4c0a0943d4f45a53ffd88ba78acd1b93e84eda4caf636e7690	Approved financial source preserved; current status and actual companion metadata
Calculation graph	6600d42e98511118ba0c17608779ad76ef7c1c56b02df8c4fb9b76c0f2056375	Exact graph bytes used by the saved-engine checks
Formula workbook	09f95d4d4b1de735fbcc5fc28a94be46ef574ff757cf35d476e850ae91fbe94f	Exact saved standalone ArtifactTool workbook
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Saved portable proof	475d9cf689b7ff781dede758d224dcdbf8570243d171325799448ad517d4e5b2	30 selected tests passed and restored; portable validation scope

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FY25_financial_EN | FY2025	FY2025 financial statements in English; Note 5 principal-versus-agent accounting policy https://www.guc-asic.com/upload/2026_01_29/8_20260129161502ryd2n2II41.pdf	2105370bb75e6d8cdbe0d844924682e4ed28cd56638a52316b40601672277521
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Private source exhibits below are provided for the owner's research review. The anonymous translated pages are not public bulletin material. HTML and Markdown embed every figure and source-image payload. The companion files are Global_Unichip_3443_TW_Deep_Dive_model.json, Global_Unichip_3443_TW_Deep_Dive_CalculationGraph.json.gz and Global_Unichip_3443_TW_Deep_Dive_FormulaModel.xlsx; their exact hashes are recorded in the provenance table. The saved proof binds the same graph and formula workbook. The approved financial values are unchanged.

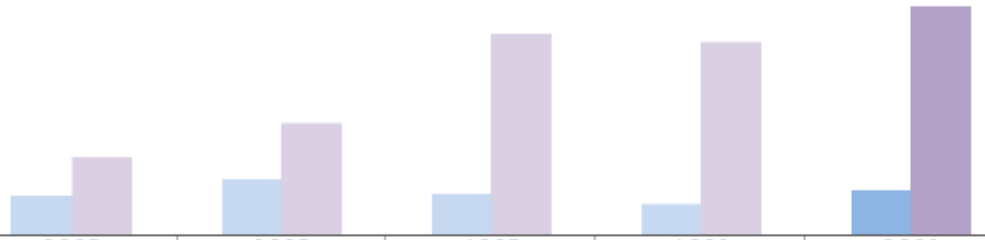
Issuer source exhibit 1 | NRE is the design ticket; turnkey is the production stream

2Q26 Revenue Breakdown

YoY

NRE & IP:+11%
Turnkey:+188%

QoQ
NRE & IP: +41%
Turnkey: +18%

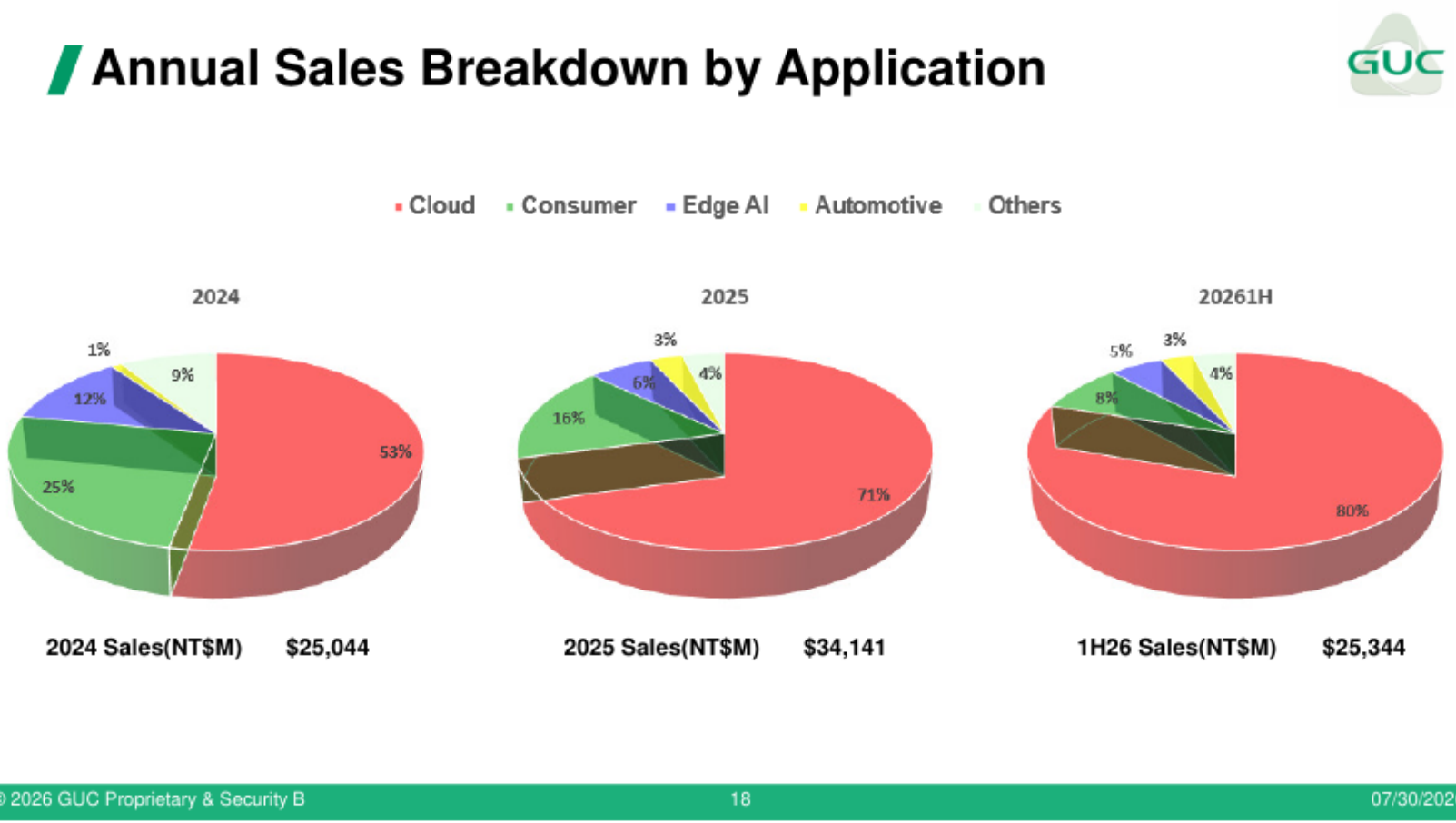


Unit:NTM	2Q25	3Q25	4Q25	1Q26	2Q26
NRE & IP	2,080	2,859	2,107	1,643	2,310
Turnkey	4,024	5,754	10,293	9,805	11,586

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Issuer source exhibit 1. GUC July 30, 2026 deck, page 10: Q2 turnkey grew 188% year over year while NRE grew 11%. Production content dominates the revenue acceleration. Original image bytes; source: Global Unichip. Financial periods and disclosure scope remain distinct from forecasts.

Issuer source exhibit 2 | Where the growth is coming from



Issuer source exhibit 2. GUC July 30 deck, page 18: cloud applications were 80% of H1 revenue. This application category does not identify individual customers. Original image bytes; source: Global Unichip. Financial periods and disclosure scope remain distinct from forecasts.

Issuer source exhibit 3 | Capacity comes with cash obligations

30. PLEDGED OR MORTGAGED ASSETS

As of June 30, 2026, December 31, 2025 and June 30, 2025, the pledged time deposits provided as collateral by GUC were as follows:

	June 30, 2026	December 31, 2025	June 30, 2025
Customs clearance	$ 40,000	$ 20,000	$ 20,000
Natural gas purchase	11,019	-	-
Land lease in Hsinchu Science Park	2,200	2,200	2,200
	$ 53,219	$ 22,200	$ 22,200

31. SIGNIFICANT CONTINGENT LIABILITIES AND UNRECOGNIZED COMMITMENTS

Under the agreement, GUC shall pay at least US$1,500 thousand to the counterparty in the period from June 2022.

Under the agreements, GUC shall pay at least US$2,100 thousand to the counterparty in the period from December 2026 to December 2029.

GUC has entered into a long-term advanced packaging capacity agreement with the counterparty. The contract service period is from July 2026 to December 2035. GUC should pay US$37,500 thousand in installments as security deposits to ensure the advanced packaging capacity supply in accordance with the contract, and GUC has paid US$12,500 thousand as of June 30, 2026. The amount of the security deposits refund will be calculated based on the annual production capacity utilization rate with the contract. However, if the certain level of production capacity utilization rate will not be achieved, the security deposits will not be refunded.

GUC has entered into a construction contract and purchase order with a certain company for the data center in Zhunan, with a total payable amount of $263,000 thousand, and GUC has paid $184,100 thousand as of June 30, 2026.

32. EXCHANGE RATE INFORMATION OF FOREIGN-CURRENCY FINANCIAL ASSETS AND LIABILITIES

The following information was aggregated by the foreign currencies other than functional currencies of the consolidated entities. The significant foreign-currency financial assets and liabilities were as follows:

(Unit: Foreign Currency in Thousands)

	Foreign Currency	Exchange Rate (Note)	Carrying Amount
June 30, 2026			
Monetary item - financial assets			
USD	$ 349,852	31.85	$ 11,144,465
Monetary item - financial liabilities			
USD	229,704	31.85	7,315,904
JPY	280,869	0.1963	55,135
RMB	4,112	4.6900	19,287
VND	4,156,388	0.00120	4,988

(Continued)

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Issuer source exhibit 3. June 30 financial statements, page 36: the packaging agreement runs July 2026-December 2035; US$37.5m security deposits are utilization-dependent and US$12.5m had already been paid. Original image bytes; source: Global Unichip. Financial periods and disclosure scope remain distinct from forecasts.

Issuer source exhibit 4 | A bigger invoice can produce a smaller profit

2Q26 Statements of Comprehensive Income

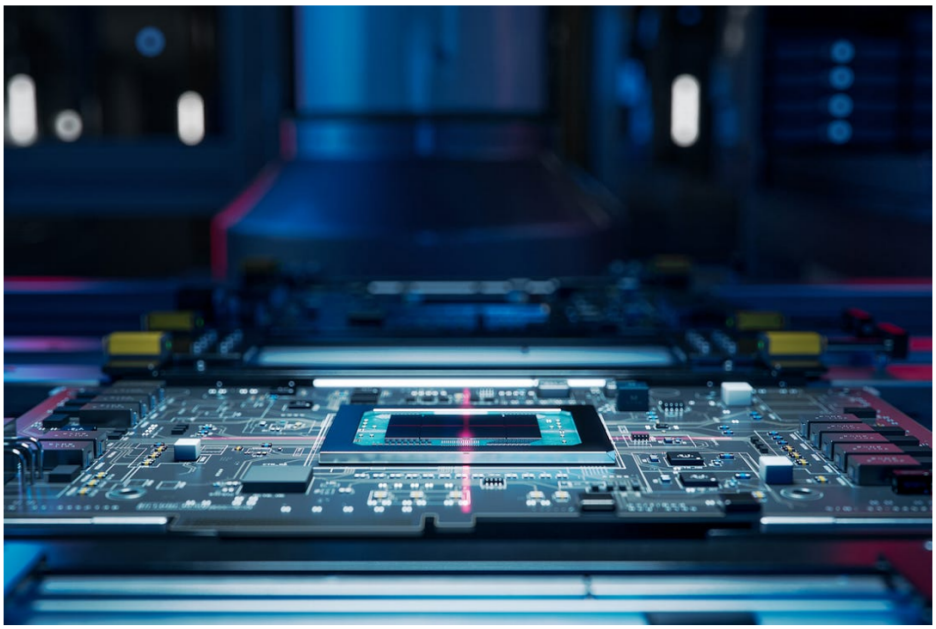
Unit: NT$M	2Q26	QoQ	1Q26	2Q25	YoY
Revenues	**13,896**	**21%**	**11,448**	**6,104**	**128%**
NRE & IP	2,310	41%	1,643	2,080	11%
Turnkey	11,586	18%	9,805	4,024	188%
Gross Margin	**2,994**	**-4%**	**3,113**	**2,034**	**47%**
%	21.5%		27.2%	33.3%	
SG&A Exp	(271)	-2%	(276)	(254)	7%
R&D Exp	(1,047)	3%	(1,013)	(837)	25%
Operating Income	**1,676**	**-8%**	**1,824**	**943**	**78%**
%	12.1%		15.9%	15.4%	
Non-Operating Items	**79**		**101**	**(15)**	
Income Tax	**(200)**		**(279)**	**(146)**	
Net Income	**1,555**	**-6%**	**1,646**	**782**	**99%**
%	11.2%		14.4%	12.8%	
EPS (NT$)	**11.61**	**-6%**	**12.28**	**5.84**	**99%**

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Issuer source exhibit 4. GUC July 30 deck, page 6: Q2 revenue increased 21% sequentially, while operating profit fell 8% as gross margin declined to 21.5%. Original image bytes; source: Global Unichip. Financial periods and disclosure scope remain distinct from forecasts.

Expert Call | GUC: 2026–2027 Hyperscaler ASIC & Supply Chain Outlook

P EQUITY RESEARCH
OCT 21, 2026 PAID



Questions:

1. What is the shipment growth of the Google CPU business in 2026 and 2027?
2. What are the main reasons 2026 Google CPU shipments exceeded expectations?
3. Considering capacity, what level are 2027 Google CPU shipments expected to reach?

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Private translated expert source | physical page 2

4. What scale are Microsoft Maia 200 shipments expected to reach in 2027?
5. How do GUC's pricing strategy and profit model differ for different types of customers?
6. How large is the margin difference between the TK1 and TK3 models?
7. Looking at the 2027 business structure, will TK1's revenue share rise substantially?
8. Which customer groups does GUC's TK1 business mainly serve?
9. What is the customer composition of the TK1 model? Why don't large customers place orders through it?
10. What are the directions for raising TK3 profit? Will advanced packaging technologies such as SoIC and CoWoS raise TK3's margin?
11. What is the specific cooperation model with Google on the ARM CPU project, and how is its value calculated?
12. How does the rebate from TSMC work specifically? Is it calculated by per-chip price?
13. What is the expected shipment volume of the Google CPU project next year?
14. What are the main drivers of the next-generation Google CPU's price doubling compared with the previous generation?
15. Please describe in detail GUC's specific supply chain management process from wafer start to final delivery.
16. What are the main cost components of GUC's overall supply chain management process?
17. In the volume production stage, does GUC take part in chip design, or does the customer (such as Google) provide a completed design?

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18. Some in the market believe that 3D IC-related work in Google's TPU v10 may be handed to GUC. Is this possible?
19. In cooperation with Google, how does GUC's division of labor differ from MediaTek's?
20. Why did Google not choose GUC in earlier projects such as TPU v8 and v9, but start considering GUC at the TPU v10 node?
21. In potential cooperation with Google, where do GUC's core IP advantages lie? Will Google choose GUC's package, or look for other IP suppliers itself?
22. What are GLink's specific application location and architecture? Is it used to connect different units such as SoC, GPU or CPU?
23. Looking to 2027, where will the incremental Microsoft and Google business mainly show up? Besides growth in chip shipments, will margins and new revenue opportunities change much?
24. How is Meta's business model different, and why can its project bring GUC higher margins?
25. What is the margin picture of the Tesla AI5 project in 2027?
26. How does the unit price of Tesla's AI5 chip compare with Google's and Microsoft's products?
27. What is the future growth trend of the China region business?
28. Who will be GUC's top three customers by revenue in 2027, and what is roughly their combined revenue share?
29. Given the Tesla AI5 project's expected 2027 shipments of 1 million units and its higher unit price, will its revenue contribution exceed Microsoft's?
30. Besides Google's business growth, what other new growth drivers are there in the future?

Summary:

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Google CPU Project Dynamics & Growth

- **Shipment Expansion:** Google CPU shipments surpassed 2026 targets (>1.2M units) due to higher end-market demand and reallocated capacity support from TSMC. Shipments are projected to grow by **60% to 70%** in 2027.
- **Capacity Realities:** While demand is doubling, market estimates of 5 million units for 2027 are considered overly optimistic due to ongoing capacity constraints.
- **Next-Gen Pricing & Design:** Next-generation Google CPUs are expected to double in price due to larger die sizes, doubled performance, and complex 2.5D/chiplet multi-die packaging (integrating two chips in one package).

Broad Hyperscaler Footprint (2027 Outlook)

- **Microsoft Maia 200:** Mass production in 2027 is expected to approach **2 million units**, exceeding early projections of ~1M units, though total scale remains slightly behind Google's volume.
- **Top 3 Revenue Drivers:** Google, Microsoft, and Tesla (AI5) will be GUC's top three customers in 2027, collectively representing roughly **70% of GUC's U.S. region revenue** (with the U.S. making up nearly 70% of total revenue).
- **Tesla AI5 Project:** Features a higher unit price than Google and Microsoft chips; however, revenue contribution will be lower than Microsoft's because Tesla orders wafers directly from TSMC, limiting GUC's income to design fees and TSMC rebates.
- **Meta & China Growth:** Meta projects will yield higher margins (~20%) on medium-volume designs. Regional revenue share in China will stabilize, yielding absolute business growth as overall GUC revenue grows.

GUC Business Models & Profit Structure

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- **TK1 vs. TK3 Models:**
  - **TK1 (Turnkey 1):** Higher double-digit margins serving mid/small enterprises and system houses without direct TSMC accounts.
  - **TK3 (Turnkey 3):** Single-digit margins providing supply chain management (SCM) for hyper-large clients (Google, Microsoft). Revenue share for TK3 will grow due to massive order scale.
- **Revenue Streams for Large Clients:** GUC earns low single-digit margins on SCM services combined with volume-based rebates from TSMC once unit delivery thresholds are met.
- **Supply Chain Management:** GUC operates a light-cost structure managing the flow from TSMC wafer start to CP testing, back-end OSAT assembly (CoWoS/3D packaging), and Final Test without bearing major material costs.

Competitive Positioning & IP Core Strengths

- **Evolving Client Relationship:** As hyperscalers like Google mature their in-house design capabilities, they move away from expensive end-to-end vendors (Broadcom, MediaTek) toward lower-cost partners like GUC to handle back-end physical design (Place & Route) and packaging.
- **IP Strengths:** GUC's primary IP edge lies in **HBM PHY**, **3D IC/stacking (GLink)**, and **CoWoS packaging IP**. Standard interface IPs like SerDes/PCIe are sourced externally by the client (e.g., Synopsys, Cadence).
- **Future Catalysts:** Next-gen growth drivers include 3D IC IP integration (e.g., TPU v10 potential cooperation), Horizon Robotics in China, and emerging clients in Japan.

Expert: experience with GUC (Global Unichip Corporation)

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Date: October 2026

Content below is translated with the use of Claude AI. Inconsistencies may appear.

Q&A

Google CPU Shipment Growth

Q1. What is the shipment growth of the Google CPU business in 2026 and 2027?

Shipments of the Google CPU business are expected to grow at least 60% to 70% in 2027. Actual 2026 shipments have already exceeded the expectation of 1.2 million units.

Q2. What are the main reasons 2026 Google CPU shipments exceeded expectations?

There are two main reasons. First, actual end-market demand is higher than originally forecast. The 2026 order forecast Google gave in the second half of 2025 was already 2.5 times the forecast from early 2025. Second, TSMC's capacity support exceeded expectations. Although TSMC initially said the large incremental demand would be hard to meet, it later reallocated capacity and ultimately basically met the order demand.

Q3. Considering capacity, what level are 2027 Google CPU shipments expected to reach?

2027 capacity has not been fully finalized; TSMC usually gives a conservative expectation first, but actual supply is often better than promised. At present, Google's demand is doubling, and capacity is not

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fully set, but figures circulating in the market such as 5 million units are too optimistic.

Microsoft Maia 200 Project

Q4. What scale are Microsoft Maia 200 shipments expected to reach in 2027?

Internal estimates are that Microsoft's shipments will be slightly lower than Google's, but the gap is not large. Although this is the product's first volume production, its scale will exceed the expectation of just over 1 million units and is expected to reach close to 2 million units.

GUC Business Model and Profit Structure

Q5. How do GUC's pricing strategy and profit model differ for different types of customers?

For very large customers such as Google and Microsoft, GUC's margin is quite low, and the main contribution is operating revenue. Profit mainly comes from supply chain management services under the TK3 (Turnkey 3) model. The company's main sources of profit are NRE (Non-Recurring Engineering, i.e., chip design and development services) and production services for small and medium customers, the latter with relatively high margins. In this way GUC balances revenue and profit between large customers and small and medium customers.

Q6. How large is the margin difference between the TK1 and TK3 models?

A: The margin difference between the two models is significant. For small and medium customers, the TK1 (Turnkey 1) model usually earns double-digit margins. The TK3 model provides only supply chain management services, with margins only in the single digits.

Q7. Looking at the 2027 business structure, will TK1's revenue share rise substantially?

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In terms of revenue share, TK1's proportion will not rise significantly and may even fall. This is because large customers' order values are huge and their growth rate and scale far exceed other businesses, so their share of total revenue expands quickly. However, in absolute amounts, both the TK1 business and NRE design service fees, the company's main business, will grow.

Q8. Which customer groups does GUC's TK1 business mainly serve?

TK1 customers mainly fall into two types: small start-ups, and system houses that have traditionally not been in the chip design field.

Q9. What is the customer composition of the TK1 model? Why don't large customers place orders through it?

TK1 customers are mainly small and medium enterprises. Large customers usually have a direct account with TSMC, so there is no need to order through the company. Even when a large customer cooperates with the company under the TK1 model, it is limited to purchasing design services, and does not place production orders through the company. Only small and medium customers that have not established a direct account with TSMC place orders through the TK1 model.

TK3 Business and Advanced Packaging

Q10. What are the directions for raising TK3 profit? Will advanced packaging technologies such as SoIC and CoWoS raise TK3's margin?

The TK3 business itself mainly earns single-digit margins through supply chain management. Cooperation involving advanced packaging such as SoIC and 3D stacking belongs to TK2, not TK3. TK2's margin is indeed higher than TK3's. Customers taking on advanced packaging are likewise mainly mid-sized companies. Although large customers have cooperation potential, the related business is expected to truly mature and ramp only in the second half of 2027 through 2028.

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Google CPU Cooperation Model and Value Calculation

Q11. What is the specific cooperation model with Google on the ARM CPU project, and how is its value calculated?

Cooperation on the Google CPU uses the TK3 model. Revenue mainly comes from two parts: first, the profit earned from providing Google with overall supply chain management services; second, the rebate received from TSMC. Because of GUC's close cooperation with TSMC, TSMC gives a certain rebate on business executed through the company, which is equivalent to sharing the resource input for customer management. Although the margins on both parts of revenue are low, both single-digit percentages, their absolute value is still considerable given the huge project scale.

Q12. How does the rebate from TSMC work specifically? Is it calculated by per-chip price?

The rebate takes the form that once total order volume exceeds a certain threshold, TSMC gives a rebate at a certain percentage of total units delivered. Specifically, it is calculated based on the total number of units delivered.

Next-Generation Google CPU

Q13. What is the expected shipment volume of the Google CPU project next year?

In 2027, the Google CPU project will be in an old/new product transition cycle, with old-model shipments declining and new-model shipments rising.

Q14. What are the main drivers of the next-generation Google CPU's price doubling compared with the previous generation?

The price doubling is driven mainly by three factors: first, the new generation's die size is larger; second, product performance has also

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doubled; finally, packaging complexity has risen significantly, for example the previous generation packaged one chip, while the new generation integrates two chips in one package. What is delivered is a complete chiplet, not a single chip, and the increase in complexity is the main reason for the price rise.

Supply Chain Management Process

Q15. Please describe in detail GUC's specific supply chain management process from wafer start to final delivery.

The whole process starts from the wafer. First, once the chip enters volume production, the company places a wafer order with TSMC. After the order is completed, wafer-level testing (CP) is performed. Next is the advanced packaging step, such as CoWoS. After advanced packaging, the chips are shipped to a back-end OSAT assembly and test house, where necessary assembly processes such as dicing are carried out. A final test (Final Test) is then performed. After passing the final test, the product can be delivered to the customer.

Q16. What are the main cost components of GUC's overall supply chain management process?

GUC's cost structure is very light, so its margin stays in the single digits (single-digit margin). Its core role is managing the whole production flow, and it does not bear major material costs itself. The highest-cost item, the wafer, is made by TSMC; CoWoS packaging and its substrate are also produced by TSMC. Shipping products to the test house for CP testing, and later to the assembly house (Assembly House) for assembly, are all done by third-party partners; the company provides no materials and is only responsible for managing the process.

Q17. In the volume production stage, does GUC take part in chip design, or does the customer (such as Google) provide a completed design?

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In the volume production stage, all earlier development work such as chip design and verification has been completed and formally released (Release to Production). So at this stage the company's job is to manage the whole production flow (Production Flow). Design work is also involved, but that happens at an earlier development stage, not in the volume production stage.

Possibility of 3D IC Cooperation

Q18. Some in the market believe that 3D IC-related work in Google's TPU v10 may be handed to GUC. Is this possible?

If 3D IC is involved, it is possible. Google may adopt the company's IP and have it assist with IP integration, and then Google would use the company's 3D IC IP for stacking. As for whether Google will use the company's overall services when finally integrating this part into the volume-production chip, the two sides are still negotiating and nothing is settled.

Division of Labor Differences with MediaTek

Q19. In cooperation with Google, how does GUC's division of labor differ from MediaTek's?

MediaTek's scope is broader than the company's. For example, the company may be responsible only for the 3D IC portion, while MediaTek's involvement may begin as early as the RTL (Register-Transfer Level) stage. In some projects, Google may subcontract part of the business to Broadcom and another part to MediaTek, usually giving Broadcom the parts with higher performance and speed requirements and MediaTek the parts with lower cost and less extreme performance requirements. In that case MediaTek may even participate from the earliest architecture and floor plan (Floor Plan) stage. By contrast, GUC usually does not step in at the initial stage of such large projects; its entry point is generally at the back end, such as the place-and-route (Place and Route) step of physical design (Physical Design). The main

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difference is that the proprietary IP GUC can offer is very limited, while companies such as MediaTek, Marvell and Broadcom have very complete IP platforms and can offer a full set of solutions including UCIe for die-to-die connection, PCIe and SerDes for chip-to-chip connection, and HBM controllers and PHY.

Why Google Chooses GUC

Q20. Why did Google not choose GUC in earlier projects such as TPU v8 and v9, but start considering GUC at the TPU v10 node?

After several product generations, Google's own design team has mastered the most core technology, and most design work can be handled by the in-house team. They therefore no longer need design companies that provide total solutions (Total Solution) like MediaTek or Broadcom. Once the in-house team can control most of the design flow, they look for lower-cost partners like GUC to outsource the remaining parts of the project and complete final productization.

GUC's Core IP Advantages

Q21. In potential cooperation with Google, where do GUC's core IP advantages lie? Will Google choose GUC's package, or look for other IP suppliers itself?

Google will combine according to need. For IP that GUC cannot provide, Google buys from other suppliers on the market, for example from Synopsys or Cadence. Google believes that handing everything to vendors like Broadcom costs too much, and that Broadcom earns far higher profit per chip than GUC. In GUC's IP portfolio, the most core and most valuable is the HBM PHY IP. In addition, the company's UCIe IP performs well but is not top-tier. The company's 3D stacking IP (called GLink) is outstanding. GUC's analog IP, such as ADC and signal-processing IP, is also somewhat competitive. However, GUC does not offer IP such as SerDes and PCIe. In advanced packaging, especially CoWoS, it also has quite good in-house IP.

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Q22. What are GLink's specific application location and architecture? Is it used to connect different units such as SoC, GPU or CPU?

GLink is not used for die-to-die interconnect inside a single die, but across the whole chiplet system, aiming to integrate all the chiplets in the system (including SoC, Service, GPU, CPU, etc.) into an organic whole. This is GUC's core area of expertise. The technology applies to both 2.5D packaging (such as CoWoS) and emerging 3D stacking packaging.

2027 Business Outlook

Q23. Looking to 2027, where will the incremental Microsoft and Google business mainly show up? Besides growth in chip shipments, will margins and new revenue opportunities change much?

Microsoft and Google revenue will grow, but margins are expected to stay stable without much change. By comparison, margins on Tesla's AI5 project and on Meta's project, which starts ramping in 2028, may be relatively higher.

Q24. How is Meta's business model different, and why can its project bring GUC higher margins?

Meta's business model is better for GUC than Microsoft's and Google's. Although its largest volume of chips is still ordered directly through TSMC, Meta has a small-to-medium-volume product for which, because the project uses a lot of GUC's design services (design service), it chooses to order through GUC. Under this model, the margin GUC earns is relatively high, expected to reach around 20%. By comparison, GUC's target margin for mid-sized customer projects is about 25%, and for small customers above 30%.

Q25. What is the margin picture of the Tesla AI5 project in 2027?

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The business model of this project is rather special. GUC mainly earns from this project fairly substantial margins on design service (design service) development fees, but wafer fabrication of the chips does not go through GUC. Because Tesla is a direct large customer of TSMC and can get better capacity support, it chooses to order directly from TSMC. The revenue GUC gets from this part is only the rebate provided by TSMC.

Q26. How does the unit price of Tesla's AI5 chip compare with Google's and Microsoft's products?

Tesla AI5's unit price is higher than Google's and Microsoft's products.

China Region Business

Q27. What is the future growth trend of the China region business?

Although exact shipment figures cannot be provided, it can be analyzed from the revenue share of GUC's internal regions. In recent years the China region's revenue percentage has declined. However, from 2026 to 2027, the China region's revenue share is expected to stay stable. Given that GUC's overall 2027 revenue will grow substantially, if the China region's share stays unchanged, its business volume will indeed rise significantly.

Q28. Who will be GUC's top three customers by revenue in 2027, and what is roughly their combined revenue share?

The top three customers in 2027 are expected to be Google, Microsoft and Tesla (as the AI5 project ramps). By region, the U.S. business's revenue share is already close to 70%. These three customers are expected to account for around 70% of U.S. region revenue.

Q29. Given the Tesla AI5 project's expected 2027 shipments of 1 million units and its higher unit price, will its revenue contribution exceed Microsoft's?

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No. Although Tesla AI5's unit price is higher, its business model is that GUC earns only a rebate rather than the full chip sales revenue, so the revenue it contributes to GUC will not exceed Microsoft's.

Future Growth Drivers

Q30. Besides Google's business growth, what other new growth drivers are there in the future?

Although Google may bring some of its business in-house in the future, GUC's growth trend remains favorable. New growth drivers keep emerging; for example, the Horizon Robotics project in the China region contributes significantly, and a company in Japan will also bring considerable business in the future. Overall, benefiting from the AI industry's development trend, the company's business will have ups and downs, but its long-term upward growth is certain.

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Complete economic law and selected control equations

The following exact source excerpt contains every economic calculation function and the complete selected parameter set. It is printed with display-only wrapping at a minimum 9 pt code font. Original unwrapped source is retained in Markdown / HTML and the audit. This report does not execute it as an independent financial evaluator.

```
import copy, itertools, math

EVENTS = ['NEW_DESIGN_CONVERSION','OPTICAL_IO_PRODUCTION','NEW_NRE_PRICING','RAMP_DELAY','REFERENCE_MARGIN_CAPTURE']

P = {
 'seasonality':[.21,.24,.26,.29],
 'annual':{'2027':[12000,83000,.60,.155,5400,1500],
           '2028':[15000,115000,.61,.165,6600,1850],
           '2029':[18000,142000,.62,.17,7800,2200],
           '2030':[21000,164000,.62,.17,8800,2500]},
 'gross_new_nre':[1000,2000,3000,4000], 'gross_new_tk':[12000,22000,30000,40000],
 'gross_optical':[300,1500,4000,7000],
 'nre_residual':.40, 'tk_residual':.30, 'optical_residual':.25,
 'design_nre_gp':.60, 'design_nre_opex':.25, 'design_tk_opex':.025,
 'optical_gp':.28,'optical_opex':.10,
 'price_eligible':.25,'price_premium':.03,'price_overlap_retention':.50,
 'price_gp':.85,'price_opex':.20,
 'probabilities':[.60,.35,.30,.30,.40],
 'delay_fraction':.15,'delay_avoidable_opex':.02,'annual_qualification_cost':250,
 'capture_eligible_reference_tk':.35,
 'capture_loss_ppt_by_year':{'2026':.02,'2027':.025,'2028':.02,'2029':.015,'2030':.01},
 'capture_delay_association':1/6,
 'branch_multiplier':1,'share_stress':1,
 'common_shares':134.011911,'potential_employee_shares':.251,'quarterly_share_allowance':.025,
 'cash':10034.146,'unrestricted_time_deposits':337.520,'operating_reserve':5000,
 'pension_claim':4.321,'multiple':20,
 'opening_ar':3933.121,'opening_inventory':14539.650,'opening_purchase_prepay':4418.131,
 'opening_ap':6591.440,'opening_customer_advances':13214.722,
 'deposit_usdm_remaining':25,'deposit_fx_assumption':32,
 'license_cash_due':402.792,'construction_commitment':78.9,
 'working_capital_revenue_ratio':.30,'tax_rate':.22,'dividend_after_tax_ratio':.40,
 'advance_loss_stress':.25,
 'source_authenticity_judgment':.60,
}

def parameters(overrides=None):
 p=copy.deepcopy(P)
 if overrides:
  for k,v in overrides.items():
   assert k in p,k;p[k]=copy.deepcopy(v)
 assert len(p['probabilities'])==5 and all(0<=x<=1 for x in p['probabilities'])
```

```
 assert p['branch_multiplier'] in [0,1], 'Preserved source control supports reference/active states 0/1'
 assert 0<=p['capture_delay_association']<=1 and 0<=p['capture_eligible_reference_tk']<=1
 assert p['share_stress']>0 and p['multiple']>=0
 return p

def joint_rm(p):
 # Construct an exact nonnegative coupling before binary-float conversion.
 # Decimal prevents cancellation producing a tiny negative cell at pM=1.
 # Excel mirrors this law with MAX(0,...) numerical closure on each cell;
 # this is probability arithmetic, never a floor on signed valuation.
 from decimal import Decimal, localcontext
 with localcontext() as ctx:
  ctx.prec=50
  r,m=[Decimal(str(x)) for x in p['probabilities'][3:5]]
  assoc=Decimal(str(p['capture_delay_association']))
  joint=r*m+assoc*(min(r,m)-r*m)
  cells={(0,0):Decimal(1)-r-m+joint,(0,1):m-joint,(1,0):r-joint,(1,1):joint}
  assert all(x>=0 for x in cells.values()) and sum(cells.values())==Decimal(1)
  return {k:(0.0 if v==0 else float(v)) for k,v in cells.items()},float(joint)

def reference_rows(p):
 rows=[]
 for i in range(18):
  n=2+i;y=2026+n//4;q=n%4+1;w=p['seasonality'][q-1]
  if i==0:nre,tk,gn,gt,rd,sga,rent=2500,16254.5,.58,.15,1120,300,4
  elif i==1:nre,tk,gn,gt,rd,sga,rent=3000,19000,.60,.15,1200,320,4
  else:
   a=p['annual'][str(y)];nre,tk=a[0]*w,a[1]*w;gn,gt=a[2:4];rd,sga=a[4]/4,a[5]/4;rent=5
  shares=p['common_shares']+p['potential_employee_shares']+p['quarterly_share_allowance']*(i+1)
  claim=p['operating_reserve']+p['pension_claim']-(p['cash']+p['unrestricted_time_deposits'])
  rr=dict(quarter=f'Q{q} {y}',year=y,q=q,weight=w,nre=nre,tk=tk,nre_gm=gn,tk_gm=gt,
          rd=rd,sga=sga,rent_reserve=rent,raw_shares=shares,shares=shares*p['share_stress'],
          claims=claim,multiple=p['multiple'],da=210 if y<2028 else 250)
  rr['reference_revenue']=nre+tk;rr['reference_gp']=nre*gn+tk*gt
  rr['reference_opex']=rd+sga+rent;rr['reference_op']=rr['reference_gp']-rr['reference_opex']
  rows.append(rr)
 return rows

def conditional_terms(r,p,design_bit=None):
 y=r['year'];i=y-2027;w=r['weight'];eligible=y>=2027
 nn=p['gross_new_nre'][i]*p['nre_residual']*w if eligible else 0
 tk=p['gross_new_tk'][i]*p['tk_residual']*w if eligible else 0
 optical=p['gross_optical'][i]*p['optical_residual']*w if eligible else 0
 design_level=p['probabilities'][0] if design_bit is None else design_bit
 price=(r['nre']+design_level*nn)*p['price_eligible']*p['price_premium']*p['price_overlap_retention'] if eligible else 0
 delay=-r['tk']*p['delay_fraction'] if eligible else 0
 design=(nn+tk,nn*p['design_nre_gp']+tk*r['tk_gm'],nn*p['design_nre_opex']+tk*p['design_tk_opex'])
```

```
 opt=(optical,optical*p['optical_gp'],optical*p['optical_opex'])
 pr=(price,price*p['price_gp'],price*p['price_opex'])
 ramp=(delay,delay*r['tk_gm'],delay*p['delay_avoidable_opex']+(p['annual_qualification_cost']*w if eligible else 0))
 return design,opt,pr,ramp

def capture_amount(r,p,ramp_bit=None):
 if r['quarter']=='Q3 2026':return 0.0
 delay=p['delay_fraction'] if r['year']>=2027 else 0
 if ramp_bit is None:
  _,joint=joint_rm(p);m=p['probabilities'][4]
  retained=1-delay*joint/m if m else 1
 else:retained=1-delay*ramp_bit
 return r['tk']*p['capture_eligible_reference_tk']*p['capture_loss_ppt_by_year'][str(r['year'])]*retained

def calculate(overrides=None):
 p=parameters(overrides);base=reference_rows(p);rows=[];bm=p['branch_multiplier']
 for r in base:
  terms=list(conditional_terms(r,p));terms.append((0,-capture_amount(r,p),0))
  bs=[{'id':k,'p':prob,'revenue':x[0],'gp':x[1],'opex':x[2]} for k,prob,x in zip(EVENTS,p['probabilities'],terms)]
  z=copy.deepcopy(r);z['branches']=bs
  for fld in ['revenue','gp','opex']:
   z[fld]=z['reference_'+fld]+sum(b['p']*b[fld]*bm for b in bs)
  z['op']=z['gp']-z['opex'];z['op_margin']=z['op']/z['revenue']
  z['capture_expected_gp_loss']=p['probabilities'][4]*capture_amount(r,p)*bm
  rows.append(z)
 for i,r in enumerate(rows[:15]):
  r['f4q_op']=sum(x['op'] for x in rows[i:i+4]);r['valuation_shares']=rows[i+3]['shares']
  r['window']=' + '.join(x['quarter'] for x in rows[i:i+4])
  # Authoritative source native method is linear, including negative perturbations.
  r['ceiling']=(r['f4q_op']*r['multiple']-rows[i+3]['claims'])/r['valuation_shares']
  r['reference_ceiling']=(sum(x['reference_op'] for x in rows[i:i+4])*r['multiple']-rows[i+3]['claims'])/r['valuation_shares']
 cells,joint=joint_rm(p);states=[]
 opening=p['cash']+p['unrestricted_time_deposits']-p['deposit_usdm_remaining']*p['deposit_fx_assumption']-p['license_cash_due']-p['construction_commitment']
 initial_wc=p['opening_ar']+p['opening_inventory']+p['opening_purchase_prepay']-p['opening_ap']-p['opening_customer_advances']
 for bits in itertools.product([0,1],repeat=5):
  weight=cells[bits[3],bits[4]]
  for bit,prob in zip(bits[:3],p['probabilities'][:3]):weight*=prob if bit else 1-prob
  cash=opening;prev=initial_wc;rr=[]
  for r in base:
   terms=conditional_terms(r,p,design_bit=bits[0]);rev=r['reference_revenue'];gp=r['reference_gp'];opex=r['reference_opex']
   for bit,x in zip(bits[:4],terms):
    rev+=bm*bit*x[0];gp+=bm*bit*x[1];opex+=bm*bit*x[2]
   loss=bm*bits[4]*capture_amount(r,p,ramp_bit=bits[3]);gp-=loss;op=gp-opex
   wc=rev*p['working_capital_revenue_ratio'];dwc=wc-prev
   capex=600 if r['year']<2028 else 700 if r['year']==2028 else 800
   tax=max(0,op)*p['tax_rate'];div=max(0,op-tax)*p['dividend_after_tax_ratio'] if r['year']>=2027 else 0
   cf=op-tax+r['da']-dwc-capex;cash+=cf-div
```

```
    rr.append(dict(quarter=r['quarter'],revenue=rev,gp=gp,opex=opex,op=op,capture_gp_loss=loss,
                   cash_end=cash,wc=wc,wc_change=dwc,cash_tax=tax,non_ROU_da_proxy=r['da'],
                   capex_and_software=capex,dividend=div,free_cash_before_dividend=cf))
    prev=wc
   minimum=min([opening]+[x['cash_end'] for x in rr]);after=min(x['cash_end'] for x in rr)-p['opening_customer_advances']*p['advance_loss_stress']
   states.append(dict(bits=list(bits),probability=weight,minimum_cash=minimum,minimum_after_advance_stress=after,
                      end_cash=cash,reserve=p['operating_reserve'],desired_reserve_gap=max(0,p['operating_reserve']-after),
                      hard_cash_gap=max(0,-after),rows=rr))
  assert math.isclose(sum(s['probability'] for s in states),1,abs_tol=1e-12)
  for i,r in enumerate(rows):
   for k in ['revenue','gp','opex','op']:
    v=sum(s['probability']*s['rows'][i][k] for s in states)
    assert math.isclose(v,r[k],abs_tol=2e-8),('Expected-state mismatch',r['quarter'],k,v,r[k])
  annual={str(y):{k:sum(r[k] for r in rows if r['year']==y) for k in ['reference_revenue','revenue','gp','opex','op','capture_expected_gp_loss']} for y in rang
e(2027,2031)}
  return dict(parameters=p,rows=rows,annual=annual,states=states,joint_R_M={f'{r}{m}':v for (r,m),v in cells.items()},joint_R_M_mass=joint)
```